Fiber Bundle Construction of Quantum Electrodynamics
Geometry, Dynamics, and Perturbations

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Abstract

We construct the minimal Lagrangian of quantum electrodynamics (QED) to understand how the electron field and the electromagnetic field couple. We start with Clifford algebra and the Spin group to construct the spinor bundle. The freedom to choose the phase at each point motivates the electromagnetic connection, whose parallel transport lets us compare spinors at different points. We obtain the coupled Dirac and Maxwell equations from a single action. In the electrostatic regime, we recover the Coulomb potential of a point charge and solve the case of a uniformly charged ball. Finally, we obtain the free propagators and the interaction vertex, and calculate a second-order contribution to electron–electron scattering.

1 Introduction

How does an atom emit light? What happens when two electrons approach each other? Quantum electrodynamics, or QED, brings these questions together in a theory of the interaction between light and charged matter. In 1865, James Clerk Maxwell showed that light could be understood as a wave of the electric and magnetic fields [2]. Among the questions still open was why atoms emit light at particular frequencies.

In 1927, Paul Dirac described light emission and absorption through processes in which quanta of radiation are created and absorbed [3]. The following year he proposed an equation for the electron compatible with special relativity [4]. The equation is written in terms of a spinor, represented in a given frame as a column of four complex numbers.

Spinors already had a mathematical history. In 1913, while studying groups of transformations, Élie Cartan found the representations now called spin representations [5]. Dirac's equation gave them a role in describing the electron.

In 1929, Hermann Weyl sought to write the Dirac equation in the curved spacetime of general relativity. He used frames chosen separately at each point [6]. If the phase of a spinor is like an angle, can we also choose at each point the reference from which we measure it? Yes, provided the electromagnetic potential changes along with that reference. The potential lets us compare neighboring spinors, and the freedom to change the reference is known as *gauge symmetry*.

In the 1940s, Sin-Itiro Tomonaga, Julian Schwinger, and Richard Feynman developed methods for extracting finite predictions from calculations that had previously given infinite results. In 1949, Freeman Dyson showed how their formulations were related, and Feynman diagrams provided a way to organize the terms in the perturbative expansion [7, 8, 9].

Decades later, the geometry of principal bundles provided a language for describing gauge fields globally. In 1975, Tai Tsun Wu and Chen Ning Yang returned to the Aharonov–Bohm effect. A charged particle can acquire a phase as it goes around a loop even if the electric and magnetic fields vanish along its path. Wu and Yang described this phase using a connection and used the monopole to show why a single connection may require several local potentials [14].

In our construction, the principal spin bundle and the spinor vector bundle play complementary roles. The first organizes the frames in which we can write spinors and the changes between them. The second, which we construct from the first, assembles the vector spaces S_x of field values, one for each point x of spacetime. A Dirac field smoothly chooses a spinor in each S_x. In Minkowski spacetime we can fix a global frame and read those choices as a single function $\psi : M \to \mathbb{C}^4$. This recovers the columns that appear in the Dirac equations.

The internal freedom of phase, described by the group $U(1)$, also acts on this field. The electromagnetic connection, written using the potential A_μ once the phase reference has been chosen, lets us transport spinors between points and compare them there. Its curvature contains the electric and magnetic fields.

First we construct the space in which the Dirac field lives, then we incorporate the rule for comparing its values. Once we add the dynamics of the electromagnetic field, a single action yields the coupled Dirac and Maxwell equations. We apply Maxwell's equations to two electrostatic examples and finally use the Lagrangian to obtain the free propagators and calculate a contribution to the scattering of two electrons.

We work in Minkowski spacetime and use natural units $\hbar = c = 1$. In the electrostatic examples we give the results in SI units.

2 From Clifford Algebra to the Spinor Bundle

Before writing the Dirac field as a function with values in $\mathbb{C}^4$, we construct the geometric space in which it takes its values. There are two steps. First we work with the Minkowski vector space and its Clifford algebra. Then we carry the construction over to the tangent spaces of a manifold using frame bundles. This distinguishes the point in spacetime, the chosen frame, and the components of the spinor.

2.1 The Clifford Algebra and Its Unit

Definition 1 (Tensor algebra)**.** Let V be a finite-dimensional real vector space and let

$$\eta : V \times V \longrightarrow \mathbb{R}, \qquad (u, v) \longmapsto \eta(u, v), \tag{1}$$

be a symmetric, nondegenerate bilinear form. That is, it is linear in each argument, satisfies $\eta(u, v) = \eta(v, u)$, and if $\eta(u, v) = 0$ for all $v \in V$, then $u = 0$. We do not require it to be positive definite. The tensor algebra of V is

$$T(V) \coloneqq \bigoplus_{k=0}^{\infty} V^{\otimes k} = \mathbb{R} \oplus V \oplus (V \otimes V) \oplus \cdots . \tag{2}$$

The product concatenates tensors: a tensor of degree r multiplied by one of degree s gives one of degree $r + s$. The unit is $1_T = (1, 0, 0, \ldots)$, the real number 1 in degree zero. Thus $v \otimes v$ and $\eta(v, v)1_T$ belong to the same algebra, even though they have different degrees.

Definition 2 (Clifford algebra)**.** The Clifford algebra of (V, η) is the quotient

$$\boxed{\mathrm{Cl}(V, \eta) \coloneqq T(V)/I_\eta, \qquad I_\eta \coloneqq \langle v \otimes v - \eta(v, v)1_T : \ v \in V \rangle,} \tag{3}$$

where I_η is the two-sided ideal generated by the indicated expressions. Its unit is the class $1 = [1_T]$. We identify V with its canonical image in the quotient and write the product without the tensor symbol. The defining relation is

$$v^2 = \eta(v, v)1, \qquad v \in V. \tag{4}$$

Expanding the same relation for $u + v$ and subtracting the relations for u and v, we obtain

$$\begin{gathered} u^2 + uv + vu + v^2 = [\eta(u, u) + 2\eta(u, v) + \eta(v, v)]1, \\ \boxed{uv + vu = 2\eta(u, v)1.} \end{gathered} \tag{5}$$

The product is constrained by these relations. In particular, orthogonal vectors anticommute.

Example 1 (Minkowski Clifford algebra)**.** For $V = \mathbb{R}^{1,3}$ we choose the ordered basis $\widehat{e} = (e_0, e_1, e_2, e_3)$ with $\eta = \mathrm{diag}(1, -1, -1, -1)$. Then

$$e_0^2 = 1, \qquad e_i^2 = -1 \ (i = 1, 2, 3), \qquad e_a e_b = -e_b e_a \ (a \neq b). \tag{6}$$

We denote this algebra by $\mathrm{Cl}_{1,3}(\mathbb{R})$. Its elements include vectors, scalars, products of vectors, and sums of these. The ordered products with distinct indices, including the unit, form a basis of dimension $2^4 = 16$. The construction preserves the parity of the number of factors. It can identify different tensor degrees, as in $e_0 e_0 = 1$, which identifies degrees two and zero.

□

2.2 The Dirac Representation and Gamma Matrices

Definition 3 (Dirac representation and gamma matrices)**.** To work with matrices, we choose a real unital representation of the algebra on the complex vector space $\mathbb{C}^4$.[1] In symbols,

$$\boxed{c : \mathrm{Cl}_{1,3}(\mathbb{R}) \longrightarrow \mathrm{End}_{\mathbb{C}}(\mathbb{C}^4), \qquad c(ab) = c(a)c(b), \quad c(1) = I_4.} \tag{7}$$

The homomorphism is linear over $\mathbb{R}$, and its values are complex-linear operators on $\mathbb{C}^4$. We write $e^\mu = \eta^{\mu\nu} e_\nu$ for the vectors of the reciprocal basis of $\widehat{e}$ and define

$$\gamma^\mu \coloneqq c(e^\mu), \qquad \gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2\eta^{\mu\nu} I_4. \tag{8}$$

The gamma matrices represent the generators. They enter after the abstract definition of the algebra. A product and a vector are represented, respectively, by

$$c(e^\mu e^\nu) = \gamma^\mu \gamma^\nu, \qquad c(v) = v_\mu \gamma^\mu \quad \text{if } v = v_\mu e^\mu. \tag{9}$$

A Minkowski vector is represented by a matrix acting on $\mathbb{C}^4$. A spinor column belongs to the space on which that matrix acts.

[1] For a real associative unital algebra A, a representation on a complex vector space W is a homomorphism of real algebras $c : A \to \mathrm{End}_{\mathbb{C}}(W)$. This means that $c(ra + sb) = r\, c(a) + s\, c(b)$ for $r, s \in \mathbb{R}$ and $a, b \in A$, and that $c(ab) = c(a)c(b)$, where the right side is composition of operators. The representation is *unital* if it also satisfies $c(1_A) = \mathrm{id}_W$. Each algebra element then acts as an operator on W compatibly with addition, multiplication, and the unit. Here $W = \mathbb{C}^4$, so the operators are complex 4×4 matrices. The word *real* refers to linearity over the scalars of the original algebra. The matrix entries can be complex. An injective representation is called *faithful*.

Example 2 (Gamma matrices from Pauli matrices)**.** Let the Pauli matrices be

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{10}$$

They satisfy $\sigma^i\sigma^j + \sigma^j\sigma^i = 2\delta_{ij}I_2$. The block matrices

$$\gamma^0 = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix}, \qquad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix} \tag{11}$$

satisfy $(\gamma^0)^2 = I_4$, $(\gamma^i)^2 = -I_4$, $\gamma^0\gamma^i + \gamma^i\gamma^0 = 0$, and the remaining Clifford relations. By the definition of the quotient, this assignment on the generators extends to a homomorphism c.

□

By allowing complex coefficients we obtain

$$\mathrm{Cl}_{1,3}(\mathbb{R}) \otimes_{\mathbb{R}} \mathbb{C} \simeq \mathrm{Mat}_4(\mathbb{C}). \tag{12}$$

The complexified Minkowski Clifford algebra is isomorphic to the algebra of all complex 4×4 matrices. The bijection preserves sums, products, and the unit. In the chosen representation, the generators e^μ correspond to the matrices γ^μ, their products correspond to matrix products, and the unit corresponds to I_4. Thus the gamma matrices and their products generate all of $\mathrm{Mat}_4(\mathbb{C})$ with complex coefficients.

To check this, take the 16 ordered products γ_I with distinct indices, including $\gamma_\varnothing = I_4$. Every nonempty product has zero trace. For an even number of factors, conjugation by a gamma matrix in the product reverses its sign. For an odd number, conjugation by a gamma matrix absent from the product reverses it. For $I \neq J$, the product $\gamma_I^{-1}\gamma_J$ reduces, up to sign, to a nonempty product, while for $I = J$ it equals I_4. Thus $\mathrm{tr}(\gamma_I^{-1}\gamma_J) = 4\delta_{IJ}$. The 16 products are therefore linearly independent and form a basis of $\mathrm{Mat}_4(\mathbb{C})$.

The representation c suffices to continue the construction of the spinor bundle. The isomorphism explains its scope and uses complexification. The real algebra and the full matrix algebra $\mathrm{Mat}_4(\mathbb{C})$ are distinct as real algebras. To construct the Spin group, we continue to work in the real algebra.

2.3 A Reflection Written Using the Clifford Product

Take $u \in V$ with $\eta(u,u) \neq 0$. As $u^2 = \eta(u,u)1$, it is invertible:

$$u^{-1} = \frac{u}{\eta(u,u)}. \tag{13}$$

Definition 4 (Orthogonal hyperplane)**.** The hyperplane orthogonal to u is $u^\perp \coloneqq \{w \in V : \eta(u,w) = 0\}$. Orthogonality is taken with respect to η, which may be indefinite.

For any $v \in V$, we define

$$a \coloneqq \frac{\eta(u,v)}{\eta(u,u)}, \qquad v_\parallel \coloneqq au, \qquad w \coloneqq v - au. \tag{14}$$

The first component belongs to $\mathrm{span}(u)$. The second is on the hyperplane because

$$\eta(u,w) = \eta(u,v) - a\eta(u,u) = \eta(u,v) - \frac{\eta(u,v)}{\eta(u,u)}\eta(u,u) = 0. \tag{15}$$

Moreover, if $bu \in u^\perp$, then $0 = b\eta(u,u)$ forces $b = 0$. This proves the unique decomposition

$$\boxed{V = \mathrm{span}(u) \oplus u^\perp, \qquad v = \underbrace{au}_{\in \mathrm{span}(u)} + \underbrace{(v - au)}_{w \in u^\perp}.} \tag{16}$$

The subscript $\|$ refers to the direction u. Relative to the hyperplane, au is the normal component and w is the tangential component. Rescaling u by a nonzero number changes a and leaves both au and w unchanged. Replacing au with the unit vector u while keeping v fixed and $w \in u^\perp$ is possible only if $a = 1$.

Definition 5 (Reflection across the hyperplane $u^\perp$)**.** We define the linear map

$$\boxed{R_u(v) \coloneqq -uvu^{-1}.} \tag{17}$$

We check that this algebraic expression is again a vector and has the stated meaning. Since $w \in u^\perp$, the Clifford relation gives $uw = -wu$. Substituting $v = au + w$, we obtain

$$\begin{aligned} -uvu^{-1} &= -u(au + w)u^{-1} \\ &= -au^2u^{-1} - uwu^{-1} \\ &= -au + wu\,u^{-1} = -au + w. \end{aligned} \tag{18}$$

Therefore,

$$\boxed{R_u(v) = -au + w = v - 2\frac{\eta(u,v)}{\eta(u,u)}u.} \tag{19}$$

The map reverses the sign of the component along u and leaves the component in $u^\perp$ fixed. For $v = au + w$ and $v' = bu + w'$, orthogonality eliminates the cross terms.

$$\begin{aligned} \eta(v, v') &= \eta(au + w, bu + w') \\ &= ab\eta(u,u) + a\underbrace{\eta(u,w')}_{0} + b\underbrace{\eta(w,u)}_{0} + \eta(w,w') \\ &= ab\eta(u,u) + \eta(w,w'), \\ \eta(R_u(v), R_u(v')) &= \eta(-au + w, -bu + w') \\ &= ab\eta(u,u) - a\underbrace{\eta(u,w')}_{0} - b\underbrace{\eta(w,u)}_{0} + \eta(w,w') \\ &= ab\eta(u,u) + \eta(w,w') = \eta(v,v'). \end{aligned} \tag{20}$$

Accordingly, for any $v, v' \in V$,

$$\boxed{\eta(R_u(v), R_u(v')) = \eta(v, v').} \tag{21}$$

The preceding identity says that R_u preserves the metric. It preserves the metric product of any two vectors. We also obtain from $R_u(au + w) = -au + w$ that

$$R_u(au) = -au, \qquad R_u(w) = w \quad (w \in u^\perp). \tag{22}$$

Thus -1 is the eigenvalue on $\mathrm{span}(u)$ and $+1$ is the eigenvalue on $u^\perp$. Since $V = \mathrm{span}(u) \oplus u^\perp$, if $n = \dim V$ we can choose a basis $(u, w_1, \ldots, w_{n-1})$ with the w_i in $u^\perp$. In that basis,

$$[R_u] = \begin{pmatrix} -1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}, \qquad \boxed{\det R_u = (-1)\underbrace{(1)\cdots(1)}_{n-1} = -1.} \tag{23}$$

Thus, reflection retains the metric and reverses a direction.

The conjugation alone $uvu^{-1} = -R_u(v)$ is not this reflection. To construct the following groups we will normalize the vectors so that $u^2 = \pm 1$.

Example 3 (Minkowski reflection in (t, x, y))**.** We work with the metric

$$\eta = \operatorname{diag}(1, -1, -1). \tag{24}$$

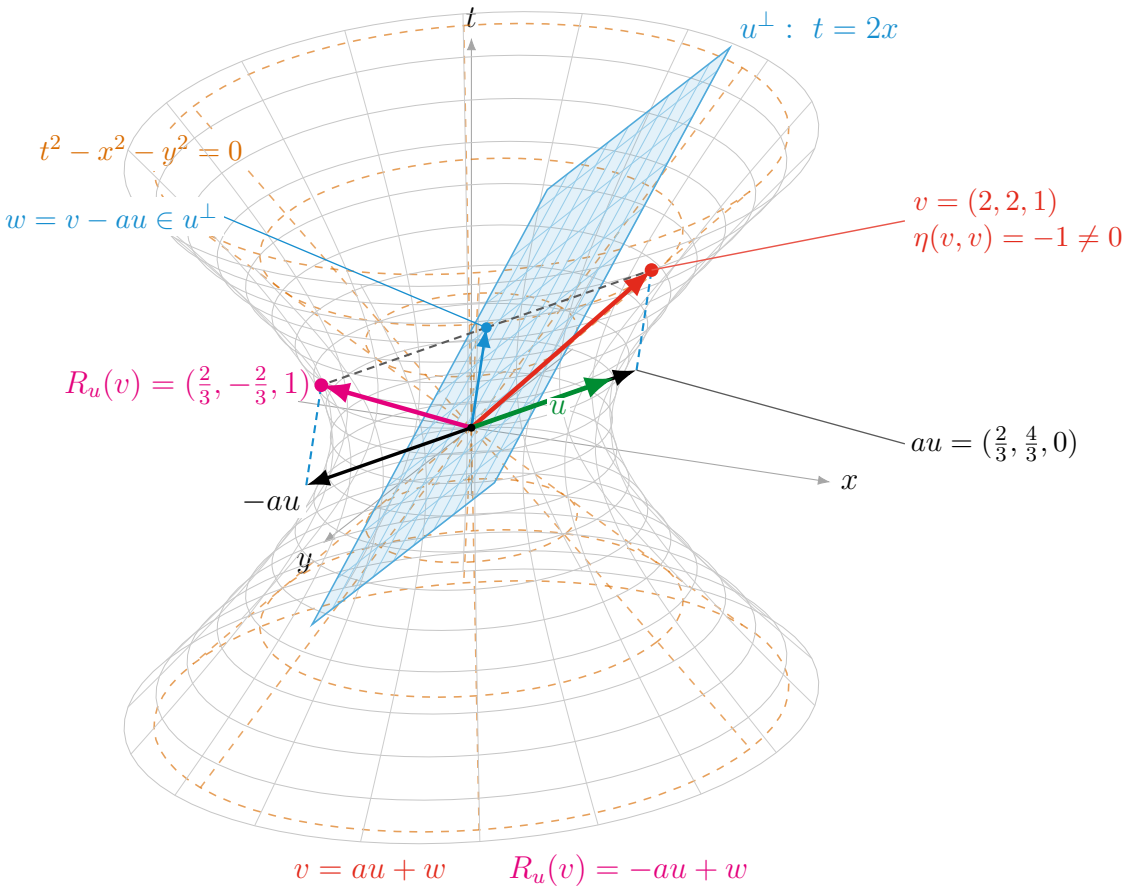
We take

$$u = \frac{1}{\sqrt{3}}(1, 2, 0), \quad \eta(u, u) = -1, \quad v = (2, 2, 1), \quad u^{\perp} = \{t = 2x\}. \tag{25}$$

The components are calculated directly:

$$\begin{aligned} a &= \frac{-2/\sqrt{3}}{-1} = \frac{2}{\sqrt{3}}, & au &= \left(\frac{2}{3}, \frac{4}{3}, 0\right), \\ w &= v - au = \left(\frac{4}{3}, \frac{2}{3}, 1\right), & R_u(v) &= -au + w = \left(\frac{2}{3}, -\frac{2}{3}, 1\right). \end{aligned} \tag{26}$$

We verify $4/3 - 2(2/3) = 0$, so w lies in the plane. The unit vector u and the component au point in the same direction, but have different lengths in the coordinates of the drawing. The parallelograms show the two sums.



$v = au + w \qquad R_u(v) = -au + w$

$u = \frac{1}{\sqrt{3}}(1, 2, 0), \quad a = \frac{2}{\sqrt{3}}, \quad \eta(u, u) = -1$

Coordinates: (t, x, y); $\quad \eta = \operatorname{diag}(1, -1, -1)$

Figure 1. Minkowski reflection: decomposition and preservation of the metric. The blue plane is $u^\perp$. The reflection fixes $w = v - au$ (blue) and reverses the component au. The dashed parallelograms show $v = au + w$ (red) and $R_u(v) = -au + w$ (hot pink). The tips of both vectors lie on the gray hyperboloid $t^2 - x^2 - y^2 = -1$. They have the same squared Minkowski norm even though their apparent lengths differ. The green arrow represents u. The light cone is drawn in dashed orange and satisfies $t^2 - x^2 - y^2 = 0$. The red tip is off the cone since $\eta(v, v) = -1 \neq 0$. Overlaps in the projection can occur between points that are distinct in spacetime. Orthogonality is with respect to the Minkowski metric.

□

2.4 Even Products, the Spin Group, and Its Two Representations

Definition 6 (Pin group)**.** The group $\mathrm{Pin}(1,3)$ is the subgroup of the invertible elements of Clifford algebra generated by unit vectors:

$$\mathrm{Pin}(1,3) \coloneqq \{u_1 \cdots u_m : \ u_j \in V, \ u_j^2 = \pm 1, \ m \geq 0\}. \tag{27}$$

The product is taken in Clifford algebra; the empty product is its unit 1.

Definition 7 (Spin group)**.** The group $\mathrm{Spin}(1,3)$ is the subgroup of $\mathrm{Pin}(1,3)$ consisting of the products of an even number of unit vectors:

$$\mathrm{Spin}(1,3) \coloneqq \{u_1 \cdots u_{2k} : \ u_j \in V, \ u_j^2 = \pm 1, \ k \geq 0\}. \tag{28}$$

The empty product is 1. Multiplying two of these elements concatenates an even number of factors, and taking the inverse reverses their order.

$$(u_1 \cdots u_{2k})^{-1} = u_{2k}^{-1} \cdots u_1^{-1}. \tag{29}$$

Each $u_j^{-1} = \pm u_j$ is still a unit vector, so the set is a group. Different expressions may represent the same algebra element. This group is neither a group of free words nor a vector space.

To connect the product to geometry, first consider n factors. By induction,

$$(R_{u_1} \circ \cdots \circ R_{u_n})(v) = (-1)^n s_n v s_n^{-1}, \qquad s_n = u_1 \cdots u_n. \tag{30}$$

For $n = 1$, this is the definition of R_u. If a composition of n reflections has this form, applying R_u on the outside gives

$$\begin{aligned} R_u\big((-1)^n s_n v s_n^{-1}\big) &= -u\big[(-1)^n s_n v s_n^{-1}\big]u^{-1} \\ &= (-1)^{n+1}(us_n)v(us_n)^{-1}, \end{aligned} \tag{31}$$

which proves the step to $n + 1$. When $n = 2k$, the signs cancel. To see the statement explicitly, take two factors and write $s = u_1 u_2$. Then

$$\begin{aligned} R_{u_1}(R_{u_2}(v)) &= -u_1(-u_2 v u_2^{-1})u_1^{-1} \\ &= (u_1 u_2)v(u_2^{-1}u_1^{-1}) = svs^{-1}. \end{aligned} \tag{32}$$

The two minus signs cancel. We call $v \mapsto svs^{-1}$ *ordinary conjugation*, which omits the minus sign in the formula for an individual reflection. This transformation preserves the metric because it composes two metric-preserving reflections. Moreover,

$$\det(R_{u_1} \circ R_{u_2}) = \det R_{u_1} \, \det R_{u_2} = (-1)(-1) = +1. \tag{33}$$

With $2k$ factors the same happens: conjugation by its product is equivalent to composing $2k$ reflections, and its determinant is $(-1)^{2k} = +1$. This does not require transformation to be identity: it can be, for example, a rotation or a boost.

Definition 8 (Connected Spin group and vector representation)**.** We define $\mathrm{Spin}^+(1,3)$ as the identity component of $\mathrm{Spin}(1,3)$. Conjugation as in equation (30), for an even number of factors, gives the homomorphism

$$\boxed{\rho : \mathrm{Spin}^+(1,3) \longrightarrow SO^+(1,3), \qquad \rho(s)(v) = svs^{-1}.} \tag{34}$$

Lemma 1 (Identity and its negative)**.** *The elements* 1 *and* -1 *belong to the same connected component of* $\mathrm{Spin}(1,3)$*. In particular,* $-1 \in \mathrm{Spin}^+(1,3)$*.*

Proof. The empty product represents 1. If e_1 is a unit spacelike vector, then $e_1 e_1 = -1$, so that -1 belongs to $\mathrm{Spin}(1,3)$ as a product of two unit vectors. To verify that it is in the identity component, we take another unit spacelike vector e_2 orthogonal to e_1 and define

$$s(t) = e_1(-\cos t\, e_1 + \sin t\, e_2), \qquad 0 \le t \le \pi. \tag{35}$$

By orthogonality, the square of the second factor is $-\cos^2 t - \sin^2 t = -1$. Therefore, $s(t)$ is a continuous path in $\mathrm{Spin}(1,3)$ whose ends are

$$s(0) = e_1(-e_1) = 1, \qquad s(\pi) = e_1 e_1 = -1. \tag{36}$$

The path image is connected and contains 1, so it is contained in the identity component. In particular, $-1 \in \mathrm{Spin}^+(1,3)$. □

By continuity, the image lies in the identity component of the Lorentz group and hence also preserves time orientation.[2] The superscript + refers to the choice of component of the group. To check the homomorphism property, we calculate

$$\begin{aligned} \rho(s_1 s_2)(v) &= s_1 s_2 v\, s_2^{-1} s_1^{-1} \\ &= \rho(s_1)\big(\rho(s_2)(v)\big). \end{aligned} \tag{37}$$

To get the Lorentz matrix $\Lambda = \rho(s)$ in the basis $\widehat{e}$, we calculate

$$s e_\nu s^{-1} = \sum_\mu e_\mu \Lambda^\mu{}_\nu. \tag{38}$$

The components of the transformed vector are $v'^\mu = \Lambda^\mu{}_\nu v^\nu$. Thus multiplication by Λ in coordinates and conjugation by s in the Clifford algebra describe the same transformation.

A spatial rotation and a *boost* are Lorentz transformations; a boost is a hyperbolic rotation in a temporal–spatial plane. A boost is not obtained by composing exclusively spatial rotations.

The Cartan–Dieudonné theorem states that every linear isometry of a real nondegenerate space of dimension n is a composition of at most n reflections with nonzero normals. Together with the properties of Spin groups, this yields the surjective double cover (34), whose kernel is $\{1, -1\}$. Gallier gives

[2] A timelike vector v is future-directed if $\eta(v,v) > 0$ and $v^0 > 0$. Preserving time orientation means that its image remains future-directed and timelike.

$$\boxed{v \text{ future-directed timelike} \quad \Longrightarrow \quad \rho(s)v \text{ future-directed timelike.}} \tag{I}$$

These transformations are called *orthochronous*. Since s belongs to the identity component, a continuous path joins it to the identity. Its image under ρ is a path of Lorentz transformations. Along this path the positive squared Minkowski norm of v is preserved. To pass from the future to the past, the time component would have to pass through zero. At that instant the squared norm would be $-|\mathbf{v}|^2 \le 0$, a contradiction. A boost can change the time coordinate while preserving time orientation.

the algebraic construction and its conventions [12]. As ρ is surjective and $\ker\rho = \{1, -1\}$, the first isomorphism theorem of groups gives

$$\rho(s) = \rho(-s), \qquad \mathrm{Spin}^+(1,3)/\{1,-1\} \xrightarrow{\sim} SO^+(1,3), \qquad [s] \longmapsto \rho(s). \tag{39}$$

The quotient recovers the Lorentz group. We constructed the Spin group earlier as a subgroup of the group of units of the Clifford algebra.

Definition 9 (Spinor representation)**.** The other representation we need is simply the restriction of c:

$$\boxed{\Delta \coloneqq c|_{\mathrm{Spin}^+(1,3)} : \mathrm{Spin}^+(1,3) \longrightarrow GL(\mathbb{C}^4).} \tag{40}$$

Its values are invertible because $c(s)c(s^{-1}) = I_4$. The two maps serve different purposes. The map $\rho(s)$ transforms vectors in V, and $\Delta(s)$ transforms columns in $\mathbb{C}^4$. They are compatible as follows.

$$\boxed{\Delta(s)c(v)\Delta(s)^{-1} = c(svs^{-1}) = c(\rho(s)v).} \tag{41}$$

Although $\rho(-1)$ is the identity on V, we have $\Delta(-1) = -I_4$. The representations are distinct. Also, Δ is generally nonunitary with respect to the usual positive Hermitian inner product on $\mathbb{C}^4$, since boosts can change that norm. The scalar phase rotation $U(1)$ introduced for electromagnetism is a different representation.

2.5 Principal bundles and orthonormal frames

We now pass from the model space V to the tangent spaces of a manifold. First we specify the structure carried by a principal bundle.

Definition 10 (Principal bundle)**.** For a Lie group H, a principal bundle with structure group H over M consists of a smooth manifold P, a smooth surjective projection $\pi_P : P \to M$, and a smooth right action

$$P \times H \longrightarrow P, \qquad (p, h) \longmapsto p \cdot h. \tag{42}$$

The action satisfies

$$p \cdot 1_H = p, \qquad (p \cdot h) \cdot k = p \cdot (hk), \qquad \pi_P(p \cdot h) = \pi_P(p). \tag{43}$$

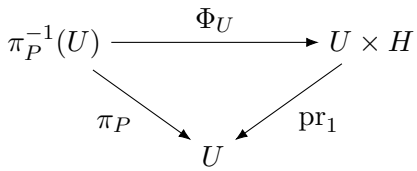
On each fiber $P_x = \pi_P^{-1}(\{x\})$ the action of H has the following properties.

$$\begin{aligned} \text{Free} \quad & \forall p \in P_x,\ \forall h \in H, \qquad p \cdot h = p \implies h = 1_H, \\ \text{Transitive} \quad & \forall p, q \in P_x,\ \exists h \in H \quad \text{such that} \quad q = p \cdot h. \end{aligned} \tag{44}$$

In addition, each point of M has a neighborhood U with a smooth trivialization

$$\Phi_U : \pi_P^{-1}(U) \longrightarrow U \times H, \tag{45}$$

Compatibility with projection is expressed by the commutative triangle



That is,

$$\boxed{\mathrm{pr}_1 \circ \Phi_U = \pi_P\big|_{\pi_P^{-1}(U)}.} \tag{46}$$

So, if $\Phi_U(p) = (x, k)$, then $x = \pi_P(p)$. Compatibility with the action requires

$$\boxed{\Phi_U(p) = (x, k) \quad \Longrightarrow \quad \Phi_U(p \cdot h) = (x, kh).} \tag{47}$$

In these coordinates the action leaves x fixed and right-multiplies k by h in H.

For $p, q \in P_x$, transitivity gives an $h \in H$ with $q = p \cdot h$. If also $q = p \cdot k$, then $p \cdot (hk^{-1}) = p$. Freeness gives $hk^{-1} = 1_H$ and hence $h = k$. The fiber behaves like the group, but has no chosen identity. Choosing a point $p \in P_x$ gives a map $h \mapsto p \cdot h$ identifying the fiber with H. A fiber need not be a vector space.

In this work the group actions on principal bundles are right actions.[3] For the general construction see Klaasse [13], section 2.1.

Definition 11 (Orthonormal frame bundle)**.** Now let (M, g) be a four-dimensional oriented and time-oriented Lorentzian manifold. An orthonormal frame over x is an ordered basis $\widehat{e} = (e_0, e_1, e_2, e_3)$ of T_xM such that $g_x(e_a, e_b) = \eta_{ab}$. We also require the chosen orientation and require e_0 to be future-directed. We define

$$P_{SO^+(1,3)} \coloneqq \{(x, \widehat{e}) : x \in M,\ \widehat{e} \text{ is a frame with these properties in } T_xM\}, \qquad \pi_{SO^+(1,3)}(x, \widehat{e}) = x. \tag{48}$$

The subscript specifies the structure group $SO^+(1,3)$.[4] Two frames in the same fiber are related by a unique $\Lambda \in SO^+(1,3)$.

$$(\widehat{e} \cdot \Lambda)_a \coloneqq \sum_{b=0}^{3} e_b \Lambda^b{}_a, \qquad (x, \widehat{e}) \cdot \Lambda = (x, \widehat{e} \cdot \Lambda). \tag{49}$$

The multiplication on the right respects the law of action

$$\boxed{(\widehat{e} \cdot \Lambda) \cdot K = \widehat{e} \cdot (\Lambda K), \qquad \Lambda, K \in SO^+(1,3).} \tag{50}$$

Indeed, component by component,

$$\big((\widehat{e} \cdot \Lambda) \cdot K\big)_\nu = \sum_{\mu,\alpha} e_\alpha \Lambda^\alpha{}_\mu K^\mu{}_\nu = \sum_\alpha e_\alpha (\Lambda K)^\alpha{}_\nu = \big(\widehat{e} \cdot (\Lambda K)\big)_\nu. \tag{51}$$

This action makes $P_{SO^+(1,3)} \to M$ a principal bundle. Its fibers contain frames. Adding two frames does not define a vector sum of orthonormal frames. Local frame fields give its trivializations.

[3] A left action is written $h \cdot p$ and satisfies $h \cdot (k \cdot p) = (hk) \cdot p$ as well as $1 \cdot p = p$. A right action is written $p \cdot h$ and satisfies $(p \cdot h) \cdot k = p \cdot (hk)$ as well as $p \cdot 1 = p$. The rule $h \cdot p \coloneqq p \cdot h^{-1}$ converts one convention into the other. The inverse accounts for the order of the factors in the action law. We use right actions on principal bundles and left actions for representations on vector spaces, as in $\Delta(h)z$.

[4] $SO^+(1,3)$ is the proper orthochronous Lorentz group. It contains spatial rotations, boosts, and their compositions, preserving space and time orientations. The Poincaré group also includes spacetime translations. Its proper orthochronous component is $\mathcal{P}_+^\uparrow = \mathbb{R}^{1,3} \rtimes SO^+(1,3)$ and acts by $x \mapsto \Lambda x + a$. The symbol $\rtimes$ denotes a *semidirect product*. An element is a pair (a, Λ), and the Lorentz transformation also acts on the translation vector. Composing the maps gives $\Lambda(\Lambda' x + a') + a = \Lambda\Lambda' x + \Lambda a' + a$, so the group law is

$$(a, \Lambda)(a', \Lambda') = (a + \Lambda a', \Lambda\Lambda'). \tag{II}$$

This law follows from composition of transformations. If the translations are zero, we recover

$$(0, \Lambda)(0, \Lambda') = (0, \Lambda\Lambda'), \tag{III}$$

the Lorentz subgroup with its usual matrix multiplication. Here we relate orthonormal frames over a fixed point, which calls for the Lorentz group.

Theorem 1 (Trivialization of the frame bundle in Minkowski spacetime)**.** *In Minkowski spacetime $M = \mathbb{R}^{1,3}$, with Cartesian coordinates (t, x, y, z) and the usual orientations, the oriented and time-oriented orthonormal frame bundle is trivial. There is an isomorphism of principal bundles $P_{SO^+(1,3)} \simeq M \times SO^+(1,3)$. It assigns to each frame the matrix of its components in the canonical Cartesian frame.*

Proof. At each point $x \in M = \mathbb{R}^{1,3}$ we take the coordinate vectors, whose components are

$$\begin{aligned} e_0 &= (1,0,0,0), \quad e_1 = (0,1,0,0), \\ e_2 &= (0,0,1,0), \quad e_3 = (0,0,0,1). \end{aligned} \tag{52}$$

These four fields form the global orthonormal frame $\widehat{e} = (e_0, e_1, e_2, e_3)$, with the chosen spatial orientation and e_0 future-directed. Now let $\widehat{f} = (f_0, f_1, f_2, f_3)$ be any frame in the fiber over x. Each vector has unique components in the preceding basis.

$$f_\nu = \sum_{\mu=0}^{3} e_\mu \Lambda^\mu{}_\nu, \qquad \Lambda = \begin{pmatrix} f_0^0 & f_1^0 & f_2^0 & f_3^0 \\ f_0^1 & f_1^1 & f_2^1 & f_3^1 \\ f_0^2 & f_1^2 & f_2^2 & f_3^2 \\ f_0^3 & f_1^3 & f_2^3 & f_3^3 \end{pmatrix}. \tag{53}$$

Thus, the columns of Λ are precisely the components of the four vectors of the frame $\widehat{f}$. Their orthonormality implies

$$\eta_{\mu\nu} = g_x(f_\mu, f_\nu) = \sum_{\alpha,\beta} \Lambda^\alpha{}_\mu \eta_{\alpha\beta} \Lambda^\beta{}_\nu, \qquad \Lambda^{\mathsf{T}} \eta \Lambda = \eta. \tag{54}$$

Because $\widehat{f}$ has the same space and time orientations as $\widehat{e}$, we have $\Lambda \in SO^+(1,3)$. Conversely, the columns of any $\Lambda \in SO^+(1,3)$ define a frame with those properties. This constructs the trivialization and its inverse.

$$\boxed{\begin{aligned} \Phi : P_{SO^+(1,3)} &\longrightarrow M \times SO^+(1,3), \\ (x, \widehat{f}) &\longmapsto (x, \Lambda), \\ \Phi^{-1}(x, \Lambda) &= (x, (\sum_{\mu} e_\mu \Lambda^\mu{}_\nu)_{\nu=0}^3). \end{aligned}} \tag{55}$$

Both maps are smooth. One reads off the Cartesian components of the vectors and the other forms vectors from those components. They also preserve the base point and respect the right action.

$$\mathrm{pr}_1 \circ \Phi = \pi_{SO^+(1,3)}, \qquad \Phi((x, \widehat{f}) \cdot K) = (x, \Lambda K) = \Phi(x, \widehat{f}) \cdot K. \tag{56}$$

Therefore, it is an isomorphism of principal bundles:

$$\boxed{P_{SO^+(1,3)} \simeq M \times SO^+(1,3).} \tag{57}$$

For an open set $U \subseteq M$, the restriction $\Phi_U = \Phi|_{\pi^{-1}_{SO^+(1,3)}(U)}$ gives exactly the local trivialization required in the definition of a principal bundle. □

This uses the fixed Cartesian coordinates in Minkowski. On a general Lorentzian manifold, one chooses a smooth orthonormal frame on each sufficiently small open set and repeats the construction locally. A global trivialization is not guaranteed. A pair (x, Λ) represents a frame over x consisting of vectors in $T_x M$.

2.6 What It Means to Choose a Spin Structure

We fix the homomorphism ρ from Definition 8, given in (34). We seek a lift of the frame bundle compatible with that homomorphism.

Definition 12 (Spin structure)**.** A spin structure over (M, g) is a pair $(P_{\mathrm{Spin}^+(1,3)}, \lambda)$, where $\pi_{\mathrm{Spin}^+(1,3)} : P_{\mathrm{Spin}^+(1,3)} \to M$ is a principal bundle with structure group $\mathrm{Spin}^+(1,3)$ and $\lambda : P_{\mathrm{Spin}^+(1,3)} \to P_{SO^+(1,3)}$ is a smooth map satisfying

$$\boxed{\pi_{SO^+(1,3)} \circ \lambda = \pi_{\mathrm{Spin}^+(1,3)}, \qquad \lambda(p \cdot s) = \lambda(p) \cdot \rho(s).} \tag{58}$$

The restriction of λ to each fiber is a two-sheeted covering compatible with ρ.

The map λ is a morphism of principal bundles over the identity of M, compatible with the homomorphism ρ. The bundles have distinct structure groups: the source has group $\mathrm{Spin}^+(1,3)$ and the target has group $SO^+(1,3)$. Compatibility with the actions therefore uses s in the source and $\rho(s)$ in the target.

The symbol $p \cdot s$ denotes the global right action of the structure group $\mathrm{Spin}^+(1,3)$ on the spin principal bundle, in the sense of Definition 10.

The two equalities in (58) express these compatibility conditions. The first says that p and $\lambda(p)$ lie over the same point of M. The second says that changing the spin frame by s and then applying λ gives the same result as first applying λ and then changing the orthonormal frame by $\rho(s)$. Thus ρ relates the groups, and λ relates their bundles compatibly with the projections and actions. These two conditions are displayed in the following commutative squares:

$$\begin{array}{ccc} P_{\mathrm{Spin}^+(1,3)} & \xrightarrow{\ \lambda\ } & P_{SO^+(1,3)} \\ \downarrow{\scriptstyle \pi_{\mathrm{Spin}^+(1,3)}} & & \downarrow{\scriptstyle \pi_{SO^+(1,3)}} \\ M & \xrightarrow[\mathrm{id}_M]{} & M \end{array}$$

Compatibility with the projection

(59)

$$\begin{array}{ccc} P_{\mathrm{Spin}^+(1,3)} \times \mathrm{Spin}^+(1,3) & \xrightarrow{\text{right action}} & P_{\mathrm{Spin}^+(1,3)} \\ \downarrow{\scriptstyle \lambda \times \rho} & & \downarrow{\scriptstyle \lambda} \\ P_{SO^+(1,3)} \times SO^+(1,3) & \xrightarrow[\text{right action}]{} & P_{SO^+(1,3)} \end{array}$$

Compatibility with the actions

(60)

In the second square, the horizontal arrows send (p, s) to $p \cdot s$ and (q, Λ) to $q \cdot \Lambda$. The two paths from (p, s) lead to $\lambda(p \cdot s)$ and $\lambda(p) \cdot \rho(s)$, respectively. Commutativity says that these agree.

The points of $P_{\mathrm{Spin}^+(1,3)}$ are called spin frames. Over a given orthonormal frame there are two spin frames, p and $p \cdot (-1)$. The spin structure is the pair consisting of the principal bundle and the morphism λ with these properties. A spinor field will be a further choice.

The existence of ρ does not guarantee that λ exists on an arbitrary M. To see this locally, choose a smooth orthonormal frame on each open set U_i, denoted by $\widehat{e}_i = (e_{i,0}, e_{i,1}, e_{i,2}, e_{i,3})$. The hat distinguishes the whole ordered basis from its individual vectors $e_{i,\mu}$. On $U_i \cap U_j$, the transition matrix

$\Lambda_{ij}(x) \in SO^+(1,3)$ satisfies

$$\widehat{e}_j(x) = \widehat{e}_i(x) \cdot \Lambda_{ij}(x), \qquad e_{j,\nu}(x) = \sum_{\mu=0}^{3} e_{i,\mu}(x)\,(\Lambda_{ij}(x))^{\mu}{}_{\nu}. \tag{61}$$

The indices i, j label the open sets and are held fixed. The index μ runs over the four vectors in the frame $\widehat{e}_i$. On a triple overlap, composing the two frame changes gives $\Lambda_{ij}\Lambda_{jk} = \Lambda_{ik}$ at each point. A spin structure requires smooth lifts $s_{ij} : U_i \cap U_j \to \mathrm{Spin}^+(1,3)$ with

$$\rho(s_{ij}) = \Lambda_{ij}, \qquad s_{ij}s_{jk} = s_{ik}. \tag{62}$$

By refining the cover so that the overlaps are contractible, we can lift each transformation separately. The signs of the lifts must be chosen so that the second equality holds on every triple overlap. This global compatibility can fail. When it holds, there may be inequivalent choices. That compatibility is the geometric content of a spin structure. See Klaasse [13], section 3.2, for the formulation of the lift and its obstructions in the Riemannian case. Here we use the corresponding Lorentzian group.

Example 4 (Spin structure in Minkowski)**.** Using the usual trivializations, for $x \in M$ and $h, s \in \mathrm{Spin}^+(1,3)$ we set

$$\boxed{P_{\mathrm{Spin}^+(1,3)} = M \times \mathrm{Spin}^+(1,3), \qquad (x,h)\cdot s = (x, hs), \qquad \lambda(x,h) \coloneqq (x, \rho(h)).} \tag{63}$$

Like this,

$$\lambda((x,h)\cdot s) = (x, \rho(hs)) = (x, \rho(h)\rho(s)) = \lambda(x,h)\cdot\rho(s). \tag{64}$$

In these coordinates, $\pi_{\mathrm{Spin}^+(1,3)}(x,h) = x$. The map sends (x,h) and $(x,-h)$ by λ to the same orthonormal frame. Minkowski spacetime is contractible and its spin structure is unique up to equivalence, although its frames and trivializations are not unique. At a fixed point the fiber consists of all pairs (x,h) with $h \in \mathrm{Spin}^+(1,3)$. Each h is an element of the group inside the Clifford algebra. The vectors in $\mathbb{C}^4$ belong to a different space.

□

2.7 The Quotient That Defines the Spinor Bundle

Recall that $\Delta : \mathrm{Spin}^+(1,3) \to GL(\mathbb{C}^4)$ is the spinor representation from (40). It assigns to each s the invertible matrix $\Delta(s) = c(s)$, which acts on columns in $\mathbb{C}^4$. We use this representation to construct the vector space over each point. Define the diagonal right action by the map

$$\begin{aligned}(P_{\mathrm{Spin}^+(1,3)} \times \mathbb{C}^4) \times \mathrm{Spin}^+(1,3) &\longrightarrow P_{\mathrm{Spin}^+(1,3)} \times \mathbb{C}^4,\\ ((p,z),s) &\longmapsto (p,z)\cdot s \coloneqq (p\cdot s, \Delta(s)^{-1}z).\end{aligned} \tag{65}$$

It's an action because

$$\begin{aligned}((p,z)\cdot s)\cdot r &= (p\cdot(sr), \Delta(r)^{-1}\Delta(s)^{-1}z)\\ &= (p\cdot(sr), \Delta(sr)^{-1}z) = (p,z)\cdot(sr).\end{aligned} \tag{66}$$

The order of the inverses is important.

Definition 13 (Associated bundle of spinors)**.** We define the associated bundle as the orbit space of this action.[5]

$$\boxed{S \coloneqq P_{\mathrm{Spin}^+(1,3)} \times_{\Delta} \mathbb{C}^4 = (P_{\mathrm{Spin}^+(1,3)} \times \mathbb{C}^4)/\,\mathrm{Spin}^+(1,3).} \tag{67}$$

[5]For a right action of a group H on a set Y, the orbit of $y \in Y$ is the set $[y] = \{y\cdot h : h \in H\}$ of all points obtained by acting on y with group elements. Two points lie in the same orbit if the action takes one to the other. The orbit space Y/H is the set of these equivalence classes. In our case, a class $[p,z]$ collects the pairs obtained from (p,z) by the diagonal action.

Its elements are classes $[p, z]$, with equality defined exactly by

$$\boxed{[p, z] = [p', z'] \quad \Longleftrightarrow \quad \exists s \in \mathrm{Spin}^+(1,3) : \; p' = p \cdot s, \quad z' = \Delta(s)^{-1} z.} \tag{68}$$

The two frames must lie over the same point. For fixed p, p' in the same fiber, the element s is unique. The pairs can differ, but the change of components compensates for the change of frame, so they define the same class. The group acts within fibers, leaving the points of M fixed.

We define the projection

$$\begin{aligned} \pi_S &: S \to M, \\ \pi_S([p, z]) &\coloneqq \pi_{\mathrm{Spin}^+(1,3)}(p). \end{aligned} \tag{69}$$

Here $\pi_{\mathrm{Spin}^+(1,3)}$ is the principal bundle projection from Definition 12. Since p and $p \cdot s$ have the same base point, π_S is well defined.

We use the right action $p \cdot s$ of $\mathrm{Spin}^+(1,3)$ on each fiber $(P_{\mathrm{Spin}^+(1,3)})_x$. Its freeness and transitivity are part of the definition of a principal bundle (Definition 10).

Lemma 2 (Unique representative in a fixed frame)**.** *Fix $x \in M$ and a frame $p \in (P_{\mathrm{Spin}^+(1,3)})_x$. Each class in S_x contains exactly one pair of the form (p, z), with $z \in \mathbb{C}^4$. In particular, the following map is bijective.*

$$\iota_p : \mathbb{C}^4 \longrightarrow S_x, \qquad z \longmapsto [p, z]. \tag{70}$$

Proof. Existence. Let $[p', z']$ be a class in S_x. Since p and p' lie in the same fiber of the spin principal bundle, transitivity of the right action on that fiber gives an $s \in \mathrm{Spin}^+(1,3)$ such that $p' = p \cdot s$. The change from frame p' to frame p is written

$$\boxed{[p', z'] = [p \cdot s, z'] = [p, \Delta(s) z'].} \tag{71}$$

To check equality, we apply s to the right side pair and we get

$$(p, \Delta(s) z') \cdot s = (p \cdot s, \underbrace{\Delta(s)^{-1} \Delta(s)}_{I_4} z') = (p', z'). \tag{72}$$

The two pairs are in the same orbit. Thus, $(p, \Delta(s) z')$ is a class representative with the chosen frame p.

Uniqueness. If $[p, z] = [p, w]$, the pairs are in the same orbit of the diagonal action. Therefore, there is $s \in \mathrm{Spin}^+(1,3)$ such that

$$(p, w) = (p, z) \cdot s = (p \cdot s, \Delta(s)^{-1} z). \tag{73}$$

The first component gives $p \cdot s = p$. Since $P_{\mathrm{Spin}^+(1,3)} \to M$ is a principal bundle, the right action of $\mathrm{Spin}^+(1,3)$ on its fiber over x is free. Thus $s = 1_{\mathrm{Spin}^+(1,3)}$. Since Δ is a group homomorphism, $\Delta(1_{\mathrm{Spin}^+(1,3)}) = I_4$. The second component then gives

$$w = \Delta(1_{\mathrm{Spin}^+(1,3)})^{-1} z = I_4^{-1} z = I_4 z = z. \tag{74}$$

□

Example 5 (Two representatives of the same spinor)**.** Let's fix a spin frame p and a column $z \in \mathbb{C}^4$. As c is linear and $c(1) = I_4$, we have $\Delta(-1) = c(-1) = -c(1) = -I_4$. The inverse of $-I_4$ is $-I_4$, so the diagonal action gives

$$(p, z) \cdot (-1) = (p \cdot (-1), \Delta(-1)^{-1} z) = (p \cdot (-1), -z). \tag{75}$$

The two pairs are in the same orbit, so that $[p, z] = [p \cdot (-1), -z]$. As the action on the frames is free, $p \cdot (-1) \neq p$. Each pair describes the same spinor in its own frame.

□

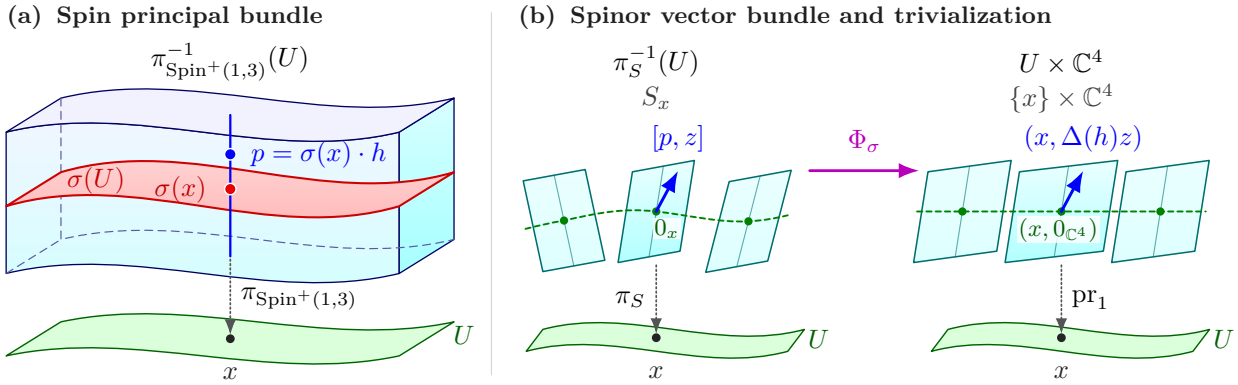


Figure 2. Section of the principal bundle and trivialization of the spinor vector bundle. (a) The red section chooses $\sigma(x)$ in the fiber of $p = \sigma(x) \cdot h$. **(b)** The planes depict the fibers of $S \to M$, of complex rank four. The green line joins their zeros. Trivialization (77) maps $[p, z]$ to $(x, \Delta(h)z)$ and $0_x = [p, 0_{\mathbb{C}^4}]$ to $(x, 0_{\mathbb{C}^4})$.

First we give each fiber a vector-space structure. Fix a frame p over x and use the bijection ι_p from (70) to transfer addition and scalar multiplication to S_x from $\mathbb{C}^4$.

$$\boxed{[p, z] + [p, w] = [p, z + w], \qquad b[p, z] = [p, bz], \quad b \in \mathbb{C}.} \tag{76}$$

Linearity of $\Delta(s)^{-1}$ makes the operations independent of the frame. Thus ι_p is an isomorphism, and its inverse gives the components in frame p. The zero of S_x is $0_x := [p, 0_{\mathbb{C}^4}]$. By (68), $[p, 0_{\mathbb{C}^4}] = [p \cdot s, 0_{\mathbb{C}^4}]$ for any change of frame over x, since $\Delta(s)^{-1} 0_{\mathbb{C}^4} = 0_{\mathbb{C}^4}$.

Now we assemble the identifications of the fibers. On an open set $U \subseteq M$ choose a smooth section $\sigma : U \longrightarrow \pi^{-1}_{\mathrm{Spin}^+(1,3)}(U) \subseteq P_{\mathrm{Spin}^+(1,3)}$ with $\pi_{\mathrm{Spin}^+(1,3)}(\sigma(x)) = x$.

Take any spinor $\xi = [p, z] \in \pi_S^{-1}(U)$ with base point $x = \pi_S(\xi)$. Since p and $\sigma(x)$ lie over x, there is a unique $h \in \mathrm{Spin}^+(1, 3)$ with $p = \sigma(x) \cdot h$. By (71) we can rewrite $\xi = [\sigma(x), \Delta(h)z]$. Bijection (70) guarantees that this expression in frame $\sigma(x)$ is unique. We define Φ_σ by taking the base point x and the column of this representative.

$$\boxed{\begin{array}{c} \Phi_\sigma : \pi_S^{-1}(U) \longrightarrow U \times \mathbb{C}^4, \\ \Phi_\sigma(\xi) := (x, \Delta(h)z). \end{array}} \tag{77}$$

Panel (b) of Figure 2 shows the projection to the base point. If we choose another frame $\sigma'(x) = \sigma(x) \cdot s(x)$, the change of components is given by

$$\boxed{[\sigma(x), z(x)] = [\sigma'(x), z'(x)], \qquad z'(x) = \Delta(s(x))^{-1} z(x).} \tag{78}$$

The change-of-frame matrix $\Delta(s(x))^{-1}$ depends smoothly on x.

Example 6 (The spinor quotient in Minkowski spacetime)**.** The global trivialization follows from (77) by taking $\sigma(x) = (x, 1_{\mathrm{Spin}^+(1,3)})$, so that $\sigma(x) \cdot h = (x, h)$. Thus,

$$\boxed{[(x, h), z] \longmapsto (x, \Delta(h)z).} \tag{79}$$

If we use another representative, we get the same:

$$\Delta(hs)\Delta(s)^{-1}z = \Delta(h)\Delta(s)\Delta(s)^{-1}z = \Delta(h)z. \tag{80}$$

On the same x, two pairs represent the same spinor if and only if they give the same column $\Delta(h)z = \Delta(h')z'$ in the frame of reference.

□

The same construction gives a vector bundle associated to any principal bundle equipped with a chosen linear representation. For example, a complex vector bundle of rank n is recovered from its frame bundle using the standard representation of $GL(n, \mathbb{C})$.

Theorem 2 (Spinor vector bundle)**.** *Let S be the quotient in (67), constructed from $P_{\mathrm{Spin}^+(1,3)}$ and Δ. With the projection in (69) and the fiber operations in (76), $\pi_S : S \to M$ is a rank-four complex vector bundle. Each smooth local section σ of the principal bundle over U induces the trivialization Φ_σ in (77).*

Proof. The operations in (76) make each S_x a complex vector space. The map Φ_σ in (77) is linear on each fiber and has inverse $(x, z) \mapsto [\sigma(x), z]$ by Lemma 2. These charts cover S because the principal bundle has local sections around every point. On an overlap, the principal trivialization induced by σ writes the other section σ' as $(x, s(x))$, so $s(x)$ is smooth. The representation Δ then gives the transition in (78). **These smooth linear transitions make $\pi_S : S \to M$ a rank-four complex vector bundle.** □

2.8 From the Bundle to the Dirac Field

A Dirac field chooses a spinor smoothly at each point. It is a section of the bundle we have just constructed.

$$\boxed{s_\psi \in \Gamma^\infty(S), \qquad s_\psi(x) \in S_x, \quad \pi_S(s_\psi(x)) = x.} \tag{81}$$

In Minkowski spacetime we use (79) to read the column $\psi(x)$ of each $s_\psi(x)$ in the global frame. This gives the function used in our calculations.

$$\boxed{s_\psi(x) = [(x, 1_{\mathrm{Spin}^+(1,3)}), \psi(x)] \longleftrightarrow (x, \psi(x)), \qquad \psi : M \longrightarrow \mathbb{C}^4.} \tag{82}$$

The complete construction is organized as

$$\boxed{(V, \eta) \;\longrightarrow\; \mathrm{Cl}(V, \eta) \;\longrightarrow\; (\mathrm{Spin}^+(1,3), \rho, \Delta), \qquad (P_{\mathrm{Spin}^+(1,3)}, \lambda) \;\longrightarrow\; S \;\longrightarrow\; s_\psi.} \tag{83}$$

The morphism λ carries the Clifford action to each fiber. Fix a spin frame p over x. Its image $\lambda(p)$ identifies $\mathbb{R}^{1,3}$ with T_xM. If $X \in T_xM$ has components $v = \lambda(p)^{-1}X$ in this frame, its action on $[p, z]$ gives $[p, c(v)z]$. The compatibility $\Delta(s)c(v)\Delta(s)^{-1} = c(\rho(s)v)$ ensures that the resulting spinor is the same after a change of frame.

In the Cartesian frame the spin connection is trivial. In the following sections we will work with ψ and the gamma matrices in that frame, and we will introduce the internal connection $U(1)$ for the electromagnetic coupling.

3 The Dirac Field and Conventions

Definition 14 (Dirac field)**.** Let (M, g) be a four-dimensional oriented and time-oriented Lorentzian manifold with a chosen spin structure in the sense of section 2.6. We use the associated bundle constructed in section 2.7, a rank-four complex vector bundle.

$$\pi_S : S \longrightarrow M. \tag{84}$$

A classical complex Dirac field is a smooth section s_ψ of this bundle. That is,

$$\boxed{s_\psi \in \Gamma^\infty(\pi_S).} \tag{85}$$

Here the space of smooth sections is defined by

$$\Gamma^\infty(\pi_S) \coloneqq \{s \in C^\infty(M, S) : \pi_S \circ s = \mathrm{id}_M\}. \tag{86}$$

The section condition means that $s_\psi(x)$ lies in the fiber $S_x \coloneqq \pi_S^{-1}(x)$ over the same point x. We also write $\Gamma^\infty(S)$ for this space of sections. In a local trivialization over an open set U, the fiber is identified with $U \times \mathbb{C}^4$, and a function $\psi_U : U \to \mathbb{C}^4$ represents the section.

We work in Minkowski spacetime $M = \mathbb{R}^{1,3}$, with its usual spin structure and metric

$$g = \mathrm{diag}(1, -1, -1, -1). \tag{87}$$

Its spinor bundle is trivializable. Fix a global trivialization and write

$$S \simeq M \times \mathbb{C}^4, \qquad \pi_S(x, v) = x. \tag{88}$$

In this identification, every section has the form

$$\boxed{s_\psi(x) = (x, \psi(x)), \qquad \psi : M \longrightarrow \mathbb{C}^4.} \tag{89}$$

Thus, section s_ψ and component function ψ are mutually determined. Once the trivialization is fixed, we will also call this function Dirac field, as usual in physics:

$$\psi(x) = \begin{pmatrix} \psi^1(x) \\ \psi^2(x) \\ \psi^3(x) \\ \psi^4(x) \end{pmatrix}. \tag{90}$$

A change of manifold can give a trivial or nontrivial bundle. The definition by sections applies in either case. The formulas below use Minkowski spacetime with the fixed trivialization.

Definition 15 (Dirac adjoint)**.** For field ψ, its Dirac adjoint is the covector field defined point-by-point by

$$\boxed{\bar\psi \coloneqq \psi^\dagger \gamma^0.} \tag{91}$$

The symbol † denotes Hermitian conjugation. In the quantized theory the fermionic components are described by operators. The functional integral uses Grassmann variables.

The gamma matrices satisfy the Clifford relations and the Hermiticity condition

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}\mathbf{1}_4, \qquad (\gamma^\mu)^\dagger = \gamma^0\gamma^\mu\gamma^0. \tag{92}$$

In particular, $(\gamma^0)^2 = \mathbf{1}_4$ and $(\gamma^i)^2 = -\mathbf{1}_4$ for $i = 1, 2, 3$. Repeated indices are summed. Greek indices run over $0, 1, 2, 3$ and Latin indices over the three spatial coordinates.

Definition 16 (Action and Lagrangian density)**.** For a classical field determined by $s_\psi \in \Gamma^\infty(\pi_S)$, the free action for a field of mass $m > 0$ over the whole manifold M is the functional

$$S_D[\psi] \coloneqq \int_M d^4x\, \bar{\psi}(x)(i\gamma^\mu \partial_\mu - m)\psi(x), \qquad \boxed{\mathcal{L}_D \coloneqq \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi.} \tag{93}$$

Here $\bar{\psi} = \psi^\dagger \gamma^0$ and the gamma matrices are fixed. Thus $\bar{\psi}$ is already determined. The action depends on ψ, its conjugate, and its first derivatives, and is not a holomorphic functional of ψ. We evaluate the density pointwise and integrate it over the whole manifold M to obtain the action. We assume sufficient integrability for the functional to be well defined.

3.1 Dependence of the Adjoint and the Variational Calculation

Although $\bar{\psi}$ is determined by ψ, we can write the action in terms of both and differentiate formally with respect to each to calculate the variation. This organizes the calculation in terms of the real and imaginary parts of each component and introduces no independent physical field.[6] The calculation below shows why this works.

For each component we write $\psi^a = u^a + iv^a$. Physical variations satisfy

$$\delta\psi^a = \delta u^a + i\delta v^a, \qquad \delta\bar{\psi} = (\delta\psi)^\dagger \gamma^0. \tag{94}$$

Take a smooth variation $\delta\psi$ with compact support in M. The first variation of the action is its derivative in the direction of this field.

$$\boxed{\delta S_D[\psi; \delta\psi] \coloneqq \lim_{\varepsilon \to 0} \frac{S_D[\psi + \varepsilon\delta\psi] - S_D[\psi]}{\varepsilon}.} \tag{95}$$

The parameter ε is real. Changing the field therefore changes its adjoint as well.

$$\overline{\psi + \varepsilon\delta\psi} = \bar{\psi} + \varepsilon\delta\bar{\psi}, \qquad \delta\bar{\psi} = (\delta\psi)^\dagger \gamma^0. \tag{96}$$

We replace directly in the quotient of the above definition:

$$\begin{aligned} \frac{S_D[\psi + \varepsilon\delta\psi] - S_D[\psi]}{\varepsilon} = \frac{1}{\varepsilon} \int_M d^4x \Big[&(\bar{\psi} + \varepsilon\delta\bar{\psi})(i\gamma^\mu \partial_\mu - m)(\psi + \varepsilon\delta\psi) \\ &- \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi \Big]. \end{aligned} \tag{97}$$

Using linearity of the derivative, multiply the two factors. The term independent of ε cancels the action we subtract. After dividing by ε, three terms remain. We single out the third and denote its integral by $I_{\delta\psi}$.

$$\begin{aligned} \frac{S_D[\psi + \varepsilon\delta\psi] - S_D[\psi]}{\varepsilon} = &\int_M d^4x\, \delta\bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi \\ &+ \int_M d^4x\, \bar{\psi}(i\gamma^\mu \partial_\mu - m)\delta\psi \\ &+ \underbrace{\varepsilon \int_M d^4x\, \delta\bar{\psi}(i\gamma^\mu \partial_\mu - m)\delta\psi}_{\varepsilon I_{\delta\psi}}. \end{aligned} \tag{98}$$

[6] The idea is the same as writing $f(z, \bar{z}) = z\bar{z}$ with $z = u + iv$. To differentiate formally with respect to z, hold $\bar{z}$ fixed to get $\partial f/\partial z = \bar{z}$. Differentiating with respect to $\bar{z}$ gives $\partial f/\partial \bar{z} = z$. These Wirtinger derivatives reorganize the derivatives with respect to the real variables u, v. The variables z and $\bar{z}$ are still related by conjugation. For ψ we proceed component by component. The fixed, invertible matrix γ^0 lets us express the conjugate through the Dirac adjoint.

We now check that the third term, $\varepsilon I_{\delta\psi}$, tends to zero. The condition we have imposed on variation is written, in our trivialization,

$$\delta\psi \in C_c^\infty(M, \mathbb{C}^4), \qquad K \coloneqq \operatorname{supp}(\delta\psi) \subset M \text{ compact.} \tag{99}$$

Outside K the variation and its derivatives vanish. Therefore,

$$I_{\delta\psi} \coloneqq \int_M d^4x\, \delta\bar{\psi}(i\gamma^\mu\partial_\mu - m)\delta\psi = \int_K d^4x\, \delta\bar{\psi}(i\gamma^\mu\partial_\mu - m)\delta\psi, \qquad |I_{\delta\psi}| < \infty. \tag{100}$$

The integrand is smooth and compactly supported, so the integral is finite and independent of ε. Consequently,

$$\lim_{\varepsilon\to 0} \varepsilon I_{\delta\psi} = 0. \tag{101}$$

Developing the two remaining terms we get the first variation, which we abbreviated as δS_D:

$$\begin{aligned}\delta S_D = \int_M d^4x\, \big[&i\,\delta\bar{\psi}\,\gamma^\mu\partial_\mu\psi + i\bar{\psi}\gamma^\mu\partial_\mu(\delta\psi) \\ &- m\,\delta\bar{\psi}\,\psi - m\bar{\psi}\,\delta\psi\big].\end{aligned} \tag{102}$$

To identify the coefficients of the first variation, we write it with derivatives removed from $\delta\psi$. Only the second term requires work.

$$T \coloneqq \int_M d^4x\, i\bar{\psi}\gamma^\mu\partial_\mu(\delta\psi). \tag{103}$$

To integrate T by parts, we use the product rule. The gamma matrices are constant, so

$$i\bar{\psi}\gamma^\mu\partial_\mu(\delta\psi) = \partial_\mu(i\bar{\psi}\gamma^\mu\delta\psi) - i(\partial_\mu\bar{\psi})\gamma^\mu\delta\psi. \tag{104}$$

We integrate this identity. The field $i\bar{\psi}\gamma^\mu\delta\psi$ has compact support, so the integral of its divergence is zero:

$$\int_M d^4x\, \partial_\mu(i\bar{\psi}\gamma^\mu\delta\psi) = 0. \tag{105}$$

Thus integration by parts gives

$$T = -\int_M d^4x\, i(\partial_\mu\bar{\psi})\gamma^\mu\delta\psi. \tag{106}$$

We replace this result in the first variation. The other three terms remain the same:

$$\begin{aligned}\delta S_D = \int_M d^4x\, \big[&i\delta\bar{\psi}\gamma^\mu\partial_\mu\psi - m\delta\bar{\psi}\psi \\ &- i(\partial_\mu\bar{\psi})\gamma^\mu\delta\psi - m\bar{\psi}\delta\psi\big].\end{aligned} \tag{107}$$

The first two terms contain $\delta\bar{\psi}$ on the left, and the last two contain $\delta\psi$ on the right. We factor these out and call their respective coefficients B and C.

$$\begin{aligned}\delta S_D = \int_M d^4x\, \Big\{\delta\bar{\psi}\ &\underbrace{(i\gamma^\mu\partial_\mu - m)\psi}_{B} \\ &+ \underbrace{\big[-i(\partial_\mu\bar{\psi})\gamma^\mu - m\bar{\psi}\big]}_{C}\ \delta\psi\Big\}.\end{aligned} \tag{108}$$

So far we have only calculated the variation: we have not yet imposed any equation of motion. The stationary action principle selects the fields for which

$$\delta S_D[\psi; \delta\psi] = 0 \qquad \text{for every } \delta\psi \in C_c^\infty(M, \mathbb{C}^4). \tag{109}$$

Recall that ψ and $\delta\psi$ are columns of size 4×1. The adjoint gives a row: $\bar{\psi} = \psi^\dagger \gamma^0$ has size 1×4. The matrices γ^μ have size 4×4, and differentiation does not change the size of a field. Therefore,

$$\begin{aligned} B &\coloneqq i\gamma^\mu(\partial_\mu \psi) - m\psi \qquad \text{is a column: } (4\times 4)(4\times 1) = 4\times 1, \\ C &\coloneqq -i(\partial_\mu \bar{\psi})\gamma^\mu - m\bar{\psi} \quad \text{is a row: } (1\times 4)(4\times 4) = 1\times 4. \end{aligned} \tag{110}$$

Both products in the integrand multiply a row by a column and give scalars.

$$\delta\bar{\psi}\, B: \ (1\times 4)(4\times 1) = 1\times 1, \qquad C\,\delta\psi: \ (1\times 4)(4\times 1) = 1\times 1. \tag{111}$$

We want to show that the preceding condition forces $B = 0_{4\times 1}$ and $C = 0_{1\times 4}$. The column B and the row C must vanish component by component. To see that this does not assume physical independence of the adjoint, we use $\delta\bar{\psi} = (\delta\psi)^\dagger \gamma^0$. Write the two row-by-column products as sums over components. For the second product, we have directly

$$C\,\delta\psi = \sum_{a=1}^{4} C_a\, \delta\psi^a. \tag{112}$$

For the first we replace the expression of $\delta\bar{\psi}$ and group the matrices with B:

$$\delta\bar{\psi}\, B = (\delta\psi)^\dagger \gamma^0 B = (\delta\psi)^\dagger (\gamma^0 B) = \sum_{a=1}^{4} \overline{\delta\psi^a}\, (\gamma^0 B)_a. \tag{113}$$

Here $\overline{\delta\psi^a}$ is the complex conjugate of the component a of the variation. Substituting both products in the integral we obtain

$$\delta S_D = \int_M d^4x \sum_{a=1}^{4} [C_a\, \delta\psi^a + (\gamma^0 B)_a\, \overline{\delta\psi^a}]. \tag{114}$$

Substitute $\delta\psi^a = \delta u^a + i\delta v^a$ and its conjugate.[7] We obtain

$$\delta S_D = \int_M d^4x \sum_{a=1}^{4} \Big[(C_a + (\gamma^0 B)_a)\delta u^a + i(C_a - (\gamma^0 B)_a)\delta v^a \Big]. \tag{115}$$

Stationarity requires $\delta S_D = 0$ for all real variations $\delta u^a, \delta v^a$. By the fundamental lemma of the calculus of variations, the coefficients vanish pointwise.

$$C_a + (\gamma^0 B)_a = 0, \qquad C_a - (\gamma^0 B)_a = 0. \tag{116}$$

Adding and subtracting the component equations gives $C = 0_{1\times 4}$ and $\gamma^0 B = 0_{4\times 1}$. Since γ^0 is invertible, we also have $B = 0_{4\times 1}$. This last equality is the Dirac equation.

$$\boxed{(i\gamma^\mu \partial_\mu - m)\psi = 0.} \tag{117}$$

Multiplying $C = 0_{1\times 4}$ by -1 gives the adjoint equation.

$$\boxed{i(\partial_\mu \bar{\psi})\gamma^\mu + m\bar{\psi} = 0.} \tag{118}$$

This justifies the abbreviated rule. Formally varying $\bar{\psi}$ yields the equation for ψ. Varying ψ yields the adjoint equation. For conjugate fields, the second equation is the adjoint of the first, so both describe the same physical dynamics.

---

[7] We can obtain the same result without splitting the variation into real and imaginary parts. Apply stationarity to two admissible variations to separate the two terms.

First we choose any $\eta \in C_c^\infty(M, \mathbb{C}^4)$. The equation (108), with $\delta\psi = \eta$, gives

$$\int_M d^4x \, (\bar{\eta} B + C\eta) = 0. \tag{IV}$$

Here $\bar{\eta} = \eta^\dagger \gamma^0$ is the Dirac adjoint of the variation.

Next choose $\delta\psi = i\eta$. This variation is also smooth and compactly supported. Taking its adjoint conjugates the factor i.

$$\overline{i\eta} = (i\eta)^\dagger \gamma^0 = -i\eta^\dagger \gamma^0 = -i\bar{\eta}. \tag{V}$$

Therefore, stationarity for this second variation requires

$$\int_M d^4x \, (-i\bar{\eta} B + iC\eta) = 0. \tag{VI}$$

Dividing this equality by i gives the two conditions

$$\begin{aligned} \int_M d^4x \, (\bar{\eta} B + C\eta) &= 0, \\ \int_M d^4x \, (-\bar{\eta} B + C\eta) &= 0. \end{aligned} \tag{VII}$$

By adding them, the terms with B are canceled and we get $2\int_M C\eta \, d^4x = 0$. By subtracting the second from the first, the terms with C cancel and we get $2\int_M \bar{\eta} B \, d^4x = 0$.

$$\int_M C\eta \, d^4x = 0, \qquad \int_M \bar{\eta} B \, d^4x = 0 \quad \text{for every } \eta \in C_c^\infty(M, \mathbb{C}^4). \tag{VIII}$$

We have separated the integrals involving B and C against arbitrary variations. Each component of η can be chosen as an independent test function. Since γ^0 is invertible, the adjoints $\bar{\eta}$ range over all smooth, compactly supported rows. The fundamental lemma of the calculus of variations forces both coefficients to vanish pointwise in every component.

$$B = 0_{4\times 1}, \qquad C = 0_{1\times 4}. \tag{IX}$$

Cancellation for one variation would be insufficient. The result follows from stationarity for all admissible variations, including η and $i\eta$, while retaining the dependence between the field and its adjoint.

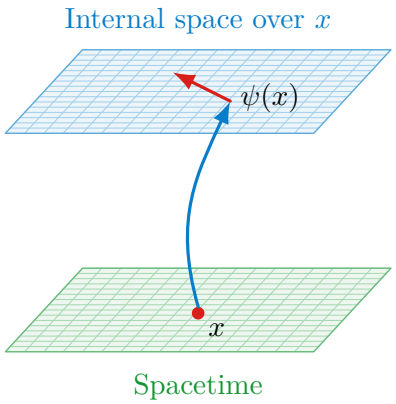


Figure 3. A field value at each point. The lower plane represents spacetime and the upper plane depicts the internal space. The blue arrow associates x with $\psi(x)$. The red arrow symbolizes the field value, which is a spinor. The upper plane depicts the fiber schematically.

3.2 Relationship to Hamiltonian density

By separating the temporal derivative into (93),

$$\mathcal{L}_D = i\psi^\dagger\dot{\psi} + \bar{\psi}i\gamma^i\partial_i\psi - m\bar{\psi}\psi. \tag{119}$$

Using the canonical differentiation convention that places the momentum to the left of $\dot{\psi}$, we obtain

$$\pi_\psi = i\psi^\dagger. \tag{120}$$

The Hamiltonian density is the product of the momentum and the time derivative of the field, minus the Lagrangian density.

$$\boxed{\mathcal{H}_D \coloneqq \pi_\psi\dot{\psi} - \mathcal{L}_D = \bar{\psi}(-i\gamma^i\partial_i + m)\psi.} \tag{121}$$

The Lagrangian is first order and the momentum conjugate to $\psi^\dagger$ vanishes. The Legendre transformation is therefore singular and gives canonical constraints.

4 Internal transformations and global symmetry

Before we introduce the transformation, let us remember what it means for a matrix to be unitary.

Definition 17 (Unitary matrix)**.** A square complex matrix U of size $n \times n$ is unitary if its adjoint is its inverse:

$$U^\dagger U = UU^\dagger = \mathbf{1}_n, \qquad \text{equivalently, } U^{-1} = U^\dagger. \tag{122}$$

Here $U^\dagger = \overline{U}^T$ is the conjugate transpose, and $\mathbf{1}_n$ is the identity of size $n \times n$. In this section we will use matrices of size 4×4.

An internal transformation changes the field at fixed coordinates. It leaves x fixed and acts independently of Lorentz transformations. Consider first a constant unitary matrix M acting on the spinor indices.

$$\psi' = M\psi, \qquad \bar{\psi}' = \psi^\dagger M^\dagger\gamma^0. \tag{123}$$

To find the conditions under which the transformation preserves the Lagrangian density, first write the Dirac adjoint explicitly.

$$\mathcal{L}_D = i\psi^\dagger\gamma^0\gamma^\mu\partial_\mu\psi - m\psi^\dagger\gamma^0\psi. \tag{124}$$

As M is constant, the derivative does not act on it. Furthermore, the adjoint of a product reverses the order of its factors:

$$\partial_\mu\psi' = M\partial_\mu\psi, \qquad (\psi')^\dagger = \psi^\dagger M^\dagger. \tag{125}$$

We replace these two expressions in the transformed density:

$$\begin{aligned}\mathcal{L}'_D &= i(\psi')^\dagger\gamma^0\gamma^\mu\partial_\mu\psi' - m(\psi')^\dagger\gamma^0\psi' \\ &= i\psi^\dagger M^\dagger\gamma^0\gamma^\mu M\partial_\mu\psi - m\psi^\dagger M^\dagger\gamma^0 M\psi.\end{aligned} \tag{126}$$

By subtracting the original expression, it remains

$$\begin{aligned}\mathcal{L}'_D - \mathcal{L}_D = i\psi^\dagger&\big(M^\dagger\gamma^0\gamma^\mu M - \gamma^0\gamma^\mu\big)\partial_\mu\psi \\ &- m\psi^\dagger\big(M^\dagger\gamma^0 M - \gamma^0\big)\psi.\end{aligned} \tag{127}$$

We want this difference to vanish for arbitrary fields without imposing the equations of motion. At a point, the value of ψ and its first derivatives can be chosen freely. The mass term therefore cannot compensate for the derivative term for all choices. The matrices multiplying the two products must vanish separately. For $m \neq 0$, this gives

$$M^\dagger\gamma^0 M = \gamma^0, \qquad M^\dagger\gamma^0\gamma^\mu M = \gamma^0\gamma^\mu. \tag{128}$$

Unitarity means that $M^\dagger = M^{-1}$; therefore, in addition to $M^\dagger M = \mathbf{1}_4$, we have $MM^\dagger = \mathbf{1}_4$. We multiply the first identity by M to the left:

$$M^\dagger\gamma^0 M = \gamma^0 \quad\Longrightarrow\quad MM^\dagger\gamma^0 M = M\gamma^0 \quad\Longrightarrow\quad \gamma^0 M = M\gamma^0. \tag{129}$$

The product of M and γ^0 is unchanged when we reverse the order. The commutator expresses this property and is defined by

$$[M, A] \coloneqq MA - AM. \tag{130}$$

We do the same operation with the second identity, for each index μ:

$$\begin{aligned} & M^\dagger\gamma^0\gamma^\mu M = \gamma^0\gamma^\mu \\ \Longrightarrow\quad & MM^\dagger\gamma^0\gamma^\mu M = M\gamma^0\gamma^\mu \\ \Longrightarrow\quad & \gamma^0\gamma^\mu M = M\gamma^0\gamma^\mu.\end{aligned} \tag{131}$$

Therefore, the two conditions are written as

$$[M, \gamma^0] = 0, \qquad [M, \gamma^0\gamma^\mu] = 0. \tag{132}$$

We still need to show that the two commutation relations imply commutation with γ^μ separately. The first lets us interchange M and γ^0. Using the second afterward gives

$$\gamma^0 M\gamma^\mu = M\gamma^0\gamma^\mu = \gamma^0\gamma^\mu M. \tag{133}$$

Multiply both sides by γ^0 on the left. Since $(\gamma^0)^2 = \mathbf{1}_4$, we obtain

$$(\gamma^0)^2 M\gamma^\mu = (\gamma^0)^2\gamma^\mu M \quad\Longrightarrow\quad M\gamma^\mu = \gamma^\mu M. \tag{134}$$

It is thus demonstrated, for each μ, that

$$\boxed{[M, \gamma^\mu] = 0.} \tag{135}$$

Example 7 (Global phase rotation)**.** A matrix that meets this condition is the identity multiplied by a constant complex phase:

$$\boxed{M = e^{i\theta}\mathbf{1}_4, \qquad \theta \in \mathbb{R}.} \tag{136}$$

This matrix commutes with all gamma matrices because it is a multiple of identity. It is also unitary:

$$M^\dagger M = (e^{-i\theta}\mathbf{1}_4)(e^{i\theta}\mathbf{1}_4) = \mathbf{1}_4. \tag{137}$$

The transformation multiplies each component of the complex vector by the same phase:

$$\psi' = e^{i\theta}\psi, \qquad \psi'^a = e^{i\theta}\psi^a, \qquad a = 1, \ldots, 4. \tag{138}$$

Multiplying a complex number by $e^{i\theta}$ preserves its modulus and rotates its phase by θ. The matrix thus rotates the phase of all four components of $\psi \in \mathbb{C}^4$ by the same angle. Since θ is constant, the rotation is the same at every point in spacetime. In addition to being constant and unitary, the matrix must satisfy the commutation relations with the gamma matrices that we derived.

□

We now focus on these phase rotations.[8]

Theorem 3 (Classification of constant internal transformations)**.** *Let γ^μ be the Dirac matrices on $\mathbb{C}^4$. A constant complex matrix M of size 4×4 is unitary and commutes with all the gamma matrices if and only if there exists $\theta \in \mathbb{R}$ such that*

$$\boxed{M = e^{i\theta}\mathbf{1}_4.} \tag{139}$$

Therefore, under these hypotheses, the only internal transformations are the phase rotations common to the four components. For a massive Dirac field, these are precisely the constant and unitary transformations that preserve the Lagrangian density for arbitrary fields.

[8] We can explain the general argument directly with gamma matrices, without introducing abstract generators. Consider the algebra $\mathcal{A}$ of all complex linear combinations of the identity and products of gamma matrices. It acts on $V = \mathbb{C}^4$ by ordinary matrix multiplication. This matrix realization is a representation of the Clifford algebra on V. The representation is of the algebra, with V as its underlying vector space.

Irreducibility means that the only subspaces W preserved by all the gamma matrices are $\{0\}$ and the whole space V.

$$\gamma^\mu W \subseteq W \quad \text{for all } \mu. \tag{X}$$

For the four-component Dirac gamma matrices, products of gamma matrices generate the full algebra of complex 4×4 matrices. The algebraic generators differ from a vector-space basis. The four matrices $\gamma^0, \gamma^1, \gamma^2, \gamma^3$ generate the algebra by products and linear combinations, so there are four algebraic generators. A complex vector-space basis consists of the identity, the four gamma matrices, the six products $\gamma^\mu\gamma^\nu$ with $\mu < \nu$, the four products of three distinct gamma matrices, and the product of all four. This gives $1 + 4 + 6 + 4 + 1 = 16$ linearly independent matrices, the dimension of the space of 4×4 matrices. The indices in each product are taken in increasing order. The anticommutation relations reduce any other product to a linear combination of these. The algebra thus has complex vector-space dimension sixteen and four algebraic generators. This proves irreducibility. If W contains a nonzero vector and is preserved by all matrices, it contains every vector in V, since a matrix can map that nonzero vector to any other vector.

In the form needed here, Schur's lemma says that if an algebra acts irreducibly on a finite-dimensional complex vector space, every linear operator commuting with all operators in the action is a multiple of the identity. The lemma does not require unitarity.

Since M commutes with every gamma matrix, it commutes with their products and linear combinations, hence with all of $\mathcal{A}$. Schur's lemma gives

$$M = z\mathbf{1}_4, \qquad z \in \mathbb{C}. \tag{XI}$$

Then we use that M is unitary:

$$\mathbf{1}_4 = M^\dagger M = |z|^2\mathbf{1}_4 \quad \Longrightarrow \quad |z| = 1. \tag{XII}$$

Thus $z = e^{i\theta}$. Schur's lemma gives a multiple of the identity, and unitarity makes the multiplier have modulus one. This classifies all possibilities under our hypotheses. The direct verification of phase rotation in the main text is independent of this classification.

Proof. If M commutes with all gamma matrices, it commutes with the matrix algebra they generate. The action on $\mathbb{C}^4$ is irreducible, as explained in the preceding note. Schur's lemma gives $M = z\mathbf{1}_4$. Unitarity requires $|z| = 1$, so $z = e^{i\theta}$ for some $\theta \in \mathbb{R}$.

Conversely, $M = e^{i\theta}\mathbf{1}_4$ commutes with all gamma matrices and is unitary, as verified above. The invariance conditions then show that it preserves the Lagrangian density. This proves both directions and excludes other matrices under the hypotheses. □

Remark 1 (Scope of the theorem)**.** The theorem concerns one massive Dirac field and matrices acting on its four components. The conclusion about phase rotations is restricted to these hypotheses. Two other situations are worth distinguishing.

- With several Dirac fields, a transformation can mix the fields as well as act on their individual components. Such mixing acts on a space larger than $\mathbb{C}^4$. Whether it is a symmetry also depends on the masses and the other terms in the theory.
- If the field is massless, the term $-m\bar{\psi}\psi$ disappears from the Lagrangian density. The condition derived from preserving that term no longer applies. The kinetic term still imposes a condition, but by itself allows other symmetries. This changes one of the hypotheses used to classify the transformations preserving the Lagrangian density.

These cases do not contradict the proven result. For the current discussion we continue to consider a single massive field and transformations on its four components.

Definition 18 (Phase group)**.** The group $U(1)$ is formed by the complex numbers of modulus one, with multiplication as operation:

$$U(1) \coloneqq \{e^{i\theta} : \theta \in \mathbb{R}\}, \qquad e^{i\theta_1}e^{i\theta_2} = e^{i(\theta_1+\theta_2)}. \tag{140}$$

The phase is defined modulo 2π. A transformation is global when the same parameter is used throughout spacetime.

For θ constant,

$$\psi' = e^{i\theta}\psi, \qquad \bar{\psi}' = e^{-i\theta}\bar{\psi}, \qquad \partial_\mu\psi' = e^{i\theta}\partial_\mu\psi. \tag{141}$$

Let us explicitly verify that this global rotation does not change the Lagrangian density or the action.

Proof of global invariance. Since θ is constant, the derivative leaves the phase factor unchanged. Substituting the preceding expressions gives

$$\begin{aligned} \mathcal{L}_D[\psi'] &= i\bar{\psi}'\gamma^\mu\partial_\mu\psi' - m\bar{\psi}'\psi' \\ &= i(e^{-i\theta}\bar{\psi})\gamma^\mu(e^{i\theta}\partial_\mu\psi) - m(e^{-i\theta}\bar{\psi})(e^{i\theta}\psi) \\ &= i\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi = \mathcal{L}_D[\psi]. \end{aligned} \tag{142}$$

In both terms the scalar phases cancel because $e^{-i\theta}e^{i\theta} = 1$. The equality holds pointwise. **The internal transformation leaves the coordinates fixed**, so the domain M and the integration measure also remain the same. Integrating gives

$$\boxed{S_D[\psi'] = \int_M d^4x\, \mathcal{L}_D[\psi'] = \int_M d^4x\, \mathcal{L}_D[\psi] = S_D[\psi].} \tag{143}$$

Thus the invariance of the Lagrangian density and that of the action for any permissible field are verified, without using the equations of motion. □

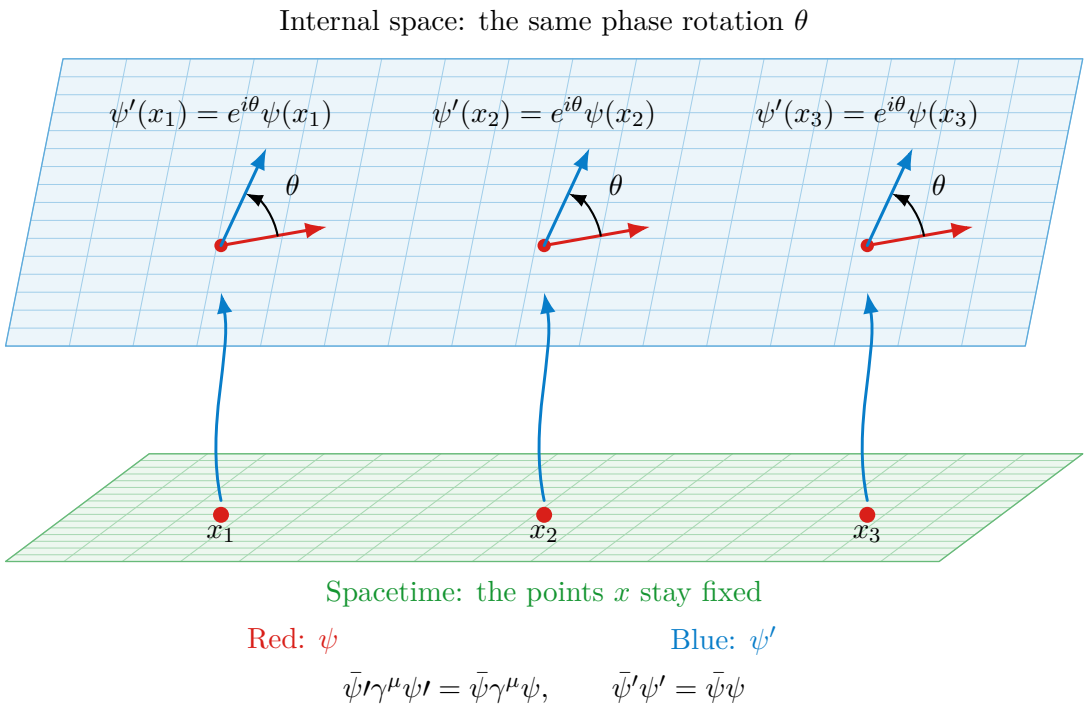


Figure 4. Global phase rotation. Red and blue arrows depict ψ and ψ' in a phase plane. Every rotation is through the same angle θ. The transformation acts on internal values and keeps the coordinates fixed. The indicated bilinears remain invariant.

5 The local phase and the failure of the ordinary derivative

Let us now make $\theta = \theta(x)$ a smooth real function. The product rule implies

$$\partial_\mu \psi' = e^{i\theta(x)} \left[\partial_\mu \psi + i(\partial_\mu \theta)\psi\right]. \tag{144}$$

To abbreviate the field products listed below, we write

$$j^\mu := \bar{\psi}\gamma^\mu\psi. \tag{145}$$

We make the substitution into the free density explicit. The adjoint transforms with the opposite phase.

$$\bar{\psi}' = \bar{\psi}e^{-i\theta(x)}. \tag{146}$$

Writing the kinetic and mass terms separately, we have

$$\mathcal{L}'_D = i\bar{\psi}'\gamma^\mu\partial_\mu\psi' - m\bar{\psi}'\psi'. \tag{147}$$

We replace the adjoint and derivative of the transformed field:

$$\mathcal{L}'_D = i\bar{\psi}e^{-i\theta}\gamma^\mu e^{i\theta}\left[\partial_\mu\psi + i(\partial_\mu\theta)\psi\right] - m\bar{\psi}e^{-i\theta}e^{i\theta}\psi. \tag{148}$$

The phases are scalar, they commute with the gamma matrices and their product is one.

$$\mathcal{L}'_D = i\bar{\psi}\gamma^\mu\partial_\mu\psi + i^2(\partial_\mu\theta)\bar{\psi}\gamma^\mu\psi - m\bar{\psi}\psi. \tag{149}$$

As $i^2 = -1$, the additional term has a negative sign. The remaining kinetic and mass terms form the original density:

$$\mathcal{L}'_D = \underbrace{i\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi}_{\mathcal{L}_D} - (\partial_\mu\theta)\underbrace{\bar{\psi}\gamma^\mu\psi}_{j^\mu}. \tag{150}$$

Thus we obtain

$$\boxed{\mathcal{L}'_D = \mathcal{L}_D - (\partial_\mu\theta)j^\mu.} \tag{151}$$

The additional term prevents invariance for arbitrary configurations. It is also seen when applying the Dirac operator to the transformed field. First we replace $\psi' = e^{i\theta(x)}\psi$:

$$(i\gamma^\mu\partial_\mu - m)\psi' = i\gamma^\mu\partial_\mu(e^{i\theta}\psi) - me^{i\theta}\psi. \tag{152}$$

By the product rule, the derivative acts both on the phase and on the field:

$$\partial_\mu(e^{i\theta}\psi) = ie^{i\theta}(\partial_\mu\theta)\psi + e^{i\theta}\partial_\mu\psi. \tag{153}$$

On substitution, the operator factor i multiplies the i obtained by differentiating the phase. Since $i^2 = -1$, we obtain

$$(i\gamma^\mu\partial_\mu - m)\psi' = -e^{i\theta}\gamma^\mu(\partial_\mu\theta)\psi + e^{i\theta}i\gamma^\mu\partial_\mu\psi - me^{i\theta}\psi. \tag{154}$$

The phase is scalar, so we can factor it out. This verifies the identity

$$(i\gamma^\mu\partial_\mu - m)\psi' = e^{i\theta}\left[(i\gamma^\mu\partial_\mu - m)\psi - \gamma^\mu(\partial_\mu\theta)\psi\right]. \tag{155}$$

If ψ satisfies the free equation, the first term inside the bracket vanishes and the following remains.

$$\boxed{(i\gamma^\mu\partial_\mu - m)\psi' = -e^{i\theta}\gamma^\mu(\partial_\mu\theta)\psi.} \tag{156}$$

A free solution is not generally transformed into another free solution.

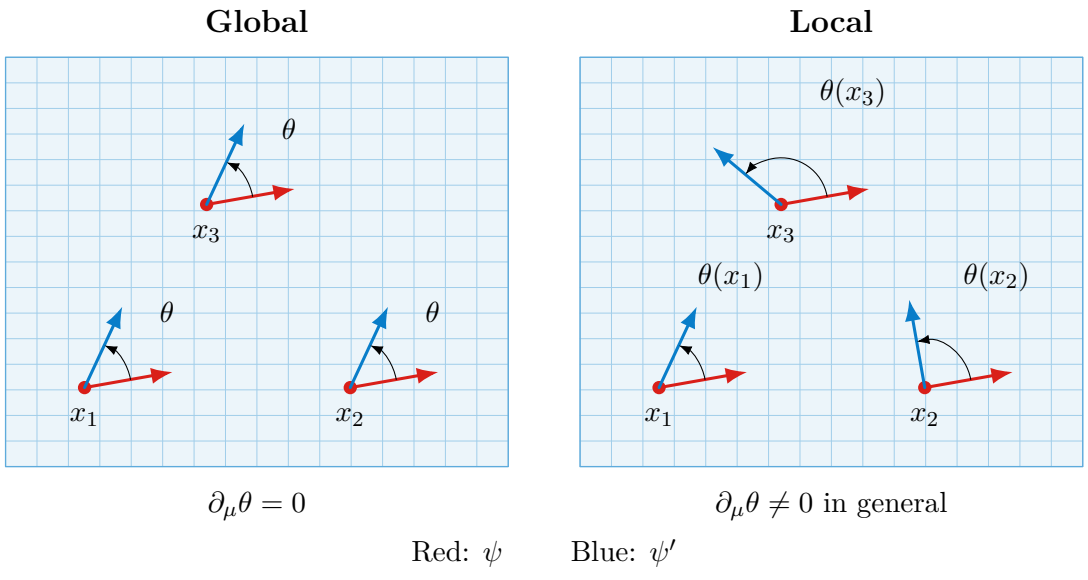


Figure 5. Global and local phase transformations. A global phase is the same everywhere, whereas local phases are chosen point by point. In the local panel, the rotation angles differ. Comparing neighboring values using ∂_μ detects the variation. The connection A_μ introduced in the next section compensates for this effect. The angles in the drawing are illustrative.

5.1 Why integrating by parts does not solve the problem

The equation (151) already expresses the difference of **Lagrangian densities**. To move to the difference of **actions**, we integrate that identity over the whole manifold M:

$$S'_D[\psi'] - S_D[\psi] = \int_M d^4x\,(\mathcal{L}'_D - \mathcal{L}_D) = -\int_M d^4x\,(\partial_\mu\theta)j^\mu. \tag{157}$$

We now integrate by parts. The product rule gives

$$\partial_\mu(\theta j^\mu) = (\partial_\mu\theta)j^\mu + \theta\partial_\mu j^\mu, \tag{158}$$

and solving for the term that occurs in the integral gives

$$-(\partial_\mu\theta)j^\mu = -\partial_\mu(\theta j^\mu) + \theta\partial_\mu j^\mu. \tag{159}$$

Substitution in the preceding integral expresses the same difference of actions as

$$S'_D - S_D = -\int_M d^4x\,\partial_\mu(\theta j^\mu) + \int_M d^4x\,\theta\partial_\mu j^\mu. \tag{160}$$

Consider the two integrals separately. The first integrates a **four-divergence**, which becomes a boundary term by the divergence theorem. In Minkowski spacetime we take bounded domains exhausting M and require the flux of θj^μ through their boundaries to tend to zero. This holds, for example, if θj^μ has compact support. Under this condition,

$$\int_M d^4x\,\partial_\mu(\theta j^\mu) = 0. \tag{161}$$

For the second integral we require ψ to be a **solution of the free Dirac equation**, with adjoint $\bar\psi = \psi^\dagger\gamma^0$. The field and adjoint equations are

$$i\gamma^\mu\partial_\mu\psi = m\psi, \qquad i(\partial_\mu\bar\psi)\gamma^\mu = -m\bar\psi. \tag{162}$$

Dividing by i gives the expressions to substitute.

$$\gamma^\mu\partial_\mu\psi = -im\psi, \qquad (\partial_\mu\bar\psi)\gamma^\mu = im\bar\psi. \tag{163}$$

Since the gamma matrices are constant in the fixed Minkowski trivialization, the product rule gives

$$\partial_\mu j^\mu = \partial_\mu(\bar\psi\gamma^\mu\psi) = (\partial_\mu\bar\psi)\gamma^\mu\psi + \bar\psi\gamma^\mu\partial_\mu\psi. \tag{164}$$

Now we replace both equations. The mass terms are canceled:

$$\boxed{\partial_\mu j^\mu = im\bar\psi\psi - im\bar\psi\psi = 0.} \tag{165}$$

Consequently, the second integral is also annulled:

$$\int_M d^4x\,\theta\partial_\mu j^\mu = 0. \tag{166}$$

If the boundary term disappears and the fields satisfy the free equations, the difference of actions is zero:

$$\boxed{S_D[\psi'] = S_D[\psi], \qquad \psi' = e^{i\theta(x)}\psi.} \tag{167}$$

We assume that the above actions and integrals are well defined; if only their difference is defined, the conclusion is expressed as zero difference.

In the variational calculation a *field configuration* is a spinor field ψ with the regularity and integrability needed to evaluate $S_D[\psi]$. The Dirac equation is imposed only by requiring stationarity of the action. We call the fields satisfying it *free solutions*. The preceding calculation gives equality of the actions on free solutions under the stated boundary conditions. **Local invariance of the action for arbitrary configurations does not follow**, since the divergence need not vanish there. In particular, even if ψ is a free solution and $S_D[\psi'] = S_D[\psi]$ holds under the stated boundary conditions, $\psi' = e^{i\theta(x)}\psi$ **generally fails to be another free solution**. Indeed, for the original solution,

$$(i\gamma^\mu\partial_\mu - m)\psi' = -e^{i\theta(x)}\gamma^\mu(\partial_\mu\theta)\psi, \tag{168}$$

and the right side can be nonzero. **Equality of action values does not guarantee stationarity.** If the phase is constant, its derivative vanishes and the transformed field remains a free solution.

Example 8 (A phase with constant gradient)**.** On a region, take $\theta(x) = k_\nu x^\nu$, with k_ν constant. Differentiate with respect to x^μ.

$$\partial_\mu\theta = \partial_\mu(k_\nu x^\nu) = k_\nu\partial_\mu x^\nu = k_\nu\delta^\nu_\mu = k_\mu. \tag{169}$$

Here $\partial_\mu x^\nu = \delta^\nu_\mu$ and the delta selects the component k_μ. We replace this result in the density difference:

$$\mathcal{L}'_D - \mathcal{L}_D = -(\partial_\mu\theta)j^\mu = -k_\mu j^\mu. \tag{170}$$

If ψ is also a free solution, the preceding identity for the Dirac operator gives

$$(i\gamma^\mu\partial_\mu - m)\psi' = -e^{i\theta}\gamma^\mu(\partial_\mu\theta)\psi = -e^{i\theta}\gamma^\mu k_\mu\psi. \tag{171}$$

For a generic k_μ the last expression is nonzero. The example is local and does not assume compact support for the phase or cancellation of the global boundary term.

□

6 The Covariant Derivative and the Electromagnetic Connection

Definition 19 (Electromagnetic covariant derivative)**.** Let $q \neq 0$ be the electric charge of the field. For the electron we choose $q = -e$, with $e > 0$. Parametrize the phase by $\theta(x) = q\alpha(x)$ and introduce

$$\boxed{D_\mu \coloneqq \partial_\mu + iqA_\mu.} \tag{172}$$

We write the local factor of $U(1)$ as

$$U(x) \coloneqq e^{iq\alpha(x)}, \qquad \psi' = U\psi. \tag{173}$$

The ordinary derivative does not preserve the field's transformation law under a local phase, because it also differentiates $U(x)$. We seek a derivative that preserves it. If ψ transforms by U, we want $D_\mu\psi$ to transform by the same U. This property is called **covariance under gauge transformations**. In the calculation, we want **the phase factor to come out intact**, without extra terms.

$$\boxed{D'_\mu(U\psi) = UD_\mu\psi.} \tag{174}$$

The index μ is a spacetime index. The transformation here acts on the internal space with the coordinates held fixed. In a more general gauge group, $U(x)$ can be a matrix acting on the internal components. Here it is the phase $e^{iq\alpha(x)}$ times the identity.

The aim is to construct an invariant kinetic term. Suppose the preceding condition holds and the adjoint transforms with the inverse phase.

$$\bar{\psi}' = \bar{\psi}U^{-1}, \tag{175}$$

then the phases cancel:

$$\boxed{\bar{\psi}' i\gamma^\mu D'_\mu \psi' = \bar{\psi}U^{-1} i\gamma^\mu U D_\mu \psi = \bar{\psi} i\gamma^\mu D_\mu \psi.} \tag{176}$$

The last equality uses commutation of the scalar phase with the gamma matrices. **The full kinetic term is invariant when we use the covariant derivative and transform the potential as well.** The two properties are distinct. The derivative $D_\mu\psi$ is *covariant* because it acquires U. When multiplied by the adjoint, which acquires U^{-1}, the factors cancel and $\bar{\psi}i\gamma^\mu D_\mu\psi$ is *invariant*. We have assumed covariance and now determine how A_μ must change to ensure it.

To deduce that transformation, we leave A'_μ unspecified and develop the left side:

$$D'_\mu(U\psi) = (\partial_\mu + iqA'_\mu)(U\psi) \tag{177}$$

$$= (\partial_\mu U)\psi + U\partial_\mu\psi + iqA'_\mu U\psi. \tag{178}$$

The derivative of the phase is

$$\partial_\mu U = iq(\partial_\mu\alpha)U. \tag{179}$$

Since U is scalar, we can take it as a common factor:

$$D'_\mu(U\psi) = U\big[\partial_\mu\psi + iq(\partial_\mu\alpha + A'_\mu)\psi\big]. \tag{180}$$

The right side of the condition we're looking for is

$$UD_\mu\psi = U\big[\partial_\mu\psi + iqA_\mu\psi\big]. \tag{181}$$

For both expressions to match for any field ψ, with $q \neq 0$, their coefficients must satisfy

$$\boxed{\partial_\mu\alpha + A'_\mu = A_\mu.} \tag{182}$$

Solving for the potential gives its transformation together with the field transformation.

$$\boxed{\psi' = e^{iq\alpha}\psi, \qquad A'_\mu = A_\mu - \partial_\mu\alpha.} \tag{183}$$

Thus, the change of potential cancels out exactly the term produced by the derivative of the phase and meets the condition of covariance that we raised at the beginning.

6.1 What Changes When the Connection Transforms

We keep both the coordinates and the bundle trivialization fixed in these calculations. They are distinct choices. The coordinates x^μ describe the points of M. The trivialization identifies the fiber at each point with $\mathbb{C}^4$, letting us write the spinor as a four-component column. Keeping that identification fixed lets us compare fields before and after transformation. A change of trivialization can leave the spacetime coordinates fixed.

We use the *active* interpretation. The field $\psi' = U\psi$ is generally a different section from ψ, and D' is a transformed connection on the same bundle. Covariance is equivalent to the operator identity

$$\boxed{D'_\mu = UD_\mu U^{-1}.} \tag{184}$$

The right side means operator composition: the derivative also acts on $U^{-1}(x)$. To see it, let us apply it to an arbitrary section η:

$$UD_\mu(U^{-1}\eta) = \partial_\mu\eta + iq(A_\mu - \partial_\mu\alpha)\eta = D'_\mu\eta. \tag{185}$$

The connections D and D' are generally distinct but gauge-equivalent. They can act differently on the same section. A gauge-compatible comparison transforms the field and the connection together.

$$(\psi, D) \longmapsto (U\psi, UDU^{-1}), \qquad D'_\mu\psi' = UD_\mu\psi. \tag{186}$$

These pairs describe the same physical situation in gauge theory because the gauge-invariant quantities agree. In the *passive* interpretation we choose a new basis, or *frame*, in each fiber. The Minkowski coordinates and the space $\mathbb{C}^4$ stay fixed. The section is written in frame $\widehat{e}(x) = (e_1(x), \dots, e_4(x))$ as

$$s(x) = \sum_{a=1}^{4} \psi^a(x) e_a(x). \tag{187}$$

When applying the reverse phase we get the frame $\widehat{e}'(x) = (e'_1(x), e'_2(x), e'_3(x), e'_4(x))$, whose vectors satisfy

$$e'_a(x) \coloneqq e^{-iq\alpha(x)} e_a(x), \tag{188}$$

the components of the same section change as

$$\psi'^a(x) = e^{iq\alpha(x)}\psi^a(x). \tag{189}$$

Both changes are compensated: the frame vectors and components are different, but the section is the same,

$$\sum_{a=1}^{4} \psi'^a(x) e'_a(x) = \sum_{a=1}^{4} \psi^a(x) e_a(x) = s(x). \tag{190}$$

We also maintain the geometric connection, which we denote here by ∇. Its operators on the components in the two frames satisfy

$$\nabla_\mu s = \sum_{a=1}^{4} (D_\mu\psi)^a e_a = \sum_{a=1}^{4} (D'_\mu\psi')^a e'_a. \tag{191}$$

Thus $D'_\mu\psi' = UD_\mu\psi$, and the expressions for the same connection are related by $D'_\mu = UD_\mu U^{-1}$. The extra term appears because the new frame depends on x, so differentiation also acts on its vectors. **In the passive interpretation the frame changes. In the active interpretation the frame is fixed and the section and connection transform.** We use the active interpretation in the main development. Haydys discusses the relation between changes of trivialization and the action of the gauge group on connections [10].

Example 9 (Gauge transformation of a flat connection)**.** In the fixed trivialization, start with the trivial internal connection and transform it.

$$A_\mu = 0, \qquad D_\mu = \partial_\mu, \qquad A'_\mu = -\partial_\mu\alpha, \qquad D'_\mu = \partial_\mu - iq\partial_\mu\alpha. \tag{192}$$

If $d\alpha \neq 0$, the operators differ. Both connections are flat. For smooth α the partial derivatives commute, so the transformed curvature is

$$\boxed{F'_{\mu\nu} \coloneqq \partial_\mu A'_\nu - \partial_\nu A'_\mu = -\partial_\mu\partial_\nu\alpha + \partial_\nu\partial_\mu\alpha = 0.} \tag{193}$$

□

This is the **internal connection** $U(1)$ of the spinor field. The Levi-Civita connection of Minkowski spacetime is unchanged.

6.2 Minimal Coupling

The gauge-invariant matter density is

$$\mathcal{L}_{\text{mat}} \coloneqq \bar{\psi}(i\gamma^\mu D_\mu - m)\psi = \mathcal{L}_D - qA_\mu\bar{\psi}\gamma^\mu\psi. \tag{194}$$

Definition 20 (Electric current and interaction term)**.** We define the electric current $J^\mu \coloneqq qj^\mu$. So,

$$\boxed{\mathcal{L}_{\text{int}} \coloneqq -J^\mu A_\mu = -q\bar{\psi}\gamma^\mu\psi A_\mu.} \tag{195}$$

The mass term is invariant because $\bar{\psi}'\psi' = \bar{\psi}\psi$. Interaction compensates exactly for variation (151), without using equations of motion.

7 Gauge Orbits of the Field and Connection

7.1 Gauge orbits and composition

Example 10 (Two pairs in the same gauge orbit)**.** We start from field ψ and trivial connection $A_\mu = 0$, $D_\mu = \partial_\mu$. For a smooth real function θ, the gauge transformation of the pair gives

$$\psi' = e^{i\theta}\psi, \qquad A'_\mu = -\frac{1}{q}\partial_\mu\theta, \qquad D'_\mu = \partial_\mu - i\partial_\mu\theta. \tag{196}$$

When applying this derivative to the transformed field, the terms containing the derivative of the phase cancel:

$$\begin{aligned} D'_\mu\psi' &= (\partial_\mu - i\partial_\mu\theta)(e^{i\theta}\psi) \\ &= e^{i\theta}\left[\partial_\mu\psi + i(\partial_\mu\theta)\psi - i(\partial_\mu\theta)\psi\right] \\ &= e^{i\theta}\partial_\mu\psi. \end{aligned} \tag{197}$$

Using $\bar{\psi}' = \bar{\psi}e^{-i\theta}$, we get

$$\boxed{\bar{\psi}'(i\gamma^\mu D'_\mu - m)\psi' = \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi = \mathcal{L}_D[\psi].} \tag{198}$$

The transformed pair still describes the free theory. As we saw in (193), A'_μ has zero curvature. The preceding equality compares the Lagrangian of the transformed pair (ψ', D') with that of the original pair (ψ, ∂). Both the field and the connection transform. The pairs belong to the same gauge orbit.

□

In the direct phase convention, the general transformation is

$$\psi' = e^{i\theta}\psi, \qquad A'_\mu = A_\mu - \frac{1}{q}\partial_\mu\theta. \tag{199}$$

Definition 21 (Gauge group and configuration space)**.** To specify what gauge equivalence identifies, consider the configuration space $\mathcal{C}$. Its elements are pairs (ψ, D) consisting of a smooth spinor field and a smooth internal connection on the fixed bundle. Write $\mathcal{A}$ for the space of these connections and define the gauge group by

$$C^\infty(M, U(1)) = \{U : M \to U(1) \mid U \text{ is smooth}\}. \tag{200}$$

Its elements are smooth functions $U : M \to U(1)$; in our Minkowski space we can write them as $U(x) = e^{i\theta(x)}$, with θ real and smooth. The operation of the group is the pointwise multiplication:

$$(U_1U_2)(x) \coloneqq U_1(x)U_2(x). \tag{201}$$

The identity is the constant function 1. The inverse takes the value $U(x)^{-1}$ at each point.

With these choices, the configuration space and the group action are

$$\mathcal{C} \coloneqq \Gamma^\infty(S) \times \mathcal{A}. \tag{202}$$

$$U \cdot (\psi, D) \coloneqq (U\psi, UDU^{-1}), \qquad U \in C^\infty(M, U(1)). \tag{203}$$

The transformation acts on both members of the pair: the field and the connection.

Definition 22 (Gauge Equivalence)**.** We define the gauge equivalence relation by

$$(\psi, D) \sim (\psi', D') \quad \Longleftrightarrow \quad \exists U \in C^\infty(M, U(1)) : \ \psi' = U\psi, \ D' = UDU^{-1}. \tag{204}$$

The relation is reflexive by the identity, symmetric by inverses, and transitive by composition. In particular, two successive phase transformations with parameters θ and η give one with parameter $\theta + \eta$.

Definition 23 (Gauge orbit and quotient space)**.** The equivalence class brings together all the pairs related in this way to a representative. It is also called **gauge orbit**:

$$\boxed{[(\psi, D)] \coloneqq \{(U\psi, UDU^{-1}) : U \in C^\infty(M, U(1))\}.} \tag{205}$$

The space of all these classes is the **quotient by the gauge group**.

$$\boxed{\mathcal{C}/C^\infty(M, U(1)) \coloneqq \{[(\psi, D)] : (\psi, D) \in \mathcal{C}\}.} \tag{206}$$

In Example 10, if we write $\theta = q\alpha$, the representatives of the same class are

$$(\psi, \partial) \sim (e^{iq\alpha}\psi, \partial - iq\,\mathrm{d}\alpha). \tag{207}$$

Here ∂ denotes the trivial connection and $\mathrm{d}\alpha$ is the one-form with components $\partial_\mu\alpha$. Fields and connections can differ while gauge-invariant quantities agree. **Equality of Lagrangians alone does not imply gauge equivalence**. There must be a transformation in $C^\infty(M, U(1))$ relating the pairs.

This construction uses the active interpretation, grouping different configurations into classes. The passive interpretation keeps the geometric section and connection fixed and changes the frame in which

we write them. The transformation laws coincide, but different things are held fixed, as in the preceding section [10].

In the fixed trivialization, $D_\mu = \partial_\mu + iqA_\mu$ represents each connection by its potential A. Thus the pairs (ψ, A) and the pairs (ψ, D) represent the same data. Keeping $[(\psi, D)]$ for the orbit, its expression in terms of potentials is

$$[(\psi, D)] \;\longleftrightarrow\; \left\{ \left(e^{i\theta}\psi,\, A - \frac{1}{q}\mathrm{d}\theta \right) : \theta \in C^\infty(M, \mathbb{R}) \right\}. \tag{208}$$

The arrow indicates the correspondence between the two ways of writing the pairs of the same orbit; it does not introduce a new orbit.

Here the internal connection does not have to be flat: the potential A is general. In notation of differential forms, its curvature is $F = \mathrm{d}A$. As $\mathrm{d}^2\theta = 0$, the gauge transformation retains that curvature:

$$\boxed{F' = \mathrm{d}A' = \mathrm{d}\left(A - \frac{1}{q}\mathrm{d}\theta \right) = \mathrm{d}A - \frac{1}{q}\mathrm{d}^2\theta = F.} \tag{209}$$

If the initial connection is flat, all representatives of its orbit have $F = 0$. If the curvature is nonzero, they all have that same F. All representatives have the same matter density from Section 6.2. The Maxwell term, which we will add later, is also invariant because it depends on F.

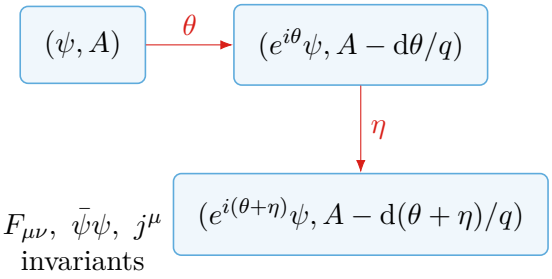


Figure 6. Representatives of the same gauge orbit. The arrows change the phase and the connection together. The family depicts a redundancy in the description.

Example 11 (Two potentials with zero electromagnetic field strength)**.** To avoid adding magnitudes of different dimensions, let's fix a length ℓ and take

$$\beta_1 := \frac{x^2 + y^2 - tx}{\ell^2}, \qquad \beta_2 := \frac{x}{\ell} - \frac{y^2}{\ell^2}. \tag{210}$$

The potentials $A^{(a)} = q^{-1}\mathrm{d}\beta_a$ have covariant components in the order (t, x, y, z)

$$A^{(1)}_\mu = \frac{1}{q}\left(-\frac{x}{\ell^2}, \frac{2x - t}{\ell^2}, \frac{2y}{\ell^2}, 0 \right), \tag{211}$$

$$A^{(2)}_\mu = \frac{1}{q}\left(0, \frac{1}{\ell}, -\frac{2y}{\ell^2}, 0 \right). \tag{212}$$

For example, $F^{(1)}_{0x} = q^{-1}[-1/\ell^2 - (-1/\ell^2)] = 0$. The other components also vanish. Choosing $\theta = \beta_1 - \beta_2$ transforms $A^{(1)}$ into $A^{(2)}$ together with the corresponding change in the spinor phase.

□

8 Curvature and Holonomy of the Electromagnetic Connection

Definition 24 (Field Tensor)**.** The electromagnetic tensor $F_{\mu\nu}$ determines the curvature of the connection by

$$[D_\mu, D_\nu]\psi = iqF_{\mu\nu}\psi, \qquad F_{\mu\nu} \coloneqq \partial_\mu A_\nu - \partial_\nu A_\mu. \tag{213}$$

The tensor is antisymmetric: $F_{\mu\nu} = -F_{\nu\mu}$. As an operator, the curvature is the two-form $iqF\,\mathbf{1}_4$, where $F = \frac{1}{2}F_{\mu\nu}\,\mathrm{d}x^\mu \wedge \mathrm{d}x^\nu = \mathrm{d}A$. When we call F the electromagnetic curvature, we suppress this fixed factor of the representation.

As partial derivatives commute,

$$F'_{\mu\nu} = F_{\mu\nu} - \partial_\mu\partial_\nu\alpha + \partial_\nu\partial_\mu\alpha = F_{\mu\nu}. \tag{214}$$

The potential A_μ can change by choosing another representative from the same gauge orbit, while the curvature $F_{\mu\nu}$ remains the same.

8.1 What the Curvature in the Drawing Represents

The corrugated mesh illustrates that a general connection may have $F_{\mu\nu} \neq 0$. It does not represent a gravitational curvature of spacetime: we continue to work on Minkowski. Phase fibres and their connection are internal structures.

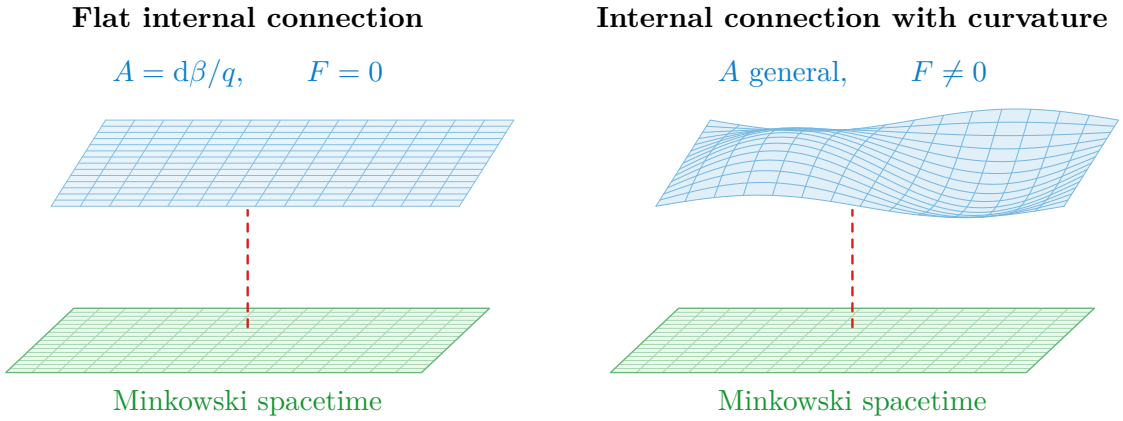


Figure 7. Flat internal connection and internal connection with curvature. The Minkowski base is flat in both cases. The upper mesh depicts the internal connection schematically: flat on the left, with nonzero curvature on the right. The undulation illustrates $F = \mathrm{d}A$. It leaves the spacetime and the fibers $\mathbb{C}^4$ undeformed.

To understand the effect of the connection, transport an internal vector around a closed loop C in Minkowski spacetime. Parametrize the path by $x(s)$, with $0 \leq s \leq 1$ and $x(0) = x(1) = x$. The transported vector $v(s)$ lies in the fiber over $x(s)$. To obtain the covariant derivative along the path, recall the chain rule for a smooth function f.

$$\frac{\mathrm{d}}{\mathrm{d}s} f(x(s)) = \frac{\mathrm{d}x^\mu}{\mathrm{d}s}\,\partial_\mu f(x(s)). \tag{215}$$

The derivative along the curve is therefore the derivative in the direction of its tangent vector. We evaluate the connection in that same direction:

$$\dot{x}^\mu(s) \coloneqq \frac{\mathrm{d}x^\mu}{\mathrm{d}s}, \qquad D_{\dot{x}} \coloneqq \dot{x}^\mu D_\mu. \tag{216}$$

Let us calculate the covariant derivative of v along the curve $x(s)$. At this stage the derivative is unrestricted. On a regular segment without self-intersections, we can locally extend $v(s)$ to a field in a neighborhood of the path and evaluate its components at $x(s)$. Use the same symbol v, so along the path $v(s) = v(x(s))$. Substitute the expression for D_μ, distribute $\dot{x}^\mu$, and apply the chain rule componentwise.

$$\begin{aligned} D_{\dot{x}} v &= \dot{x}^\mu D_\mu v|_{x(s)} \\ &= \dot{x}^\mu (\partial_\mu + iqA_\mu) v|_{x(s)} \\ &= \dot{x}^\mu \partial_\mu v(x(s)) + iqA_\mu(x(s))\dot{x}^\mu v(s) \\ &= \frac{\mathrm{d}}{\mathrm{d}s} v(x(s)) + iqA_\mu(x(s))\dot{x}^\mu v(s) \\ &= \frac{\mathrm{d}v(s)}{\mathrm{d}s} + iqA_\mu(x(s))\dot{x}^\mu(s)\, v(s). \end{aligned} \tag{217}$$

The step from the third line to the fourth uses the chain rule above. The result involves only the values $v(s)$ and their derivative along the path. The calculation gives the covariant derivative in the tangent direction. We impose parallel transport in the next step.

$$\boxed{\frac{Dv}{\mathrm{d}s} \coloneqq D_{\dot{x}} v = \underbrace{\frac{\mathrm{d}v}{\mathrm{d}s}}_{\text{change of components}} + \underbrace{iqA_\mu(x(s))\dot{x}^\mu(s)\, v(s)}_{\text{connection term}}.} \tag{218}$$

The chain rule explains the first term. In the second, $A_\mu(x(s))\dot{x}^\mu(s)$ is the potential evaluated in the direction of the path. Both terms act on the vector $v(s)$ defined along the curve. The final formula defines the induced connection on the pullback bundle over the interval, x^*S. It applies to v along the curve without a global extension to S, including at self-intersections or points where $\dot{x} = 0$.

Parallel transport requires this covariant derivative to vanish. The ordinary derivative can be nonzero because the connection term compensates for the phase change. The parallel transport condition is

$$\frac{\mathrm{d}v}{\mathrm{d}s} + iqA_\mu(x(s))\frac{\mathrm{d}x^\mu}{\mathrm{d}s}\, v(s) = 0. \tag{219}$$

Solving for the derivative gives the transport equation

$$\frac{\mathrm{d}v}{\mathrm{d}s} = -iqA_\mu(x(s))\dot{x}^\mu(s)\, v(s). \tag{220}$$

The coefficient is scalar and multiplies all four components equally. For the initial value $v(0)$, propose the solution

$$v(s) = \exp\left[-iq\int_0^s \mathrm{d}t_1\, A_\mu(x(t_1))\dot{x}^\mu(t_1)\right] v(0). \tag{221}$$

To verify it, use the chain rule and the fundamental theorem of calculus.

$$\begin{aligned} \frac{\mathrm{d}v}{\mathrm{d}s} &= \left[-iq\frac{\mathrm{d}}{\mathrm{d}s}\int_0^s \mathrm{d}t_1\, A_\mu(x(t_1))\dot{x}^\mu(t_1)\right] \exp\left[-iq\int_0^s \mathrm{d}t_1\, A_\mu(x(t_1))\dot{x}^\mu(t_1)\right] v(0) \\ &= -iqA_\mu(x(s))\dot{x}^\mu(s) \exp\left[-iq\int_0^s \mathrm{d}t_1\, A_\mu(x(t_1))\dot{x}^\mu(t_1)\right] v(0) \\ &= -iqA_\mu(x(s))\dot{x}^\mu(s)\, v(s). \end{aligned} \tag{222}$$

The expression satisfies, therefore, the transport equation. It also satisfies the initial value, because in $s = 0$ the integral is zero:

$$v(s)\big|_{s=0} = e^0 v(0) = v(0). \tag{223}$$

The same solution describes the transport between neighboring points. For an infinitesimal displacement from x to $x + dx$, the transport factor is

$$U(x + dx, x) = 1 - iqA_\mu(x)dx^\mu + O(dx^2). \tag{224}$$

The value $U(x + dx, x)\psi(x)$ belongs to the fiber over $x + dx$, so we can compare it to $\psi(x + dx)$. Expanding also the field to the first order, we obtain

$$\begin{aligned} \delta\psi &\coloneqq \psi(x + dx) - U(x + dx, x)\psi(x) \\ &= dx^\mu \partial_\mu \psi(x) + iqA_\mu(x)dx^\mu \psi(x) + O(dx^2) \\ &= dx^\mu D_\mu \psi(x) + O(dx^2). \end{aligned} \tag{225}$$

The covariant derivative measures the change in the field after its values are compared by transport with the connection. The field ψ can differ from a parallel-transported field. The quantity $D_{\dot{x}}\psi$ measures precisely that departure.

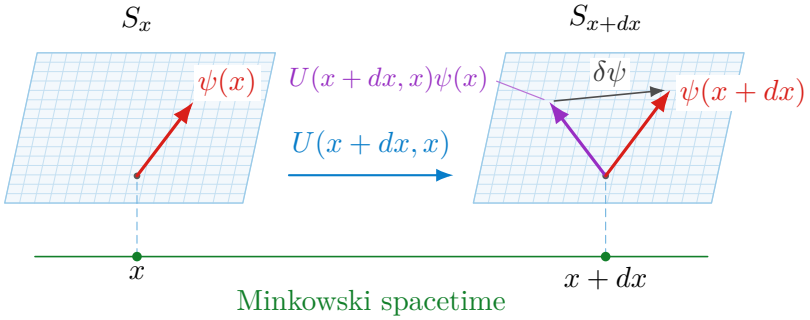


Figure 8. Transport between neighboring fibers and the covariant derivative. Transport takes $\psi(x)$ from S_x to S_{x+dx}. There we compare $U(x + dx, x)\psi(x)$ with $\psi(x + dx)$, which belong to the same fiber. The gray arrow connects the tip of the transported vector to the tip of the field vector in the arrival fiber. It represents their difference $\delta\psi = \psi(x + dx) - U(x + dx, x)\psi(x)$, which to first order is $dx^\mu D_\mu \psi(x)$. The translucent planes are schematic views into fibers $S_x \simeq \mathbb{C}^4$ and $S_{x+dx} \simeq \mathbb{C}^4$. The arrows omit some of the spinor components. Drawing them in a plane does not restrict the general field to a fixed phase plane.

At the end of the path, $s = 1$, the parameter integral is the line integral over C.

$$v(1) = \exp\left[-iq \int_0^1 \mathrm{d}t_1 \, A_\mu(x(t_1))\dot{x}^\mu(t_1)\right] v(0) = \exp\left(-iq \oint_C \mathrm{d}x^\mu \, A_\mu\right) v(0). \tag{226}$$

Solving the differential equation determines the transport factor. We call this factor $U(C)$.

Definition 25 (Transport factor and holonomy)**.** For a closed oriented loop C, define the transport factor of the internal connection $U(1)$ by

$$\boxed{U(C) \coloneqq \exp\left(-iq \oint_C \mathrm{d}x^\mu \, A_\mu\right).} \tag{227}$$

This is the **holonomy of the connection around** C. It relates the initial vector to the vector after traversing the loop.

$$v(1) = U(C)v(0). \tag{228}$$

The integral that appears is a line integral of the 1-form $A = A_\mu \, \mathrm{d}x^\mu$:

$$\oint_C \mathrm{d}x^\mu \, A_\mu = \int_0^1 \mathrm{d}t_1 \, A_\mu(x(t_1)) \frac{\mathrm{d}x^\mu(t_1)}{\mathrm{d}t_1}. \tag{229}$$

At each point we contract the potential with the tangent vector to the path. For real A_μ and q the integral is real and $U(C)$ has unit modulus. Define the accumulated angle by

$$\Theta_C := -q \oint_C A, \qquad U(C) = e^{i\Theta_C}, \qquad |U(C)| = 1. \tag{230}$$

The angle Θ_C determines the final phase. Angles differing by $2\pi n$, with $n \in \mathbb{Z}$, yield the same factor. This angle is a phase, as distinct from a variation of the field.

$$\boxed{v_{\text{final}} = U(C) v_{\text{inicial}} = e^{i\Theta_C} v_{\text{inicial}}.} \tag{231}$$

All four components are multiplied by the same phase. As an operator on the fiber, transport is $U(C)\mathbf{1}_4$. It preserves the norm and records the cumulative effect of the connection along the entire path rather than the curvature at a single point.

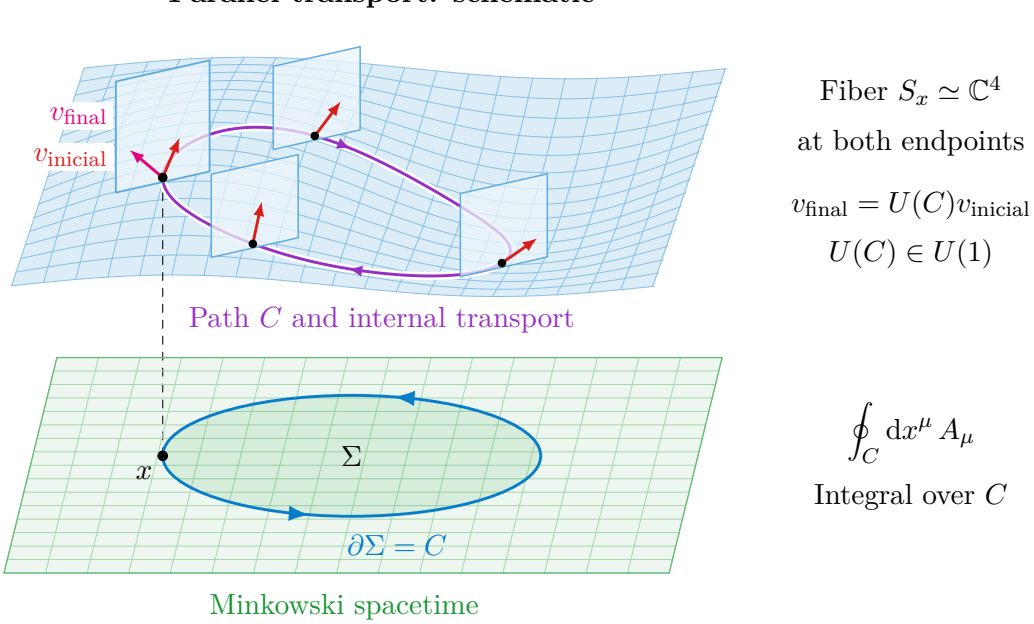


Figure 9. Holonomy around a closed loop. The loop C bounds Σ in Minkowski spacetime. The upper panel depicts transport along the same loop, with small windows showing phase planes of the fibers. In fiber S_x, the initial (red) and final (hot pink) arrows have the same origin and length: $v_{\text{final}} = e^{i\Theta_C} v_{\text{inicial}}$. The upper curve depicts the base path. The horizontal lift in the total space can fail to close. The mesh illustrates the connection schematically, and the intermediate phases are illustrative.

The surface Σ lies in Minkowski spacetime and has the closed base loop C as its boundary. If A is smooth and defined on a region containing the surface, Stokes' theorem with compatible orientations gives

$$\partial\Sigma = C, \qquad \oint_C A = \int_\Sigma \mathrm{d}A = \int_\Sigma F = \frac{1}{2} \int_\Sigma F_{\mu\nu} \, \mathrm{d}\Sigma^{\mu\nu}. \tag{232}$$

This lets us compute the holonomy from the curvature flux.

$$U(C) = \exp\left(-iq \int_\Sigma F\right) = \exp\left(-\frac{iq}{2} \int_\Sigma F_{\mu\nu} \, \mathrm{d}\Sigma^{\mu\nu}\right). \tag{233}$$

Stokes' theorem applies to the closed base loop even when the lift is open.

When does exactly the same vector return? The exponential $e^{i\Theta_C}$ equals 1 precisely when the angle is an integer multiple of 2π.

$$\boxed{U(C) = 1 \quad \Longleftrightarrow \quad \Theta_C = 2\pi n, \qquad n \in \mathbb{Z}.} \tag{234}$$

When the above Stokes conditions are met, this is equivalent to

$$-q \int_\Sigma F = 2\pi n, \qquad n \in \mathbb{Z}. \tag{235}$$

If $U(C) = 1$, each component is multiplied by 1 and therefore,

$$v_{\text{final}} = 1\, v_{\text{inicial}} = v_{\text{inicial}}. \tag{236}$$

This describes the final result. The vector may have changed along the path. For a nonzero initial vector, $U(C) \neq 1$ gives a different final vector in the same fiber, with the same norm and a different phase. The zero vector remains zero for any holonomy.

If $F = 0$ on Σ, the flux is zero and $U(C) = 1$. The same can happen with nonzero curvature. Contributions to the flux can cancel, or the total flux can produce an angle of $2\pi n$. Thus $U(C) = 1$ **for a particular loop does not imply flatness**. For a global potential of the form $A = \mathrm{d}\beta/q$, the integral over any closed loop is zero and the holonomy is 1. For a more general flat connection, this holds for loops bounding a surface in the region under consideration. Topology can prevent a loop from bounding such a surface.

Example 12 (The Euclidean plane in polar coordinates)**.** A change of coordinates leaves curvature unchanged. In polar coordinates $\mathrm{d}s^2 = \mathrm{d}r^2 + r^2\mathrm{d}\phi^2$, the geometric connection has nonzero coefficients even though the Riemann tensor of the Euclidean plane vanishes.[9] This geometric connection and the internal connection $U(1)$ are distinct.

□

9 The Full Lagrangian and Equations of Motion

The connection and its curvature are already defined. To give dynamics to the electromagnetic field, we now introduce its Lagrangian density.

We choose Maxwell's Lagrangian density as

$$\boxed{\mathcal{L}_{\text{EM}} \coloneqq -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}.} \tag{237}$$

Definition 26 (Potentials and electromagnetic fields)**.** Write the four-potential as $A^\mu = (\varphi, \mathbf{A})$, with $\varphi \coloneqq A^0$ and $\mathbf{A} \coloneqq (A^1, A^2, A^3)$. Here φ is the **electric scalar potential** and $\mathbf{A}$ the **magnetic vector potential**. We define the electric and magnetic fields in terms of these potentials by

$$\boxed{\mathbf{E} \coloneqq -\boldsymbol{\nabla}\varphi - \partial_t \mathbf{A}, \qquad \mathbf{B} \coloneqq \boldsymbol{\nabla} \times \mathbf{A}.} \tag{238}$$

[9] We are referring to the full Riemann tensor. The Ricci tensor and scalar curvature are obtained by contraction.

$$R_{\mu\nu} \coloneqq R^{\rho}{}_{\mu\rho\nu}, \qquad R \coloneqq g^{\mu\nu} R_{\mu\nu}. \tag{XIII}$$

If the Riemann tensor vanishes, so do the Ricci tensor and scalar curvature. The converse need not hold. The condition $R = 0$ does not force the Riemann tensor to vanish. Even the Ricci tensor may vanish while the Riemann tensor is nonzero, as in Schwarzschild spacetime outside the source. All three vanish in the Euclidean plane of the example. The nonzero connection coefficients in polar coordinates come from the choice of coordinates.

We check how these fields appear in the curvature. By lowering indices with the metric fixed, we have $A_0 = \varphi$ and $A_i = -A^i$. By replacing them in the definition of $F_{\mu\nu}$, we obtain

$$F_{0i} = \partial_t(-A^i) - \partial_i\varphi = -\partial_t A^i - \partial_i\varphi = E^i, \tag{239}$$

$$F_{ij} = \partial_i(-A^j) - \partial_j(-A^i) = -\partial_i A^j + \partial_j A^i = -\epsilon_{ijk}B^k. \tag{240}$$

Here $i, j, k \in \{1, 2, 3\}$ and ϵ_{ijk} is the totally antisymmetric symbol, with $\epsilon_{123} = 1$ and $(\boldsymbol{\nabla} \times \mathbf{A})^k = \epsilon_{kij}\partial_i A^j$. The mixed time–space components of F contain the electric field, and its spatial components contain the magnetic field.

Therefore $F_{\mu\nu}F^{\mu\nu} = 2(\mathbf{B}^2 - \mathbf{E}^2)$ and $\mathcal{L}_{\text{EM}} = (\mathbf{E}^2 - \mathbf{B}^2)/2$. The density of electromagnetic energy is $(\mathbf{E}^2 + \mathbf{B}^2)/2$.

Remark 2 (Why we chose Maxwell's term)**.** Gauge invariance allowed us to introduce the potential A_μ into the covariant derivative and determine how it should be transformed. But that still does not establish how the electromagnetic field itself evolves: we need to add a term describing its dynamics.

Maxwell's term is invariant because it is constructed with $F_{\mu\nu}$, which does not change under gauge transformations. However, **Gauge invariance, by itself, does not require the choice of only that term**. For example, an expression proportional to the square of $F_{\mu\nu}F^{\mu\nu}$ would also be gauge-invariant.

We choose Maxwell's term to obtain the usual electromagnetic dynamics. It gives a local Lorentz-invariant Lagrangian quadratic in $F_{\mu\nu}$ that produces Maxwell's equations. This choice adds a physical criterion to the symmetry requirement.

Adding the Maxwell term to the matter density from Section 6.2 gives

$$\boxed{\begin{aligned}\mathcal{L}_{\text{QED}} &\coloneqq -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\gamma^\mu D_\mu - m)\psi \\ &= -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi - q\bar{\psi}\gamma^\mu\psi A_\mu.\end{aligned}} \tag{241}$$

In the interaction term we have written the current explicitly using its earlier definition $j^\mu \coloneqq \bar{\psi}\gamma^\mu\psi$. The part containing ψ and $\bar{\psi}$ is called *fermionic*.[10] We can also read the decomposition term by term.

$$\mathcal{L}_{\text{QED}} = \underbrace{i\bar{\psi}\gamma^\mu\partial_\mu\psi}_{\text{fermionic kinetic term}} \underbrace{-m\bar{\psi}\psi}_{\text{mass}} \underbrace{-q\bar{\psi}\gamma^\mu\psi A_\mu}_{\text{interaction}} \underbrace{-\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu}}_{\text{electromagnetic kinetic term}}. \tag{242}$$

Each term is a Lorentz scalar. The form with D_μ makes the gauge invariance of the density manifest. The expanded expression separates free matter, interaction, and electromagnetic dynamics. The complete action is $S_{\text{QED}}[\psi, A] \coloneqq \int_M \mathrm{d}^4x\, \mathcal{L}_{\text{QED}}$.

We now vary the electromagnetic potentials A_μ in the **full QED action**, keeping the field ψ and hence its adjoint $\bar{\psi}$ fixed. Take a real, smooth, compactly supported variation δA.

$$A_\mu \longmapsto A_\mu + \varepsilon\delta A_\mu, \qquad \psi \longmapsto \psi, \qquad \bar{\psi} \longmapsto \bar{\psi}. \tag{243}$$

The subscript A indicates which variable we are varying:

$$\delta_A S_{\text{QED}}[\psi, A; \delta A] \coloneqq \frac{\mathrm{d}}{\mathrm{d}\varepsilon} S_{\text{QED}}[\psi, A + \varepsilon\delta A]\Big|_{\varepsilon=0}. \tag{244}$$

[10] Here we are discussing the Dirac field ψ. The term "fermionic" refers to the particles described by its quantization: spin-1/2 fermions such as the electron and their antiparticles such as the positron. Fermions are particles with half-integer spin that obey Fermi–Dirac statistics. The field ψ describes these particles. In this article, "fermionic term" means the term associated with the Dirac field.

We use the whole Lagrangian, not just Maxwell. Dirac's free term does not depend on A_μ and its variation is zero. The other two terms do contribute; with $J^\nu = q\bar{\psi}\gamma^\nu\psi$ fixed in this variation,

$$\delta_A F_{\mu\nu} = \partial_\mu \delta A_\nu - \partial_\nu \delta A_\mu, \tag{245}$$

$$\delta_A \mathcal{L}_{\text{QED}} = -\frac{1}{2} F^{\mu\nu} \delta_A F_{\mu\nu} - J^\nu \delta A_\nu \tag{246}$$

$$= -F^{\mu\nu} \partial_\mu \delta A_\nu - J^\nu \delta A_\nu. \tag{247}$$

Integrate over the whole manifold and integrate the first term by parts. Compact support of δA makes the boundary term vanish.

$$\delta_A S_{\text{QED}}[\psi, A; \delta A] = \int_M \mathrm{d}^4x \, [\partial_\mu F^{\mu\nu} - J^\nu] \delta A_\nu. \tag{248}$$

Stationarity of the complete action with respect to potentials A_ν, maintaining ψ and $\bar{\psi}$ fixed, requires

$$\boxed{\frac{\delta S_{\text{QED}}}{\delta A_\nu(x)} = 0 \quad \Longleftrightarrow \quad \partial_\mu F^{\mu\nu}(x) = J^\nu(x).} \tag{249}$$

Varying the potentials in the full action produces Maxwell's equations with the Dirac current as the source. The source $J^\nu = q\bar{\psi}\gamma^\nu\psi$ comes from the interaction term already present in the Lagrangian.

Varying A_μ gives its sourced equation. Varying the Dirac field, as we do below, gives its equation coupled to A_μ. **The same action describes both fields and their interaction**. The potential enters the Dirac equation, and the Dirac field supplies the charge and current in Maxwell's equations. Holding ψ fixed during this variation gives the equation for A_μ. A solution of the full system must satisfy both field equations.

Applying ∂_ν to this equation gives a vanishing contraction by symmetry of $\partial_\nu\partial_\mu$ and antisymmetry of $F^{\mu\nu}$. We obtain the compatibility condition

$$\partial_\nu J^\nu = \partial_\nu \partial_\mu F^{\mu\nu} = 0. \tag{250}$$

On the other hand, the definition $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ implies an identity other than the conservation of the current. To obtain it, we calculate

$$\partial_\lambda F_{\mu\nu} = \partial_\lambda \partial_\mu A_\nu - \partial_\lambda \partial_\nu A_\mu, \tag{251}$$

$$\partial_\mu F_{\nu\lambda} = \partial_\mu \partial_\nu A_\lambda - \partial_\mu \partial_\lambda A_\nu, \tag{252}$$

$$\partial_\nu F_{\lambda\mu} = \partial_\nu \partial_\lambda A_\mu - \partial_\nu \partial_\mu A_\lambda. \tag{253}$$

We add these expressions and group by the component of the potential on which derivatives act:

$$\begin{aligned} &\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} \\ &\quad = \underbrace{(\partial_\lambda\partial_\mu - \partial_\mu\partial_\lambda)A_\nu}_{0} + \underbrace{(\partial_\mu\partial_\nu - \partial_\nu\partial_\mu)A_\lambda}_{0} + \underbrace{(\partial_\nu\partial_\lambda - \partial_\lambda\partial_\nu)A_\mu}_{0}. \end{aligned} \tag{254}$$

The partial derivatives commute for smooth potentials, yielding the **Bianchi identity**.

$$\boxed{\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} = 0.} \tag{255}$$

The electric current $J^\mu = q\bar{\psi}\gamma^\mu\psi$ has already been defined in section 6.2 and appears here as a source in Maxwell. We now identify its temporal and spatial components with charge density and electric current density:

$$\rho \coloneqq J^0 = q\bar{\psi}\gamma^0\psi = q\psi^\dagger\psi, \qquad \mathbf{J} \coloneqq (J^1, J^2, J^3), \qquad J^i \coloneqq q\bar{\psi}\gamma^i\psi. \tag{256}$$

The function ρ is electric charge density, and $\mathbf{J}$ is electric current density. The latter is distinct from the magnetic field $\mathbf{B}$ and appears as a source in the Ampère–Maxwell law. The temporal equality uses $\bar{\psi} = \psi^\dagger \gamma^0$ and $(\gamma^0)^2 = 1$. These expressions and the equations in this section are in natural units.

Start with the equation of motion obtained by varying the full action. First work in terms of the potentials. We identify the electric and magnetic fields afterward.

$$\partial_\mu F^{\mu\nu} = J^\nu, \qquad F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu. \tag{257}$$

With $A^\mu = (\varphi, \mathbf{A})$ and $g = \mathrm{diag}(1, -1, -1, -1)$, we have $\partial^0 = \partial_t$ and $\partial^i = -\partial_i$.

$$F^{i0} = -\partial_i \varphi - \partial_t A^i, \quad F^{0i} = \partial_t A^i + \partial_i \varphi, \quad F^{ji} = -\partial_j A^i + \partial_i A^j. \tag{258}$$

For $\nu = 0$, substitute directly and sum over the spatial indices.

$$\rho = J^0 = \partial_\mu F^{\mu 0} = \sum_{i=1}^{3} \partial_i(-\partial_i \varphi - \partial_t A^i) \tag{259}$$

$$= -\boldsymbol{\nabla}^2 \varphi - \partial_t(\boldsymbol{\nabla} \cdot \mathbf{A}) \tag{260}$$

$$= \boldsymbol{\nabla} \cdot \underbrace{\left(-\boldsymbol{\nabla}\varphi - \partial_t \mathbf{A}\right)}_{\mathbf{E}}. \tag{261}$$

The indicated group is $\mathbf{E}$. We obtain **Gauss's law for electricity**.

$$\boxed{\boldsymbol{\nabla} \cdot \mathbf{E} = \rho.} \tag{262}$$

For $\nu = i$, the same substitution gives, component by component,

$$J^i = \partial_t F^{0i} + \sum_{j=1}^{3} \partial_j F^{ji} \tag{263}$$

$$= \partial_t(\partial_t A^i + \partial_i \varphi) + \sum_{j=1}^{3} \partial_j(-\partial_j A^i + \partial_i A^j) \tag{264}$$

$$= \partial_t^2 A^i + \partial_i \partial_t \varphi - \boldsymbol{\nabla}^2 A^i + \partial_i(\boldsymbol{\nabla} \cdot \mathbf{A}). \tag{265}$$

We gather the three components into a vector equation and use the identity $\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \mathbf{A}) = \boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \mathbf{A}) - \boldsymbol{\nabla}^2 \mathbf{A}$:

$$\mathbf{J} = \partial_t^2 \mathbf{A} + \boldsymbol{\nabla}\partial_t \varphi - \boldsymbol{\nabla}^2 \mathbf{A} + \boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \mathbf{A}) \tag{266}$$

$$= \boldsymbol{\nabla} \times \underbrace{(\boldsymbol{\nabla} \times \mathbf{A})}_{\mathbf{B}} - \partial_t \underbrace{\left(-\boldsymbol{\nabla}\varphi - \partial_t \mathbf{A}\right)}_{\mathbf{E}}. \tag{267}$$

Recognizing the indicated groups as $\mathbf{B}$ and $\mathbf{E}$ gives the **Ampère–Maxwell law**.

$$\boxed{\boldsymbol{\nabla} \times \mathbf{B} - \partial_t \mathbf{E} = \mathbf{J}.} \tag{268}$$

We have not imposed that J^μ vanish, so the source is still $J^\mu = q\bar{\psi}\gamma^\mu\psi$. The calculation uses only the definition of F and commutation of derivatives of smooth potentials, and applies in any gauge.

The source $J^\mu = q\bar{\psi}\gamma^\mu\psi$ is therefore the same electric four-current that appears in Maxwell's equations. Its components ρ and $\mathbf{J}$ satisfy the Gauss and Ampère–Maxwell laws just derived.

The other two laws follow from the Bianchi identity, which we recall before expanding its components.

$$\boxed{\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} = 0.} \tag{269}$$

For indices $(\lambda, \mu, \nu) = (1, 2, 3)$, we replace only the definition $F_{ij} = \partial_i A_j - \partial_j A_i$, with $A_i = -A^i$:

$$0 = \partial_1 F_{23} + \partial_2 F_{31} + \partial_3 F_{12} \tag{270}$$
$$= \partial_1(-\partial_2 A^3 + \partial_3 A^2) + \partial_2(-\partial_3 A^1 + \partial_1 A^3) + \partial_3(-\partial_1 A^2 + \partial_2 A^1) \tag{271}$$
$$= -\partial_1\partial_2 A^3 + \partial_1\partial_3 A^2 - \partial_2\partial_3 A^1 + \partial_2\partial_1 A^3 - \partial_3\partial_1 A^2 + \partial_3\partial_2 A^1. \tag{272}$$

The six terms cancel in pairs. At the end we recognize the components of the curl $\mathbf{B} \coloneqq \boldsymbol{\nabla} \times \mathbf{A}$.

$$\partial_2 A^3 - \partial_3 A^2 = B^1, \qquad \partial_3 A^1 - \partial_1 A^3 = B^2, \qquad \partial_1 A^2 - \partial_2 A^1 = B^3. \tag{273}$$

Therefore, the same identity is written

$$0 = -\partial_1 B^1 - \partial_2 B^2 - \partial_3 B^3 = -\boldsymbol{\nabla} \cdot \mathbf{B}. \tag{274}$$

This gives **Gauss's law for magnetism**, expressing the absence of magnetic charges.

$$\boxed{\boldsymbol{\nabla} \cdot \mathbf{B} = 0.} \tag{275}$$

To obtain Faraday's law, we take the indices $(0, 1, 2)$ and replace F again in terms of potentials:

$$0 = \partial_t F_{12} + \partial_1 F_{20} + \partial_2 F_{01} \tag{276}$$
$$= \partial_t(-\partial_1 A^2 + \partial_2 A^1) + \partial_1(\partial_2 \varphi + \partial_t A^2) + \partial_2(-\partial_t A^1 - \partial_1 \varphi) \tag{277}$$
$$= -\partial_t\partial_1 A^2 + \partial_t\partial_2 A^1 + \partial_1\partial_2 \varphi + \partial_1\partial_t A^2 - \partial_2\partial_t A^1 - \partial_2\partial_1 \varphi. \tag{278}$$

For smooth potentials the derivatives commute, so the terms cancel in pairs. Regrouping the same expression using the earlier definitions gives the substitutions shown under the terms.

$$0 = \partial_t \underbrace{(-\partial_1 A^2 + \partial_2 A^1)}_{-B^3} + \partial_1 \underbrace{(\partial_2 \varphi + \partial_t A^2)}_{-E^2} + \partial_2 \underbrace{(-\partial_t A^1 - \partial_1 \varphi)}_{E^1}. \tag{279}$$

Therefore, the same account is written

$$0 = -\partial_t B^3 - \partial_1 E^2 + \partial_2 E^1 \quad \Longleftrightarrow \quad \partial_t B^3 + (\boldsymbol{\nabla} \times \mathbf{E})^3 = 0. \tag{280}$$

The pairs $(2, 3)$ and $(3, 1)$ give the other two components. Combining them gives

$$\boxed{\boldsymbol{\nabla} \times \mathbf{E} + \partial_t \mathbf{B} = 0.} \tag{281}$$

This is **Faraday's law**. The two sourced laws follow from variation of the action and the two homogeneous laws from the Bianchi identity. Together they are Maxwell's four equations in our conventions.

$$\boxed{\begin{aligned} &\boldsymbol{\nabla} \cdot \mathbf{E} = \rho, \quad \boldsymbol{\nabla} \times \mathbf{B} - \partial_t \mathbf{E} = \mathbf{J}, \\ &\boldsymbol{\nabla} \cdot \mathbf{B} = 0, \quad \boldsymbol{\nabla} \times \mathbf{E} + \partial_t \mathbf{B} = 0, \end{aligned}} \tag{282}$$

with $J^\mu = (\rho, \mathbf{J})$.

9.1 The Coupled Dirac Equations

Variation with respect to $\bar{\psi}$ and ψ produces

$$(i\gamma^\mu D_\mu - m)\psi = 0, \qquad i(\bar{\psi}\overleftarrow{D}_\mu)\gamma^\mu + m\bar{\psi} = 0, \tag{283}$$

with $\bar{\psi}\overleftarrow{D}_\mu \coloneqq \partial_\mu \bar{\psi} - iqA_\mu \bar{\psi}$. Multiplying these equations by the corresponding fields gives cancellation of the mass and potential terms in $\partial_\mu j^\mu$. The ordinary current j^μ is conserved and gauge-invariant.

10 Background Field and Dynamical Field

10.1 Hamiltonian form of the Dirac equation

Start with the coupled Dirac equation obtained in section 9.1.

$$(i\gamma^\mu D_\mu - m)\psi = 0. \tag{284}$$

We isolate the time derivative to identify the operator governing the evolution of ψ. First rewrite the operator acting on the field by substituting $D_\mu = \partial_\mu + iqA_\mu$ and distributing $i\gamma^\mu$. Since $i^2 = -1$, we obtain

$$i\gamma^\mu D_\mu - m = i\gamma^\mu(\partial_\mu + iqA_\mu) - m = i\gamma^\mu\partial_\mu - q\gamma^\mu A_\mu - m. \tag{285}$$

We now separate the temporal index $\mu = 0$ from the spatial indices $i = 1, 2, 3$. We use $A^\mu = (\varphi, \mathbf{A})$, so that $A_0 = \varphi$ and $A_i = -A^i$:

$$i\gamma^\mu\partial_\mu - q\gamma^\mu A_\mu - m = i\gamma^0\partial_t + i\gamma^i\partial_i - q\gamma^0\varphi + q\gamma^i A^i - m. \tag{286}$$

We replace this expanded operator in the Dirac equation:

$$[i\gamma^0\partial_t + i\gamma^i\partial_i - q\gamma^0\varphi + q\gamma^i A^i - m]\psi = 0. \tag{287}$$

We define $\alpha^i \coloneqq \gamma^0\gamma^i$ to shorten the spatial terms. We multiply by γ^0, use $(\gamma^0)^2 = \mathbf{1}_4$ and clear the temporal derivative:

$$i\partial_t\psi = [\boldsymbol{\alpha}\cdot(-i\boldsymbol{\nabla} - q\mathbf{A}) + \gamma^0 m + q\varphi]\psi. \tag{288}$$

Define the Dirac Hamiltonian in the potential A by

$$\boxed{h[A] \coloneqq \boldsymbol{\alpha}\cdot(-i\boldsymbol{\nabla} - q\mathbf{A}) + \gamma^0 m + q\varphi, \qquad i\partial_t\psi = h[A]\psi.} \tag{289}$$

The matter Hamiltonian density is $\psi^\dagger h[A]\psi$. It includes both the vector and scalar potentials and reduces to the free operator when $A = 0$. The expression also holds in the coupled system, where A must satisfy Maxwell's equations as well and is not independent input.

10.2 External Potential, Backreaction, and Radiation

In the full system ψ and A_μ are unknowns that jointly satisfy the Dirac and Maxwell equations. The potential enters the Dirac equation, and the current $J^\mu = q\bar{\psi}\gamma^\mu\psi$ is a source of the electromagnetic field. We call the effect of the Dirac field's current on the electromagnetic field $F_{\mu\nu}$, which comprises $\mathbf{E}$ and $\mathbf{B}$, **backreaction**. The current acts as a source in the equation for the potential A_μ, and the resulting potential in turn enters the evolution of ψ.

If the potential $A^{\text{ext}}_\mu(x)$ is given, we solve only the Dirac equation in that potential, or equivalently $i\partial_t\psi = h[A^{\text{ext}}]\psi$, with the corresponding initial data. The potential need not satisfy Maxwell's equations with the current of that same ψ, since we omit backreaction. This approximation can describe, for example, the response to a field from external sources when the system under study changes it negligibly. The given potential may depend on time.

We already held A fixed while varying ψ to derive the Dirac equation. The additional choice in this approximation is **to leave the equation for A unsolved**.

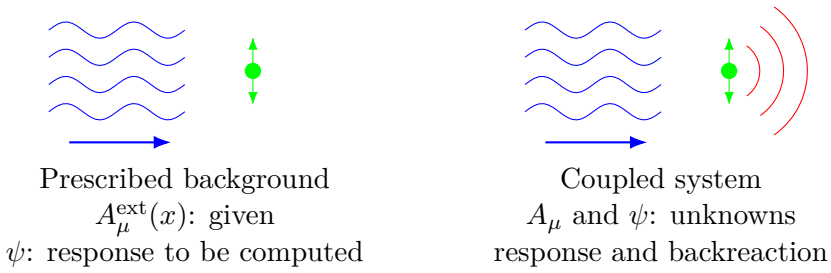


Figure 10. Prescribed external potential and coupled system. The left panel prescribes the external potential and calculates the response of the Dirac field while omitting the change in the potential due to its current. The given potential can vary in space and time. In the right panel the fields are solved jointly. The outgoing wave depicts the electromagnetic field produced by the source. Omitting backreaction in the background approximation does not imply that an accelerated charge cannot radiate.

Remark 3 (Interpretation of waves)**.** The drawing gives a classical interpretation of the coupling. Photon emission in the quantum theory requires states, initial conditions, and an amplitude calculation. A drawn trajectory cannot establish emission. The background approximation is useful when the change to the external field produced by the system under study is negligible.

11 Examples and Physical Interpretation

Example 13 (A Pure Gauge Potential)**.** We start from the representative with $A_\mu = 0$. Then $F_{\mu\nu} = 0$: there is no electric or magnetic field, and $D_\mu = \partial_\mu$. A local gauge transformation simultaneously changes the field and potential:

$$\psi' = e^{iq\alpha}\psi, \qquad A'_\mu = -\partial_\mu\alpha, \qquad D'_\mu = \partial_\mu - iq\partial_\mu\alpha. \tag{290}$$

The transformed potential can be nonzero, but it is **pure gauge**. For smooth α,

$$F'_{\mu\nu} = \partial_\mu A'_\nu - \partial_\nu A'_\mu = -\partial_\mu\partial_\nu\alpha + \partial_\nu\partial_\mu\alpha = 0. \tag{291}$$

Thus the electric and magnetic fields also vanish in this representative. The prime distinguishes the representatives. The field ψ' is determined by ψ and α, so it is not an additional unknown. We can write the gauge transformation of the derivative entirely in terms of ψ.

$$\begin{aligned} D'_\mu\psi' &= (\partial_\mu - iq\partial_\mu\alpha)(e^{iq\alpha}\psi) \\ &= iq(\partial_\mu\alpha)e^{iq\alpha}\psi + e^{iq\alpha}\partial_\mu\psi - iq(\partial_\mu\alpha)e^{iq\alpha}\psi \\ &= e^{iq\alpha}\partial_\mu\psi. \end{aligned} \tag{292}$$

The term from differentiating the phase cancels exactly against the connection term. In particular,

$$(i\gamma^\mu D'_\mu - m)(e^{iq\alpha}\psi) = e^{iq\alpha}(i\gamma^\mu\partial_\mu - m)\psi. \tag{293}$$

If ψ satisfies the free Dirac equation, $e^{iq\alpha}\psi$ satisfies the Dirac equation with the pure-gauge potential A'. They describe the same physical situation when α is smooth and single-valued on the region considered. Here we solve the Dirac equation in a prescribed pure-gauge background. To solve the full Dirac–Maxwell system, $F = 0$ would also require $J^\mu = 0$ by Maxwell's equation. For a classical complex field of nonzero charge this forces $\psi = 0$, since $J^0 = q\psi^\dagger\psi$. The transformed field obeys the equation with the transformed connection, which can differ from the ordinary derivative.

□

Remark 4 (Electron charge and sign of the interaction)**.** Here q is the electric charge associated with the Dirac field coupling.[11] For the electron we take $q = -e$. Our conventions give

$$D_\mu = \partial_\mu - ieA_\mu, \qquad \mathcal{L}_{\text{int}} = +e\bar{\psi}\gamma^\mu\psi A_\mu, \qquad J^\mu = -e\bar{\psi}\gamma^\mu\psi. \tag{294}$$

In particular, the charge density of the Dirac field is $\rho = -e\psi^\dagger\psi$. In the next example we keep the charge $-e$ and prescribe a classical point source in place of solving for ψ.

Example 14 (Point charge at rest and the Coulomb field)**.** We want to model a point charge $-e$ at rest at the origin and find the electric potential it produces. To represent it, use the charge density

$$\boxed{\rho(\mathbf{x}) \coloneqq -e\delta^{(3)}(\mathbf{x}).} \tag{295}$$

Here $\delta^{(3)}$ is the three-dimensional Dirac delta. **This places the entire charge $-e$ at the origin**. It is the source we impose to determine the potential. The charge is at rest, so the spatial current is zero.

We seek an electrostatic solution with no external magnetic field and vanishing potential at infinity. In the notation of the theory, we take the potentials

$$A^0 = \varphi(\mathbf{x}), \qquad A^1 = A^2 = A^3 = 0, \qquad \mathbf{E} = -\boldsymbol{\nabla}\varphi, \qquad \mathbf{B} = 0. \tag{296}$$

The components A^1, A^2, A^3 form the **electromagnetic vector potential $\mathbf{A}$**, also called the magnetic vector potential because $\mathbf{B} = \boldsymbol{\nabla} \times \mathbf{A}$. In this electrostatic case with no magnetic field, we switch it off by choosing $\mathbf{A} = 0$. This gauge choice is compatible with the problem. The condition $\mathbf{B} = 0$ also admits representatives with a vector potential other than $\mathbf{A} = 0$. Thus φ is the time component of the electromagnetic potential and the potential function for the electric field. With a prescribed source we solve the Maxwell sector without replacing the Dirac field ψ by a delta.

Use the time component of Maxwell's equations obtained by varying the potentials in the QED action, namely Gauss's law. For this example we use SI units, with e in coulombs, distances in meters, and the vacuum permittivity ε_0 in farads per meter. We convert the electrostatic equation to those units before substituting charge values in coulombs. The identification $A^0 = \varphi$ above uses $c = 1$. With time coordinate $x^0 = ct$, the usual SI four-potential is $A^\mu_{\text{SI}} = (\varphi/c, \mathbf{A}_{\text{SI}})$. Substituting the source and potentials gives

$$\boldsymbol{\nabla} \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0} \quad \Longrightarrow \quad -\nabla^2\varphi = \frac{\rho}{\varepsilon_0} = -\frac{e}{\varepsilon_0}\delta^{(3)}(\mathbf{x}). \tag{297}$$

We solve this equation explicitly. The point source and the conditions are spherically symmetric, so we seek $\varphi = \varphi(r)$ with $r \coloneqq |\mathbf{x}|$. Outside the origin, $\rho = 0$. Using the Laplacian of a radial function, Poisson's equation becomes

$$\nabla^2\varphi = \frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\left(r^2\frac{\mathrm{d}\varphi}{\mathrm{d}r}\right) = 0, \qquad r > 0. \tag{298}$$

Multiply by r^2 and integrate once. Solve for the derivative and integrate again.

$$\frac{\mathrm{d}}{\mathrm{d}r}(r^2\varphi') = 0 \quad \Longrightarrow \quad r^2\varphi' = C \quad \Longrightarrow \quad \varphi' = \frac{C}{r^2} \quad \Longrightarrow \quad \varphi(r) = -\frac{C}{r} + B. \tag{299}$$

[11] Electric charge determines how a particle produces and responds to the electromagnetic field. The electron has charge $-e$ and the positron $+e$, with $e > 0$. The SI unit is the coulomb, symbol C. One coulomb is the charge carried by a current of one ampere for one second, $1\,\text{C} = 1\,\text{A}\,\text{s}$. The elementary charge has the exact value $e = 1.602176634 \times 10^{-19}\,\text{C}$. See the BIPM definition of the ampere. In the natural units and electromagnetic normalization of this text, q is a dimensionless coupling. The value in coulombs must be converted before using it in our formulas. Gauge symmetry determines the form of the coupling and leaves the value of the charge undetermined.

Here C and B are constants of integration. The condition $\varphi(r) \to 0$ as $r \to \infty$ fixes $B = 0$, giving $\varphi(r) = -C/r$. To determine C, **return to the full Poisson equation, including the point source at the origin**.

$$-\nabla^2 \varphi = -\frac{e}{\varepsilon_0}\delta^{(3)}(\mathbf{x}). \tag{300}$$

We substitute directly $\varphi = -C/r$ in that equation. By linearity of Laplacian, we obtain

$$-\nabla^2\left(-\frac{C}{r}\right) = C\nabla^2\frac{1}{r} = -\frac{e}{\varepsilon_0}\delta^{(3)}(\mathbf{x}). \tag{301}$$

To solve for C, integrate this equality over a ball $B_R \coloneqq \{\mathbf{x} : |\mathbf{x}| < R\}$ containing the origin, with $R > 0$. The integral of the delta over the ball is one, so

$$C\int_{B_R} \mathrm{d}^3x\, \nabla^2\frac{1}{r} = -\frac{e}{\varepsilon_0}\int_{B_R} \mathrm{d}^3x\, \delta^{(3)}(\mathbf{x}) = -\frac{e}{\varepsilon_0}. \tag{302}$$

Continue the calculation on the boundary of the ball using Gauss's theorem.[12] On the sphere $r = R$, use spherical coordinates

$$\begin{cases} x = R\sin\theta\cos\beta, \\ y = R\sin\theta\sin\beta, \qquad 0 \le \theta \le \pi, \quad 0 \le \beta < 2\pi. \\ z = R\cos\theta, \end{cases} \tag{303}$$

Here β is the azimuthal angle, distinct from the potential φ. To calculate the gradient, note that $1/r$ depends only on the radial coordinate, so its angular derivatives vanish. In spherical coordinates only the radial term remains.

$$\boldsymbol{\nabla}\frac{1}{r} = \mathbf{e}_r\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{1}{r}\right) = -\frac{\mathbf{e}_r}{r^2}, \qquad r > 0. \tag{304}$$

On the sphere $r = R$, the outward normal is $\mathbf{n} = \mathbf{e}_r$ and has length one, so $\boldsymbol{\nabla}(1/r)\cdot\mathbf{n} = -1/R^2$. Using the area element $\mathrm{d}S = R^2\sin\theta\,\mathrm{d}\theta\,\mathrm{d}\beta$, substitute these expressions directly into the integral.

$$\begin{aligned} -\frac{e}{\varepsilon_0} &= C\int_{B_R} \mathrm{d}^3x\, \nabla^2\frac{1}{r} \\ &= C\int_{\partial B_R} \mathrm{d}S\, \boldsymbol{\nabla}\frac{1}{r}\cdot\mathbf{n} \\ &= C\int_0^{2\pi}\mathrm{d}\beta\int_0^{\pi}\mathrm{d}\theta\, R^2\sin\theta\left(-\frac{1}{R^2}\right) \\ &= -C\left(\int_0^{2\pi}\mathrm{d}\beta\right)\left(\int_0^{\pi}\mathrm{d}\theta\,\sin\theta\right) = -C(2\pi)(2) = -4\pi C. \end{aligned} \tag{305}$$

Solving this equality gives

$$\boxed{C = \frac{e}{4\pi\varepsilon_0}.} \tag{306}$$

[12] **Gauss's theorem**, also called the divergence theorem, holds for a bounded domain $\Omega \subset \mathbb{R}^3$ with piecewise smooth boundary and a vector field $\mathbf{F}$ of class C^1 in a neighborhood of $\overline{\Omega}$.

$$\int_\Omega \mathrm{d}V\, \boldsymbol{\nabla}\cdot\mathbf{F} = \int_{\partial\Omega}\mathrm{d}S\,\mathbf{F}\cdot\mathbf{n}. \tag{XIV}$$

Here $\mathbf{n}$ is the outward unit normal. It has length one and is perpendicular to the tangent plane to the boundary at its regular points. The surface integral gives the net outward flux. The dot product $\mathbf{F}\cdot\mathbf{n}$ selects the component perpendicular to the surface, positive when it points outward and negative when it points inward. In our calculation $\mathbf{F} = \boldsymbol{\nabla}(1/r)$ points inward, so its flux is -4π. The field is singular at the origin, and the equality requires the distributional interpretation described in the text.

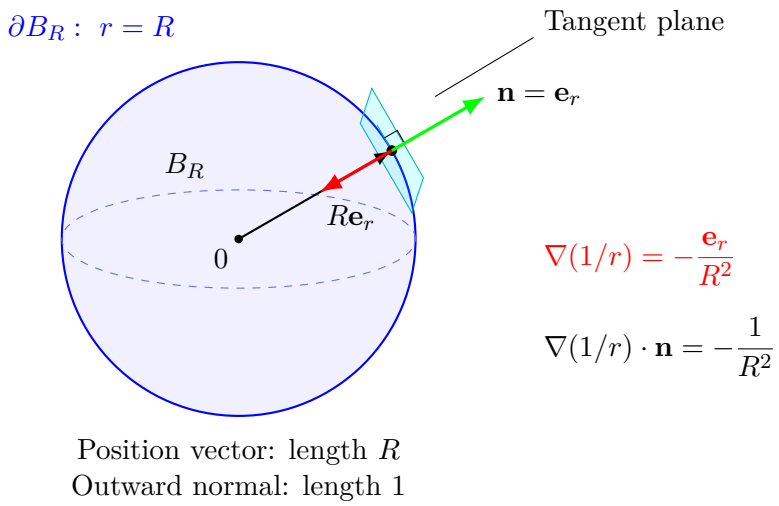


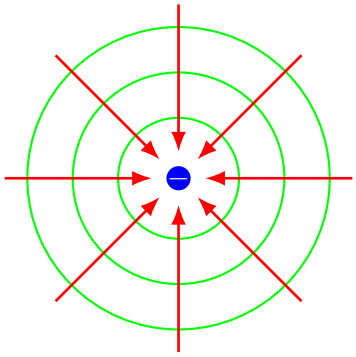
Figure 11. Outward normal and flux through a sphere. The ball B_R and its boundary sphere ∂B_R are drawn schematically. At the marked point, the position vector is $R\mathbf{e}_r$ and the outward unit normal is $\mathbf{e}_r$. The cyan plane is tangent to the sphere. The green normal is perpendicular to it and points outward. The gradient of $1/r$ points inward, so its outward flux is negative. Arrow lengths are illustrative. The area enters through $\mathrm{d}S$, while the unit normal is independent of the radius R.

The first equality is distributional because $1/r$ is singular at the origin. One can justify it using smooth approximations to the indicator of the ball, since the gradient is regular near its boundary. The hypotheses of the classical divergence theorem do not hold on the entire ball.

The condition at infinity fixes B, and the point source in Poisson's equation fixes C. Any $R > 0$ works for the second step because all charge is at the origin and the factors R^2 cancel. The factor ε_0 was already present in the SI Poisson equation and is not an integration constant.

Therefore, in SI units, the solution is

$$\boxed{\varphi(\mathbf{x}) = -\frac{e}{4\pi\varepsilon_0 r}, \qquad \mathbf{E}(\mathbf{x}) = -\frac{e}{4\pi\varepsilon_0 r^2}\hat{\mathbf{r}}, \qquad \mathbf{B} = 0 \quad (r > 0).} \tag{307}$$



$\rho(\mathbf{x}) = -e\delta^{(3)}(\mathbf{x})$

$-\nabla^2\varphi = \rho/\varepsilon_0$

Equipotentials: $r =$ constant

$\mathbf{E} = -\boldsymbol{\nabla}\varphi$

Field pointing toward the charge $-e$

Figure 12. Solution of Poisson's equation for a negative point charge. The drawing is a spatial cross section. The green circles cut the spheres on which the potential φ is constant. The red arrows give the direction of the electric field $\mathbf{E} = -\boldsymbol{\nabla}\varphi$, perpendicular to these surfaces. Their lengths are schematic, while the magnitude decreases as $1/r^2$.

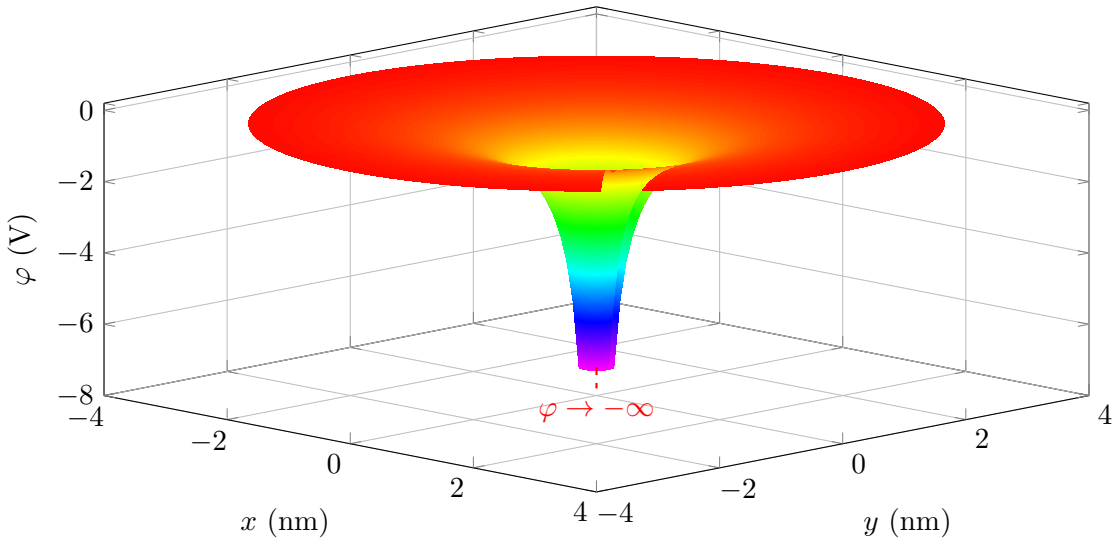


Figure 13. Coulomb potential on a spatial cross section. The potential in SI units is $\varphi(x, y, 0) = -e/(4\pi\varepsilon_0\sqrt{x^2+y^2})$. The horizontal axes give distances in nanometers ($1\,\mathrm{nm} = 10^{-9}\,\mathrm{m}$), and the height gives the potential in volts. At 1 nm from the origin, $\varphi \simeq -1.44\,\mathrm{V}$. This is the $z = 0$ section of the three-dimensional potential. The point $r < 0.20\,\mathrm{nm}$ is omitted so that the surface is visible. At the origin the potential tends to $-\infty$ and has no finite minimum. Away from the origin it approaches zero from below.

The electric field points toward the negative charge. It is time-independent and its magnitude decreases in space as $1/r^2$. It is singular at the origin, where Maxwell's equations are interpreted distributionally. The formulas already include ε_0. With e in coulombs and r in meters, φ is in volts and $\mathbf{E}$ in volts per meter.

With $C = e/(4\pi\varepsilon_0)$ and $B = 0$, the potential is determined explicitly: $\varphi(r) = -e/(4\pi\varepsilon_0 r)$, the Coulomb potential of the electron charge in the classical point-charge model. The graph shows its values on the spatial section $z = 0$, with height in volts.

Varying the QED action with respect to A_μ gives Maxwell's equations. In electrostatics, Gauss's law gives Poisson's equation. For the source $\rho = -e\delta^{(3)}$, the conditions determine the Coulomb potential. **We thus recover the classical result in the electromagnetic sector of the theory**. The calculation solves Maxwell's equations for a prescribed point source. A quantum state of the electron or a joint Dirac–Maxwell solution would require further equations.

□

Example 15 (Uniformly charged ball)**.** Distribute a total charge Q uniformly throughout a ball of radius $R > 0$ at rest. The density ρ_0 is charge per unit volume. It is constant inside the ball, so

$$Q = \rho_0 \,\mathrm{Vol}(B_R) = \rho_0 \frac{4\pi R^3}{3} \quad \Longrightarrow \quad \rho_0 = \frac{3Q}{4\pi R^3}. \tag{308}$$

In SI, Q is measured in coulombs and ρ_0 in $\mathrm{C/m^3}$. The charge occupies the whole volume, in contrast with a point charge. Define the distribution by

$$\rho(r) := \begin{cases} \rho_0, & 0 \le r \le R, \\ 0, & r > R, \end{cases} \qquad \rho_0 := \frac{3Q}{4\pi R^3}, \qquad \mathbf{J} = 0. \tag{309}$$

The value of ρ exactly at $r = R$ does not affect volume integrals. There is no surface charge. In SI we seek a spherically symmetric electrostatic solution regular at the origin with $\varphi \to 0$ at infinity. Choose $\mathbf{A} = 0$, so $\mathbf{E} = -\boldsymbol{\nabla}\varphi$ and $\mathbf{B} = 0$.

Use the same Gauss law obtained by varying the action. Substituting the electrostatic field gives Poisson's equation.

$$\boldsymbol{\nabla}\cdot\mathbf{E}=\frac{\rho}{\varepsilon_0},\qquad \mathbf{E}=-\boldsymbol{\nabla}\varphi\quad\Longrightarrow\quad \nabla^2\varphi=-\frac{\rho}{\varepsilon_0}. \tag{310}$$

Spherical symmetry leads us to seek $\varphi=\varphi(r)$, with $r=\sqrt{x^2+y^2+z^2}$. We need $\nabla^2\varphi$ in terms of derivatives with respect to r. For $r>0$, the chain rule gives

$$\partial_i r=\frac{x_i}{r},\qquad \partial_i\varphi(r)=\varphi'(r)\frac{x_i}{r},\qquad \varphi'(r)\coloneqq\frac{\mathrm{d}\varphi}{\mathrm{d}r}. \tag{311}$$

Therefore the Laplacian in Poisson's equation is

$$\nabla^2\varphi=\sum_{i=1}^{3}\partial_i^2\varphi=\sum_{i=1}^{3}\partial_i\left(\varphi'(r)\frac{x_i}{r}\right). \tag{312}$$

To expand this sum, differentiate both factors. Here x_i is the corresponding Cartesian coordinate, with $i\in\{1,2,3\}$. This three-dimensional calculation uses the spatial Euclidean metric. For each fixed i, with no summation, first differentiate $x_i/r=x_i(1/r)$. Since r depends on x_i, the chain rule gives

$$\partial_i\left(\frac{1}{r}\right)=-\frac{1}{r^2}\partial_i r=-\frac{1}{r^2}\frac{x_i}{r}=-\frac{x_i}{r^3}. \tag{313}$$

We now apply the product rule and replace that result:

$$\partial_i\left(\frac{x_i}{r}\right)=\underbrace{(\partial_i x_i)}_{1}\frac{1}{r}+x_i\,\partial_i\left(\frac{1}{r}\right) \tag{314}$$

$$=\frac{1}{r}+x_i\left(-\frac{x_i}{r^3}\right)=\frac{1}{r}-\frac{x_i^2}{r^3}. \tag{315}$$

The other factor is derived by the same chain rule:

$$\partial_i\varphi'(r)=\varphi''(r)\partial_i r=\varphi''(r)\frac{x_i}{r}. \tag{316}$$

With both derivatives already calculated, we get the second derivative of potential:

$$\partial_i\left(\varphi'(r)\frac{x_i}{r}\right)=\varphi''(r)\frac{x_i^2}{r^2}+\varphi'(r)\left(\frac{1}{r}-\frac{x_i^2}{r^3}\right). \tag{317}$$

We substitute this result in the sum of Laplacian. As $\sum_{i=1}^{3}x_i^2=r^2$, we obtain

$$\nabla^2\varphi=\varphi''(r)\underbrace{\sum_{i=1}^{3}\frac{x_i^2}{r^2}}_{1}+\varphi'(r)\underbrace{\sum_{i=1}^{3}\left(\frac{1}{r}-\frac{x_i^2}{r^3}\right)}_{3/r-1/r=2/r} \tag{318}$$

$$=\varphi''(r)+\frac{2}{r}\varphi'(r). \tag{319}$$

We now have the radial Laplacian in Poisson's equation. To integrate the equation, write $\varphi''+2\varphi'/r$ as a single derivative. The product rule verifies the equality.

$$\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\big(r^2\varphi'(r)\big)=\frac{1}{r^2}\big(2r\varphi'(r)+r^2\varphi''(r)\big)=\frac{2}{r}\varphi'(r)+\varphi''(r)=\nabla^2\varphi. \tag{320}$$

We now return to Poisson and replace the expression we have just calculated:

$$\boxed{\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}(r^2\varphi'(r)) = -\frac{\rho(r)}{\varepsilon_0}, \qquad r > 0.} \tag{321}$$

Density takes different values inside and out, so we solve this same equation in two regions:

$$\boxed{\begin{aligned} &\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}(r^2\varphi'_{\mathrm{int}}) = -\frac{\rho_0}{\varepsilon_0}, \quad 0 < r < R, \\ &\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}(r^2\varphi'_{\mathrm{ext}}) = 0, \qquad r > R. \end{aligned}} \tag{322}$$

The subscripts int and ext distinguish the interior and exterior expressions for the same potential. Solve each case and then match the solutions at $r = R$.

Case 1: inside the ball ($0 < r < R$). Here $\rho = \rho_0$. We multiply the first equation by r^2 and integrate twice:

$$r^2\varphi'_{\mathrm{int}} = -\frac{\rho_0 r^3}{3\varepsilon_0} + C_1, \qquad \varphi_{\mathrm{int}} = -\frac{\rho_0 r^2}{6\varepsilon_0} - \frac{C_1}{r} + C_2. \tag{323}$$

The term $-C_1/r$ diverges at the origin if $C_1 \neq 0$. It would add a point charge at the center on top of the uniform density. Its total charge would be finite, although its density would be singular. The model excludes a point charge and requires a finite potential at $r = 0$, so we impose $C_1 = 0$.

With $C_1 = 0$, the inner potential remains, for now,

$$\boxed{\varphi_{\mathrm{int}}(r) = -\frac{\rho_0 r^2}{6\varepsilon_0} + C_2.} \tag{324}$$

Regularity leaves C_2 undetermined. The value of the potential at the center is $\varphi_{\mathrm{int}}(0) = C_2$. We will determine it by matching to the exterior solution at $r = R$.

Case 2: outside of the ball ($r > R$). Here $\rho = 0$. We multiply the second equation by r^2 and integrate twice:

$$\frac{\mathrm{d}}{\mathrm{d}r}(r^2\varphi'_{\mathrm{ext}}) = 0 \quad \Longrightarrow \quad r^2\varphi'_{\mathrm{ext}} = -K \quad \Longrightarrow \quad \varphi'_{\mathrm{ext}} = -\frac{K}{r^2}. \tag{325}$$

The second integration gives

$$\boxed{\varphi_{\mathrm{ext}}(r) = \frac{K}{r} + C_3.} \tag{326}$$

The condition $\varphi(r) \to 0$ when $r \to \infty$ fixes $C_3 = 0$, so that

$$\boxed{\varphi_{\mathrm{ext}}(r) = \frac{K}{r}.} \tag{327}$$

We still need K in the exterior solution and C_2 in the interior solution.

Matching the solutions at $r = R$. With no surface charge, the normal component of the electric field is continuous. By symmetry this is equivalent to matching the radial derivatives of the potential. We also match the potential itself.

$$\varphi'_{\mathrm{int}}(R) = \varphi'_{\mathrm{ext}}(R) \quad \Longrightarrow \quad -\frac{\rho_0 R}{3\varepsilon_0} = -\frac{K}{R^2} \quad \Longrightarrow \quad K = \frac{\rho_0 R^3}{3\varepsilon_0} = \frac{Q}{4\pi\varepsilon_0}, \tag{328}$$

$$\varphi_{\mathrm{int}}(R) = \varphi_{\mathrm{ext}}(R) \quad \Longrightarrow \quad -\frac{\rho_0 R^2}{6\varepsilon_0} + C_2 = \frac{\rho_0 R^2}{3\varepsilon_0} \quad \Longrightarrow \quad C_2 = \frac{\rho_0 R^2}{2\varepsilon_0}. \tag{329}$$

The potential is thus determined:

$$\boxed{\varphi(r) = \begin{cases} \dfrac{Q}{8\pi\varepsilon_0 R}\left(3 - \dfrac{r^2}{R^2}\right), & 0 \le r \le R, \\ \dfrac{Q}{4\pi\varepsilon_0 r}, & r \ge R. \end{cases}} \tag{330}$$

The function φ is the **electric scalar potential**. The electric field is obtained from its gradient. For a radial potential, the chain rule used above gives

$$\boldsymbol{\nabla}\varphi = \varphi'(r)\left(\frac{x}{r}, \frac{y}{r}, \frac{z}{r}\right) = \varphi'(r)\hat{\mathbf{r}}, \qquad \mathbf{E} = -\varphi'(r)\hat{\mathbf{r}} \quad (r > 0). \tag{331}$$

Inside the ball, differentiate the potential just determined.

$$\varphi'_{\text{int}}(r) = \frac{Q}{8\pi\varepsilon_0 R}\frac{\mathrm{d}}{\mathrm{d}r}\left(3 - \frac{r^2}{R^2}\right) = -\frac{Qr}{4\pi\varepsilon_0 R^3}, \tag{332}$$

$$\mathbf{E}_{\text{int}}(\mathbf{x}) = -\varphi'_{\text{int}}(r)\hat{\mathbf{r}} = \frac{Qr}{4\pi\varepsilon_0 R^3}\hat{\mathbf{r}} = \frac{Q}{4\pi\varepsilon_0 R^3}\mathbf{x}. \tag{333}$$

The last expression is defined at the origin and gives $\mathbf{E}(0) = 0$. Outside the ball,

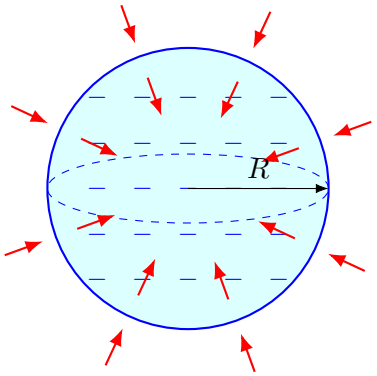
$$\varphi'_{\text{ext}}(r) = \frac{Q}{4\pi\varepsilon_0}\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{1}{r}\right) = -\frac{Q}{4\pi\varepsilon_0 r^2}, \tag{334}$$

$$\mathbf{E}_{\text{ext}}(\mathbf{x}) = -\varphi'_{\text{ext}}(r)\hat{\mathbf{r}} = \frac{Q}{4\pi\varepsilon_0 r^2}\hat{\mathbf{r}}. \tag{335}$$

The vector potential remains $\mathbf{A} = 0$, so $\mathbf{B} = \boldsymbol{\nabla} \times \mathbf{A} = 0$. Collecting the fields gives

$$\boxed{\mathbf{E}(\mathbf{x}) = \begin{cases} \dfrac{Q}{4\pi\varepsilon_0 R^3}\mathbf{x}, & r \le R, \\ \dfrac{Q}{4\pi\varepsilon_0 r^2}\hat{\mathbf{r}}, & r \ge R, \end{cases} \qquad \mathbf{B} = 0.} \tag{336}$$

At the center $\mathbf{E} = 0$ and the potential is finite. Inside, the magnitude grows linearly with r. Outside, it decreases as $1/r^2$. The expressions agree at the boundary. The second derivative of the potential can jump, corresponding to the jump in charge density.



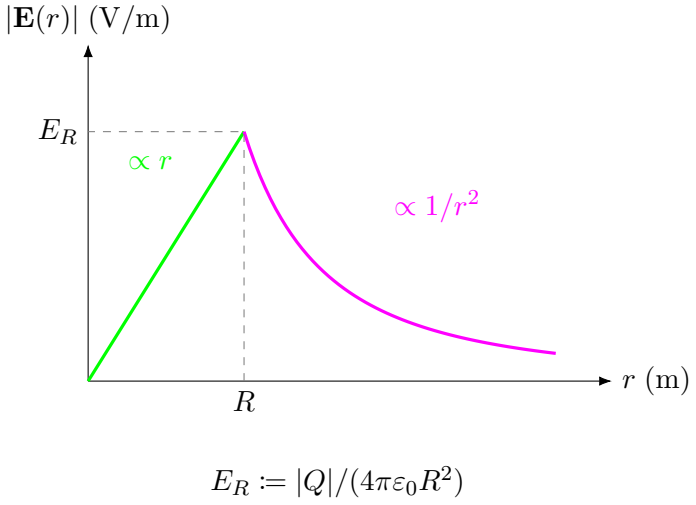
Charge distributed throughout the volume
$Q = -e$, $\rho_0 = -3e/(4\pi R^3)$

Figure 14. Uniformly charged ball with $Q = -e$. The minus signs indicate negative charge throughout the volume. The red arrows show the direction of the field toward the center. Their lengths are schematic.

The next graph gives the **magnitude of the electric field**. The horizontal axis is distance r from the center in meters, and the vertical axis is $|\mathbf{E}(r)|$ in V/m. We mark the radius R and the field magnitude E_R at the boundary. The preceding expressions give

$$E_R \coloneqq \frac{|Q|}{4\pi\varepsilon_0 R^2}, \qquad |\mathbf{E}(r)| = \begin{cases} \dfrac{|Q|}{4\pi\varepsilon_0 R^3}\, r, & 0 \le r \le R, \\ \dfrac{|Q|}{4\pi\varepsilon_0 r^2}, & r \ge R. \end{cases} \tag{337}$$

The scale is symbolic. No numerical values are assigned to Q or R. Inside the ball the field grows linearly with r. Outside it decays as the inverse square of the distance.



$$E_R \coloneqq |Q|/(4\pi\varepsilon_0 R^2)$$

Figure 15. Electric field magnitude of the charged ball. Inside ($0 \le r < R$) it rises linearly from zero (green), reaches E_R at the boundary ($r = R$), and decreases outside as $1/r^2$ (magenta). The curve is continuous and changes slope at the boundary.

For $Q = -e$, the potential is negative and the field points toward the center. **Outside the ball we recover exactly the Coulomb potential of the point charge $-e$.** The distributed charge removes the interior singularity. This is a classical model of extended charge. As in the previous example, we prescribe the source and solve Maxwell's equations without calculating ψ.

□

12 From the Lagrangian to Quantum Theory

We have constructed the QED Lagrangian and derived its equations of motion. The electrostatic examples use the classical electromagnetic sector with prescribed sources. To develop the quantum theory we also need to specify how the fields are quantized and their correlations calculated. This section outlines the route to perturbative calculations, leaving the full quantization for later.

The starting point is to separate the free part and the interaction:

$$\mathcal{L}_{\text{QED}} = \underbrace{\bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}}_{\mathcal{L}_0} + \underbrace{(-q\bar{\psi}\gamma^\mu\psi A_\mu)}_{\mathcal{L}_{\text{int}}}. \tag{338}$$

First we study free propagation and then we incorporate interaction through an expansion in the coupling q. In that expansion, the propagators and vertices will be the elements with which we construct Feynman diagrams.

From the Dirac operator to the propagator. The free equation contains the operator $i\gamma^\mu\partial_\mu - m$. We seek a matrix-valued Green function that acts as its inverse, with the convention

$$(i\gamma^\mu\partial_\mu - m)S_F(x-y) = i\delta^{(4)}(x-y)I_4. \tag{339}$$

In the quantum theory, S_F is the Feynman propagator, a time-ordered correlation of two Dirac fields. The preceding equation relates it to the free operator, and the Feynman condition selects one of the possible inverses. The propagator enters the calculation of correlations and amplitudes, as distinct from a particular field ψ.

To construct this inverse, use the Fourier transform. The function $S_F(p)$ is the Fourier transform of $S_F(x-y)$. They represent the same propagator in position space and momentum space. The difference $z = x - y$ is the displacement between two spacetime points. Position integrals run over $M = \mathbb{R}^{1,3}$. Write $\mathcal{P} := (\mathbb{R}^{1,3})^*$ for the space of covariant four-momenta, identified with $\mathbb{R}^4$ by the coordinates p_μ. The pairing $p \cdot z = p_\mu z^\mu$ appears in the exponential. The variables p, k, ℓ belong to the same space $\mathcal{P}$. In our convention the forward transform is

$$S_F(p) = \int_M \mathrm{d}^4 z\, e^{ip\cdot z} S_F(z). \tag{340}$$

The inverse transform recovers the position-space propagator.

$$S_F(x-y) = \int_{\mathcal{P}} \frac{\mathrm{d}^4 p}{(2\pi)^4}\, e^{-ip\cdot(x-y)} S_F(p), \qquad \partial_\mu e^{-ip\cdot x} = -ip_\mu e^{-ip\cdot x}. \tag{341}$$

To see how the Green function equation transforms, set $z = x - y$ and substitute the Fourier representation of $S_F(z)$. The derivative acts on the exponential, since $S_F(p)$ is independent of z.

$$(i\gamma^\mu\partial_\mu - m)S_F(z) = \int_{\mathcal{P}} \frac{\mathrm{d}^4 p}{(2\pi)^4}\, (i\gamma^\mu\partial_\mu - m)e^{-ip\cdot z} S_F(p) \tag{342}$$

$$= \int_{\mathcal{P}} \frac{\mathrm{d}^4 p}{(2\pi)^4}\, e^{-ip\cdot z} (\underbrace{i\gamma^\mu(-ip_\mu)}_{\gamma^\mu p_\mu} - m) S_F(p). \tag{343}$$

The change is therefore $\partial_\mu \mapsto -ip_\mu$, or equivalently $i\partial_\mu \mapsto p_\mu$. We also write the delta on the right side by its representation of Fourier:

$$\delta^{(4)}(z) = \int_{\mathcal{P}} \frac{\mathrm{d}^4 p}{(2\pi)^4}\, e^{-ip\cdot z}. \tag{344}$$

Let us now gather the two substitutions in the equation (339), without changing the equation that we are solving:

$$\underbrace{(i\gamma^\mu\partial_\mu - m)S_F(z)}_{\text{lado izquierdo de (339)}} = \underbrace{i\delta^{(4)}(z)I_4}_{\text{lado derecho de (339)}}. \tag{345}$$

The preceding calculation expresses the left side using $S_F(p)$. The delta representation gives the right side using the same exponential. Substitute both results to obtain

$$\int_{\mathcal{P}} \frac{\mathrm{d}^4 p}{(2\pi)^4}\, e^{-ip\cdot z}(\gamma^\mu p_\mu - m)S_F(p) = \int_{\mathcal{P}} \frac{\mathrm{d}^4 p}{(2\pi)^4}\, e^{-ip\cdot z}\, iI_4. \tag{346}$$

To recover an equality in momentum space, apply the forward transform to both sides. Multiplication by $e^{i\ell\cdot z}$ followed by integration over z is understood in the sense of tempered distributions. We use

$$\int_M \mathrm{d}^4 z\, e^{i(\ell-p)\cdot z} = (2\pi)^4 \delta^{(4)}(\ell - p). \tag{347}$$

The integral over z produces the delta on both sides and cancels the factor $(2\pi)^{-4}$ in the measure. Thus,

$$\begin{aligned} \int_{\mathcal{P}} \mathrm{d}^4 p\, \delta^{(4)}(\ell - p)(\gamma^\mu p_\mu - m) S_F(p) &= \int_{\mathcal{P}} \mathrm{d}^4 p\, \delta^{(4)}(\ell - p)\, i I_4 \\ \implies \quad (\gamma^\mu \ell_\mu - m) S_F(\ell) &= i I_4. \end{aligned} \tag{348}$$

In this operation the delta reproduces each distribution in the variable ℓ. Relabeling ℓ as p gives

$$\boxed{(\gamma^\mu p_\mu - m) S_F(p) = i I_4.} \tag{349}$$

This is the same Green function equation in momentum space. Fourier transformation replaces differentiation by multiplication, so we can find the inverse by matrix algebra.

To solve the equation, invert the matrix $\gamma^\mu p_\mu - m I_4$. We use Feynman slash notation to abbreviate the contraction with the Dirac matrices.

$$\not{p} \coloneqq \gamma^\mu p_\mu, \qquad (\not{p} - m I_4) S_F(p) = i I_4. \tag{350}$$

The Clifford relation gives the square of $\not{p}$. Since $p_\mu p_\nu$ is symmetric, we can exchange the indices and average the two expressions.

$$\not{p}^2 = \gamma^\mu \gamma^\nu p_\mu p_\nu = \frac{1}{2}(\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu) p_\mu p_\nu \tag{351}$$

$$= g^{\mu\nu} p_\mu p_\nu I_4 = p^2 I_4. \tag{352}$$

We multiply now by the matrix with the opposite sign in the mass term. Cross terms are canceled:

$$(\not{p} - m I_4)(\not{p} + m I_4) = \not{p}^2 - m^2 I_4 = (p^2 - m^2) I_4. \tag{353}$$

When $p^2 \neq m^2$, we can divide by the scalar $p^2 - m^2$.[13] This gives the inverse and, on substitution into the previous equation, the propagator away from the poles.

$$(\not{p} - m I_4)^{-1} = \frac{\not{p} + m I_4}{p^2 - m^2}, \qquad S_F(p) = \frac{i(\not{p} + m I_4)}{p^2 - m^2}. \tag{354}$$

So far we have performed an algebraic inversion. The momenta with $p^2 = m^2$ make the denominator vanish. The Green function equation alone leaves the treatment of the poles undetermined. The Feynman prescription makes the additional choice.

$$\boxed{S_F(p) = \lim_{\epsilon \to 0^+} \frac{i(\not{p} + m I_4)}{p^2 - m^2 + i\epsilon} = \lim_{\epsilon \to 0^+} \frac{i(\gamma^\mu p_\mu + m I_4)}{p^2 - m^2 + i\epsilon}.} \tag{355}$$

The limit is distributional. **The term $i\epsilon$ selects the Green function independently of the matrix multiplication.**

[13] This condition lets us invert the matrix $\not{p} - m I_4$. Multiplying it by $\not{p} + m I_4$ gives $(p^2 - m^2) I_4$. If $p^2 = m^2$, there are nonzero spinors $u(p)$ satisfying $(\not{p} - m I_4) u(p) = 0$, the free Dirac equation in momentum space. The matrix has a nontrivial kernel and is singular. These physical momenta are still included. Ordinary inversion applies away from them, and the Feynman prescription then defines the propagator as a distribution including the singularity.

The prescription corresponds to time ordering of the propagator. The parameter ϵ is distinct from the permittivity ε_0 in the electrostatic examples.

We have obtained $S_F(p)$, the Dirac field propagator in momentum space with the Feynman prescription. To recover the position-space propagator sought at the outset, substitute this result in the inverse transform.

$$\boxed{S_F(x-y) = \lim_{\epsilon\to 0^+} \int_{\mathcal{P}} \frac{\mathrm{d}^4 p}{(2\pi)^4}\, e^{-ip\cdot(x-y)} \frac{i(\gamma^\mu p_\mu + mI_4)}{p^2 - m^2 + i\epsilon}.} \tag{356}$$

The limit is distributional. This integral expression determines the position-space propagator and satisfies the initial Green function equation. We started from a differential equation, solved it by Fourier transformation in momentum space, and returned to position space. Expressing the integral in special functions would be a further step.

The electromagnetic field propagator. So far we have worked with the Dirac field. Now seek the propagator of the potential A_μ by the same method. Identify its free operator, pass to momentum space, and invert it. This calculation uses only the free Maxwell term. The interaction with ψ enters afterward. The function we seek is

$$D_{\mu\nu}(x-y) \coloneqq \langle 0|T\{A_\mu(x)A_\nu(y)\}|0\rangle_0. \tag{357}$$

Start with the Maxwell action. Expand the curvature by substituting its definition and distributing the products.

$$\begin{aligned} F_{\mu\nu}F^{\mu\nu} &= (\partial_\mu A_\nu - \partial_\nu A_\mu)(\partial^\mu A^\nu - \partial^\nu A^\mu) \\ &= (\partial_\mu A_\nu)(\partial^\mu A^\nu) - (\partial_\mu A_\nu)(\partial^\nu A^\mu) \\ &\quad - (\partial_\nu A_\mu)(\partial^\mu A^\nu) + (\partial_\nu A_\mu)(\partial^\nu A^\mu) \\ &= 2(\partial_\mu A_\nu)(\partial^\mu A^\nu) - 2(\partial_\mu A_\nu)(\partial^\nu A^\mu). \end{aligned} \tag{358}$$

In the last equality we interchange the dummy indices μ, ν in the last two terms. The factor $-1/4$ in the action thus becomes $-1/2$.

$$S_{\mathrm{M}} = -\frac{1}{4}\int_M \mathrm{d}^4x\, F_{\mu\nu}F^{\mu\nu} = -\frac{1}{2}\int_M \mathrm{d}^4x\, [(\partial_\mu A_\nu)(\partial^\mu A^\nu) - (\partial_\mu A_\nu)(\partial^\nu A^\mu)]\,. \tag{359}$$

Integrate each of the two terms by parts. Assume boundary conditions allowing the boundary integrals to be omitted.

$$\int_M \mathrm{d}^4x\, (\partial_\mu A_\nu)(\partial^\mu A^\nu) = -\int_M \mathrm{d}^4x\, A_\nu \Box A^\nu, \tag{360}$$

$$\int_M \mathrm{d}^4x\, (\partial_\mu A_\nu)(\partial^\nu A^\mu) = -\int_M \mathrm{d}^4x\, A_\nu \partial^\nu \partial_\mu A^\mu. \tag{361}$$

By replacing these results in the action, the signs remain

$$\begin{aligned} S_{\mathrm{M}} &= \frac{1}{2}\int_M \mathrm{d}^4x\, \left[A_\nu \Box A^\nu - A_\nu \partial^\nu \partial_\mu A^\mu\right] \\ &= \frac{1}{2}\int_M \mathrm{d}^4x\, A_\mu \left[g^{\mu\nu}\Box - \partial^\mu\partial^\nu\right] A_\nu. \end{aligned} \tag{362}$$

Here we use $A^\mu = g^{\mu\nu}A_\nu$, the constancy of the metric, and the freedom to rename summation indices. This identifies the operator

$$S_{\mathrm{M}} = \frac{1}{2}\int_M \mathrm{d}^4x\, A_\mu K^{\mu\nu} A_\nu, \qquad K^{\mu\nu} \coloneqq g^{\mu\nu}\Box - \partial^\mu\partial^\nu. \tag{363}$$

This is the operator we want to invert. It annihilates potentials of the form $A_\nu = \partial_\nu \alpha$ because the derivatives commute.

$$K^{\mu\nu}\partial_\nu \alpha = \partial^\mu \Box \alpha - \partial^\mu \Box \alpha = 0. \tag{364}$$

Thus K has a nontrivial kernel. Distinct potentials can give the same result under the operator. Indeed,

$$K^{\mu\nu}(A_\nu - \partial_\nu \alpha) = K^{\mu\nu} A_\nu. \tag{365}$$

Therefore KA does not determine a unique potential. An inverse of K on the full space of potentials does not exist.

These directions correspond to gauge freedom. Consider the equivalence relation

$$A_\mu \sim A'_\mu \quad \text{if} \quad A'_\mu = A_\mu - \partial_\mu \alpha. \tag{366}$$

Each class contains potentials representing the same $F_{\mu\nu}$. Passing to these classes removes gauge redundancy. Constructing a propagator still requires boundary conditions and a prescription for the poles. In the covariant calculation below, gauge fixing handles the redundancy and yields a propagator for the potential components in the chosen gauge.

To construct the propagator in a covariant formulation, add the gauge-fixing term with $\xi \neq 0$.

$$\mathcal{L}_{\text{gf}} \coloneqq -\frac{1}{2\xi}(\partial_\mu A^\mu)^2. \tag{367}$$

Add its integral to the Maxwell action. To see the effect on the operator, integrate by parts again.

$$S_{\text{gf}} = -\frac{1}{2\xi}\int_M \mathrm{d}^4x\, (\partial^\mu A_\mu)(\partial^\nu A_\nu) \tag{368}$$

$$= \frac{1}{2\xi}\int_M \mathrm{d}^4x\, A_\mu \partial^\mu \partial^\nu A_\nu. \tag{369}$$

The quadratic action used to construct the propagator is therefore

$$S_{\text{M}} + S_{\text{gf}} = \frac{1}{2}\int_M \mathrm{d}^4x\, A_\mu \left(K^{\mu\nu} + \frac{1}{\xi}\partial^\mu \partial^\nu\right) A_\nu. \tag{370}$$

The new operator is, precisely,

$$K_\xi^{\mu\nu} = g^{\mu\nu}\Box - \left(1 - \frac{1}{\xi}\right)\partial^\mu \partial^\nu. \tag{371}$$

Choose **Feynman gauge**, $\xi = 1$. The second term vanishes, leaving $K_1^{\mu\nu} = g^{\mu\nu}\Box$. The propagator is the inverse with the Feynman condition. In our convention it satisfies

$$g^{\mu\nu}\Box_x D_{\nu\rho}(x-y) = i\delta^\mu{}_\rho \delta^{(4)}(x-y). \tag{372}$$

Now we use Fourier, like before:

$$D_{\mu\nu}(z) = \int_{\mathcal{P}} \frac{\mathrm{d}^4k}{(2\pi)^4}\, e^{-ik\cdot z} D_{\mu\nu}(k), \qquad \Box e^{-ik\cdot z} = -k^2 e^{-ik\cdot z}. \tag{373}$$

To transform the full propagator equation, set $z = x - y$ so that $\partial_x = \partial_z$. The first derivative of the exponential gives $-ik_\alpha$ and the second gives

$$\Box e^{-ik\cdot z} = g^{\alpha\beta}(-ik_\alpha)(-ik_\beta)e^{-ik\cdot z} = -k^2 e^{-ik\cdot z}. \tag{374}$$

We substitute simultaneously the transform of $D_{\nu\rho}$ and the representation of Fourier of the delta on the right side:

$$\begin{aligned} g^{\mu\nu}\Box D_{\nu\rho}(z) &= \int_{\mathcal{P}} \frac{\mathrm{d}^4k}{(2\pi)^4} e^{-ik\cdot z}[-k^2 g^{\mu\nu} D_{\nu\rho}(k)], \\ i\delta^{\mu}{}_{\rho}\delta^{(4)}(z) &= \int_{\mathcal{P}} \frac{\mathrm{d}^4k}{(2\pi)^4} e^{-ik\cdot z}\, i\delta^{\mu}{}_{\rho}. \end{aligned} \tag{375}$$

Equate the integrals. As in the Dirac case, multiplication by $e^{i\ell\cdot z}$ and integration over z produces $(2\pi)^4\delta^{(4)}(\ell - k)$, reproducing each distribution in the variable ℓ. Relabeling ℓ as k gives

$$-k^2 g^{\mu\nu} D_{\nu\rho}(k) = i\delta^{\mu}{}_{\rho}. \tag{376}$$

To solve for the propagator, contract with $g_{\sigma\mu}$ to lower the free index μ.

$$\begin{aligned} -k^2 \underbrace{g_{\sigma\mu} g^{\mu\nu}}_{\delta_{\sigma}{}^{\nu}} D_{\nu\rho}(k) &= i\, \underbrace{g_{\sigma\mu}\delta^{\mu}{}_{\rho}}_{g_{\sigma\rho}}, \\ -k^2 D_{\sigma\rho}(k) &= i g_{\sigma\rho}. \end{aligned} \tag{377}$$

For $k^2 \neq 0$, we divide by $-k^2$ and rename the free indices:

$$D_{\mu\nu}(k) = -\frac{i g_{\mu\nu}}{k^2}. \tag{378}$$

The singularity appears now in $k^2 = 0$, corresponding to the massless field. As before, algebraic inversion does not fix how to treat those poles. We incorporate Feynman's prescription:

$$\boxed{D_{\mu\nu}(k) = \lim_{\epsilon\to 0^+} \frac{-i g_{\mu\nu}}{k^2 + i\epsilon}, \qquad \xi = 1.} \tag{379}$$

This gives the photon propagator in momentum space. Its two indices correspond to the two fields A that it relates, and the denominator k^2 corresponds to a massless field. Substituting the preceding momentum-space expression into the inverse transform gives the **position-space propagator of the electromagnetic potential** in Feynman gauge.

$$\boxed{D_{\mu\nu}(x-y) = \lim_{\epsilon\to 0^+} \int_{\mathcal{P}} \frac{\mathrm{d}^4k}{(2\pi)^4}\, e^{-ik\cdot(x-y)} \frac{-i g_{\mu\nu}}{k^2 + i\epsilon}.} \tag{380}$$

The limit is distributional, and the expression satisfies the Green function equation of the Feynman-gauge operator. We now have both free propagators, S_F for the Dirac field and $D_{\mu\nu}$ for the electromagnetic potential. Gauge fixing permits the inversion. Physical predictions are independent of the choice of gauge.

12.1 From the Interaction to the Vertex

We have the free propagators for both fields. To incorporate the interaction term $\mathcal{L}_{\text{int}}$, we introduce the path integral. In natural units ($\hbar = 1$), correlations are represented formally as integrals over field configurations with weight e^{iS}, normalized by the same integral with no insertions. Schematically,

$$\langle T\mathcal{O}\rangle = \frac{\int_{\mathfrak{F}} \mathcal{D}A\, \mathcal{D}\bar{\psi}\, \mathcal{D}\psi\; \mathcal{O}\, e^{i(S_0 + S_{\text{int}})}}{\int_{\mathfrak{F}} \mathcal{D}A\, \mathcal{D}\bar{\psi}\, \mathcal{D}\psi\; e^{i(S_0 + S_{\text{int}})}}. \tag{381}$$

Here $\mathfrak{F}$ denotes the formal configuration space of $(A, \bar{\psi}, \psi)$, with the gauge-fixing term included in the action. The fermionic integral is a Berezin integral, distinct from ordinary integration on a finite-dimensional manifold. The symbol $\mathcal{O}$ is the product of fields whose correlation we seek, and T denotes time ordering. Angle brackets denote a vacuum expectation value.

Example 16 (Free correlation of two Dirac fields)**.** By choosing two Dirac field insertions in free theory, we regain the propagator

$$\mathcal{O} = \psi(x)\bar{\psi}(y) \quad \Longrightarrow \quad \langle 0|T\{\psi(x)\bar{\psi}(y)\}|0\rangle_0 = S_F(x-y). \tag{382}$$

□

The functional representation allows us to include the interaction. The free part S_0 includes gauge fixing, and fermionic integration uses anticommuting Grassmann variables.[14] The vacuum condition and the Feynman prescription are understood. The functional representation is a quantization step beyond the classical equations.

In the perturbative expansion, expand the interaction factor.

$$e^{iS_{\text{int}}} = 1 + iS_{\text{int}} + \frac{(iS_{\text{int}})^2}{2!} + \cdots, \qquad S_{\text{int}} = -q\int_M \mathrm{d}^4x\, \bar{\psi}\gamma^\mu\psi A_\mu. \tag{383}$$

This expansion is related to Dyson's expansion. The operator formulation uses a time-ordered exponential, whereas here we expand the interaction weight of the path integral. The first term uses the free theory, and the next adds one interaction insertion.

Focus on the **first-order term** iS_{int} of the expansion. Since S_{int} contains a factor q, this term is first order in the coupling and gives one interaction insertion, or vertex. To identify its coefficient, substitute the interaction action already obtained.

$$iS_{\text{int}} = i\left[-q\int_M \mathrm{d}^4x\, \bar{\psi}\gamma^\mu\psi A_\mu\right] = \int_M \mathrm{d}^4x\, \bar{\psi}\,(-iq\gamma^\mu)\,\psi A_\mu. \tag{384}$$

The factor q is already in the QED Lagrangian, and i comes from the linear term of the expansion of $e^{iS_{\text{int}}}$. Their product gives the coefficient $-iq$.

We can write the same expression showing its components:

$$iS_{\text{int}} = \int_M \mathrm{d}^4x \sum_{a,b=1}^{4} \bar{\psi}_a(x) \underbrace{(-iq\gamma^\mu)_{ab}}_{\text{interaction coefficient}} \psi_b(x)A_\mu(x). \tag{385}$$

The symbols a, b index the four spinor components in a basis of the internal fiber. The integration variables and fields are unchanged. The spacetime index μ is also summed. This notation makes the matrix product $\bar{\psi}\gamma^\mu\psi$ explicit. The earlier expansion contains the same information.

For perturbative correlations, Wick's theorem writes free correlations as field contractions, each contributing a propagator. The insertion brings together $\bar{\psi}$, ψ, and A_μ at one point. **There are three factors in a single interaction**. The factor $\bar{\psi}$ denotes the Dirac adjoint, and A_μ the electromagnetic potential. In the fermionic integral, ψ and $\bar{\psi}$ are independent integration variables as described in the preceding note. They still describe a single species of Dirac field.

We represent this insertion with two fermion lines and a photon line. The coefficient that accompanies the vertex is

$$\boxed{V^\mu = -iq\gamma^\mu.} \tag{386}$$

[14] For the Dirac field, $\bar{\psi} = \psi^\dagger\gamma^0$. In the variational calculation we treat ψ and $\bar{\psi}$ as independent variables to obtain the Dirac and adjoint equations. In the fermionic path integral these symbols denote two independent families of Grassmann variables, and $\mathcal{D}\bar{\psi}\,\mathcal{D}\psi$ is a Berezin integration measure. This does not introduce two additional physical fields. Counting the two equations of motion alone does not justify the measure. We can abbreviate the two measures by a single symbol defined to include both families. Omitting one family would change the integral. See *Advanced Quantum Field Theory*, section 2.3.1 on Gaussian integrals of Grassmann variables [11].

The figure shows the **elementary QED vertex**, a building block of Feynman diagrams. The lines correspond to the field factors, and the central point carries the matrix coefficient $-iq\gamma^\mu$. The arrows indicate fermion flow. Reading the diagram as a physical process requires specifying initial and final states. Each power of S_{int} in the expansion introduces more insertions. Connecting them by propagators constructs the perturbative diagrams. In momentum space, integrating over each interaction point also gives a delta function imposing momentum conservation at the vertex.

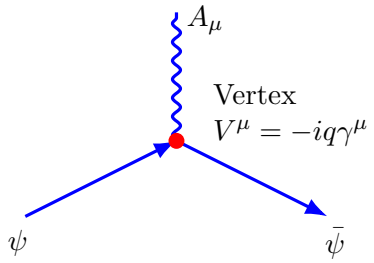


Figure 16. Elementary QED vertex. The red point represents an interaction insertion and the adjacent label gives its matrix factor $V^\mu = -iq\gamma^\mu$. The two fermion lines and the wavy photon line correspond to the three field factors in the insertion. The arrows indicate fermion flow and the coefficient $-iq\gamma^\mu$ comes from $i\mathcal{L}_{\text{int}}$.

Second order: two connected vertices. To see how to build a diagram from these pieces, take the next term in the expansion.

$$\frac{(iS_{\text{int}})^2}{2!} = \frac{(-iq)^2}{2!}\int_{M\times M} \mathrm{d}^4x\,\mathrm{d}^4y\,[\bar{\psi}(x)\gamma^\mu\psi(x)A_\mu(x)][\bar{\psi}(y)\gamma^\nu\psi(y)A_\nu(y)]. \tag{387}$$

The two red points correspond to the two interaction insertions of this second-order term, one at x and one at y. Associate the point x with matrix factor $V^\mu = -iq\gamma^\mu$, and the point y with factor $V^\nu = -iq\gamma^\nu$. Two vertices thus give an order-q^2 contribution. The positions x, y are integrated over. To obtain a diagram, we must also specify the correlation and the contractions between its field factors.

Example 17 (The t channel of electron–electron scattering)**.** Consider scattering of two electrons and contract $A_\mu(x)$ from the first insertion with $A_\nu(y)$ from the second. This contraction contributes the free photon propagator.

$$\langle 0|T\{A_\mu(x)A_\nu(y)\}|0\rangle_0 = D_{\mu\nu}(x-y). \tag{388}$$

The remaining fermionic factors connect to the external lines of the two electrons. This gives two vertices joined by an internal photon line. Summing equivalent assignments of the two interaction points duplicates the contribution, which cancels the factor 1/2! from the expansion. The resulting diagram has no additional automatic factor of 1/2.

To obtain a scattering contribution from these insertions, choose the external states explicitly: two incoming electrons with momenta p_1, p_2 and two outgoing electrons with momenta p_1', p_2'. These momenta are inputs for the process. The external states satisfy $p_i^2 = (p_i')^2 = m^2$ and overall conservation gives $p_1 + p_2 = p_1' + p_2'$.

In this channel the first electron line connects p_1 with p_1', and the second connects p_2 with p_2'. The momentum transfer is

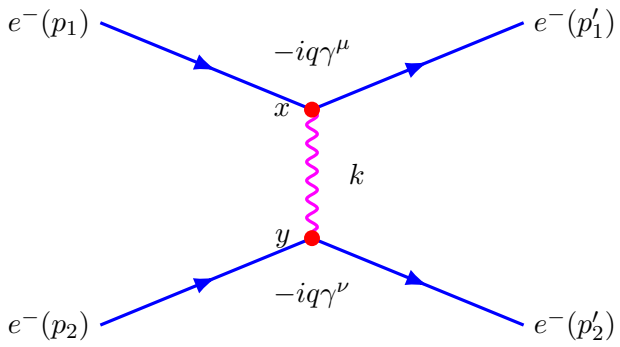
$$k = p_1 - p_1' = p_2' - p_2. \tag{389}$$

The channel takes its name from the invariant $t = k^2$. After passing to momentum space and amputating the external propagators, assign a spinor $u(p)$ to each incoming electron and its adjoint $\bar{u}(p')$

to each outgoing electron. The contribution of the t channel, with the overall momentum-conservation delta omitted, is

$$i\mathcal{M}_t = \underbrace{[\bar{u}(p_1')(-iq\gamma^\mu)u(p_1)]}_{\text{first fermion line}} \quad \underbrace{D_{\mu\nu}(k)}_{\text{internal photon}} \quad \underbrace{[\bar{u}(p_2')(-iq\gamma^\nu)u(p_2)]}_{\text{second fermion line}}. \tag{390}$$

Here $i\mathcal{M}_t$ is the contribution to the amplitude from the chosen channel. The formula combines one vertex on each fermion line with the propagator joining them. The figure shows these factors with the external momenta just introduced.



Channel t: order q^2

Figure 17. Electron–electron scattering by one-photon exchange: the t channel at second order in QED. Straight lines represent the external electrons. The wavy line carries momentum transfer $k = p_1 - p_1' = p_2' - p_2$ and contributes the photon propagator $D_{\mu\nu}(k)$ to the amplitude. The red points correspond to the two insertions of $(iS_{\text{int}})^2/2!$ at x and y, over whose positions we integrate. The wavy line connects the factors $A_\mu(x)$ and $A_\nu(y)$ from those insertions. Each vertex contributes $-iq\gamma^\mu$ with the corresponding index. The arrows indicate fermion flow. The diagram is schematic and does not depict spatial trajectories.

For identical electrons, the amplitude also includes the channel that exchanges the final electrons, with the relative fermionic sign. The figure shows the t channel contribution to that amplitude.

□

The Maxwell term is quadratic in A_μ and gives no elementary three- or four-photon vertices. We have combined the ingredients in one second-order channel. Full amplitudes, cross sections, and quantum corrections are beyond the scope of this text.

13 Conclusions

Clifford algebra, Spin, and the associated bundle explain the Dirac field as a section and show how its components depend on the frame. In Minkowski spacetime a global trivialization lets us calculate with functions valued in $\mathbb{C}^4$ while retaining that geometric interpretation.

The Lagrangian density of the massive Dirac field and its action are invariant under global phase rotations $U(1)$. Making the phase local creates an extra term that the ordinary derivative cannot compensate for. Introducing $D_\mu = \partial_\mu + iqA_\mu$ and transforming the potential together with the spinor cancels that term for any field configuration.

The curvature of this connection is $F_{\mu\nu}$. With Maxwell dynamics, the construction gives the minimal QED Lagrangian and the coupled Dirac and Maxwell equations. The distinction between conservation

on solutions and invariance of the action explains the need for the gauge field and why the free theory lacks local gauge symmetry.

A gradient connection rewrites the free field locally and has zero curvature. With the Maxwell term, the potential responds to the Dirac current. For prescribed electrostatic sources the same equations recover the Coulomb potential and the potential of a uniformly charged ball.

In perturbation theory, the free operators give the propagators and the interaction term gives the vertex. Joining two vertices with a photon propagator yields the t channel contribution to electron–electron scattering.

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